Capture plan, second real data set

80 layouts x 4 rounds = 320 scenes (5 readings each). One page per layout, in capture order.

Arena centre (the + on every page): Motive x = -0.28 m, y = -0.58 m. Mark it on the floor with tape.

Box positions are given from that centre in cm (dx along Motive x, dy along Motive y) and in Motive metres.

Fastest way: place the boxes roughly, then run python -m tools.place_check --layout Rxx : it reads OptiTrack and

says, live, how far every box (and, for the random family, every robot) is from its target. Within 5 cm is fine.

Triangles: box orientation, by eye. Blue circles: robots with heading. Red hatched: obstacle (about 30 cm).

Yellow: white object that is not a box. Dotted blue circle: the circle the robots stand on.

Days

day 1: R01-R16 (16 layouts; boxes 5)

day 2: R17-R36 (20 layouts; boxes 3, 5)

day 3: R37-R52 (16 layouts; boxes 7)

day 4: R53-R68 (16 layouts; boxes 5, 7)

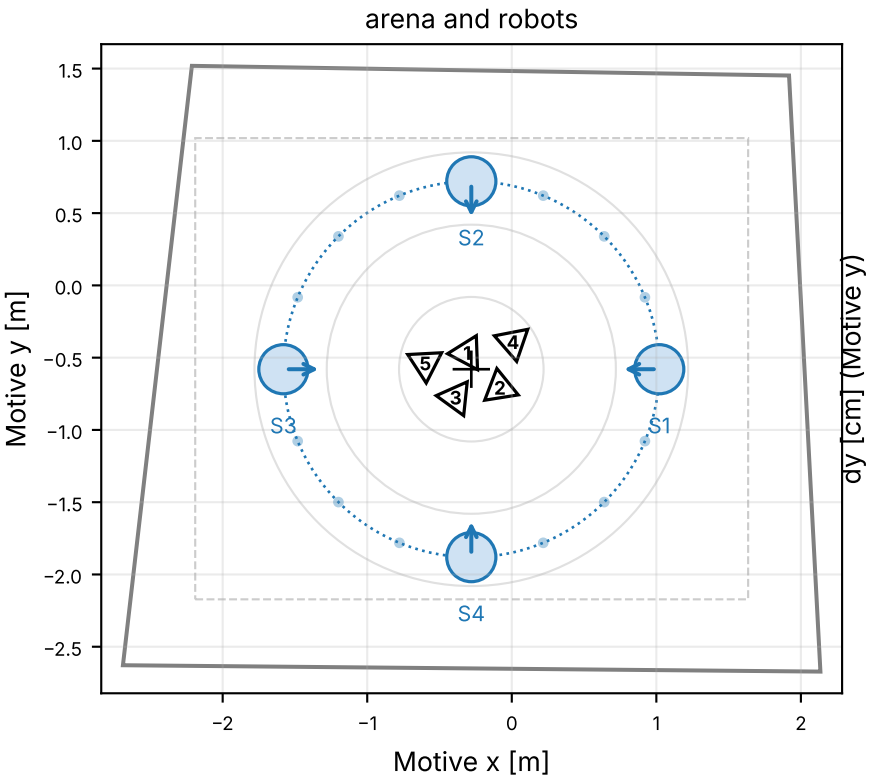
day 5: R69-R80 (12 layouts; boxes 1, 3, 5, 7)

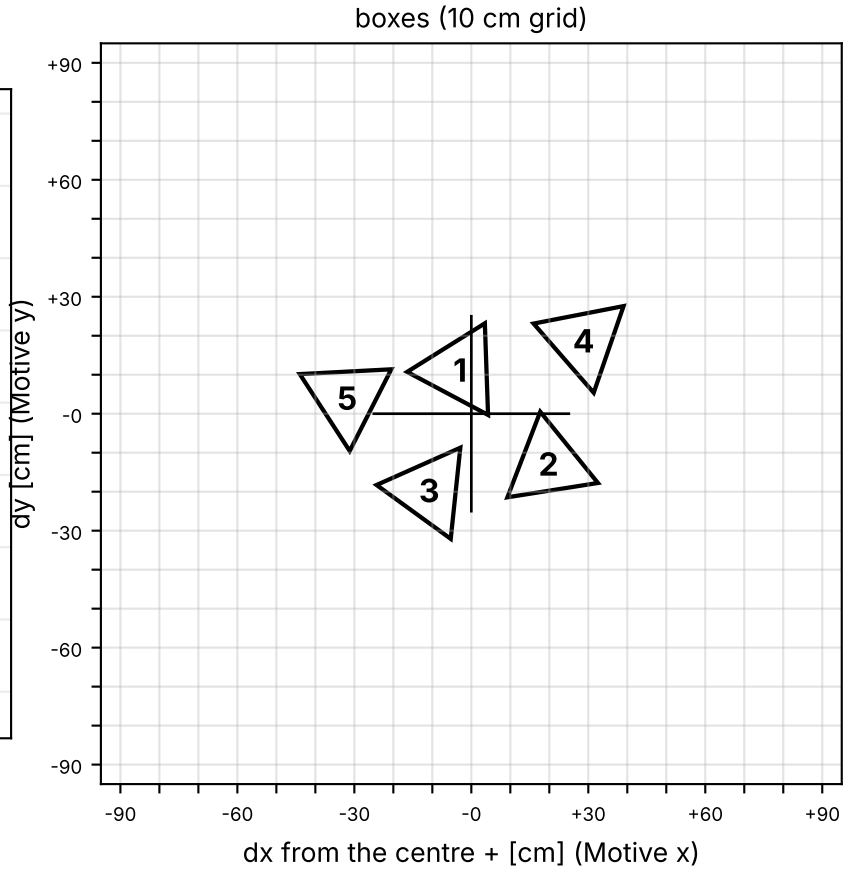
Rules

- Once, before starting: find which way Motive +x and +y point on the floor (slide a box and watch place_check) and mark them with tape arrows at the centre.
- After the first 2-3 layouts of day 1: commit and push, and wait for the check before going on.
- place_check: Enter starts the capture of that layout (launch_all with the right options).
- --box lists only the boxes on the floor (ids 11-17 in this order); the others out of the arena.
- Keep --camera-offset-forward-m 0.12 (the pipeline corrects the lens offset).
- Obstacle layouts: the robots drive along the circle; keep an eye on them near the obstacles.
- End of each day: git add data/survey_v2 docs/capture_plan_v2.csv, commit, push.

R01 day 1 A base (test)

5 boxes, fitta, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-3	+11
2	+20	-13
3	-11	-20
4	+29	+19
5	-32	+4

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

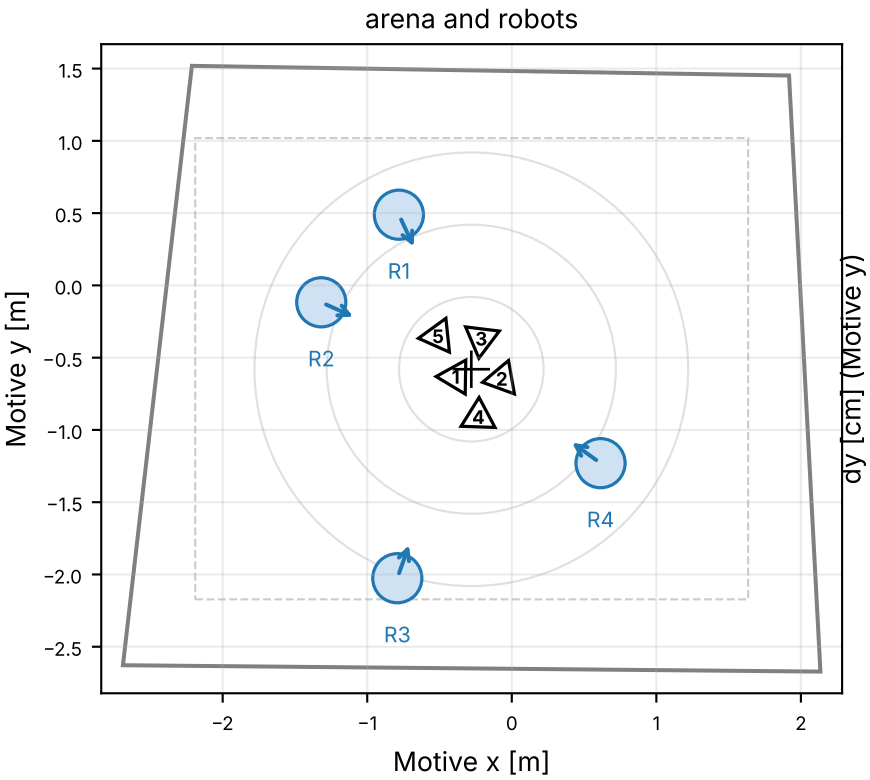
Place and capture: python -m tools.place_check --layout R01

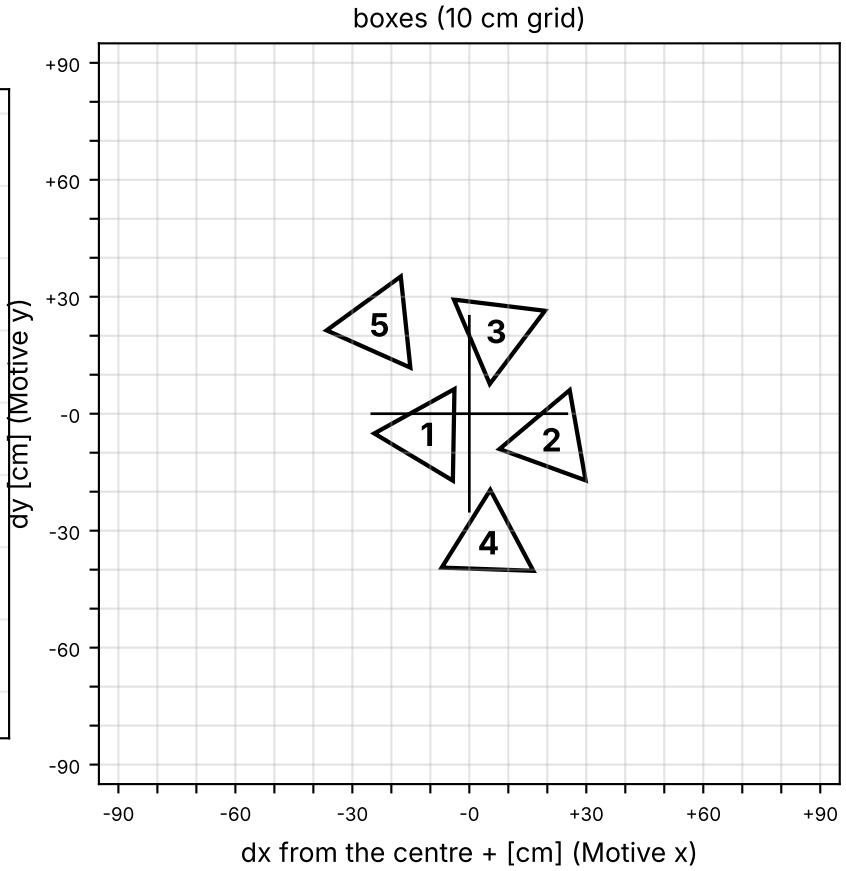
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R01 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R02 day 1 A base (test)

5 boxes, fitta, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-11	-5
2	+21	-7
3	+7	+21
4	+5	-33
5	-23	+23

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	-0.78	+0.49
R2	-1.32	-0.12
R3	-0.79	-2.03
R4	+0.61	-1.23

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

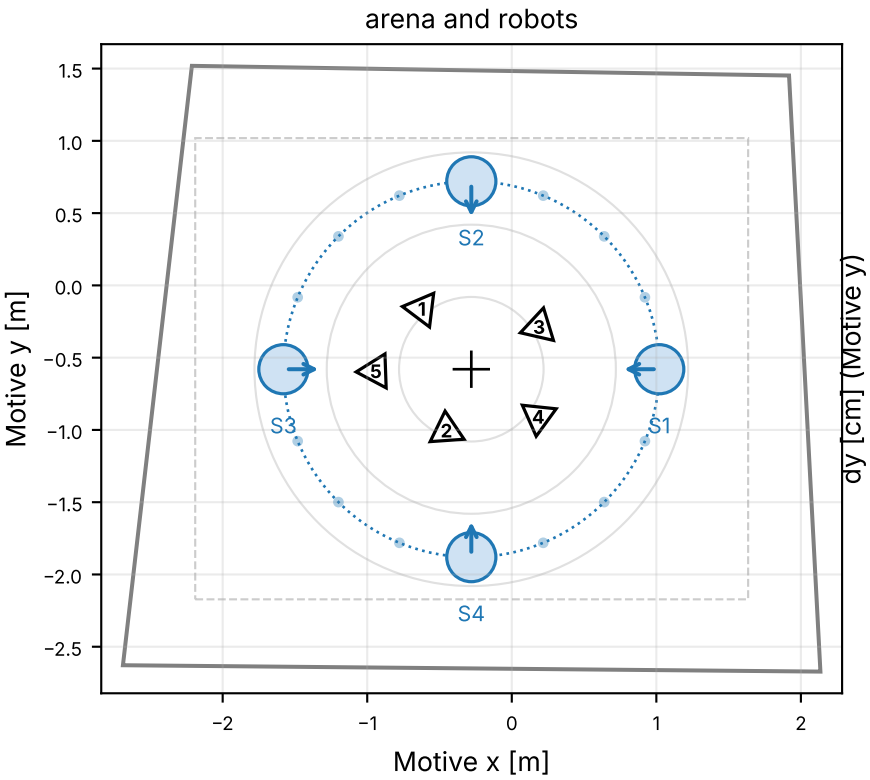
Place and capture: `python -m tools.place_check --layout R02`

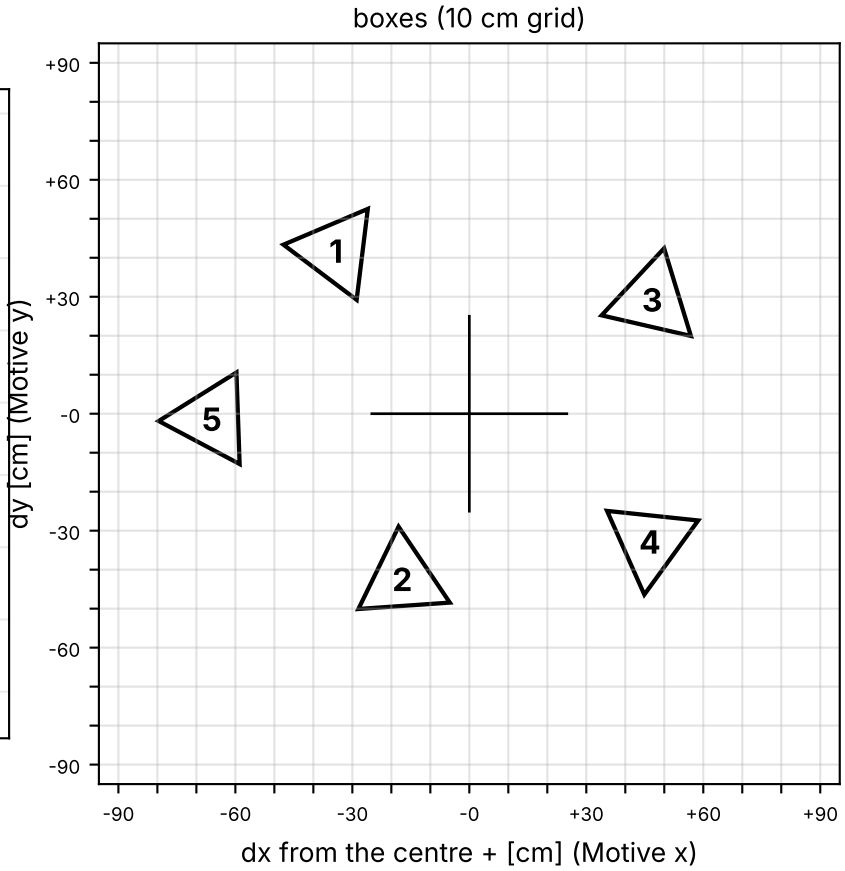
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R02 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R03 day 1 A base (test)

5 boxes, sparsa, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-34	+42
2	-17	-42
3	+47	+29
4	+46	-33
5	-66	-1

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

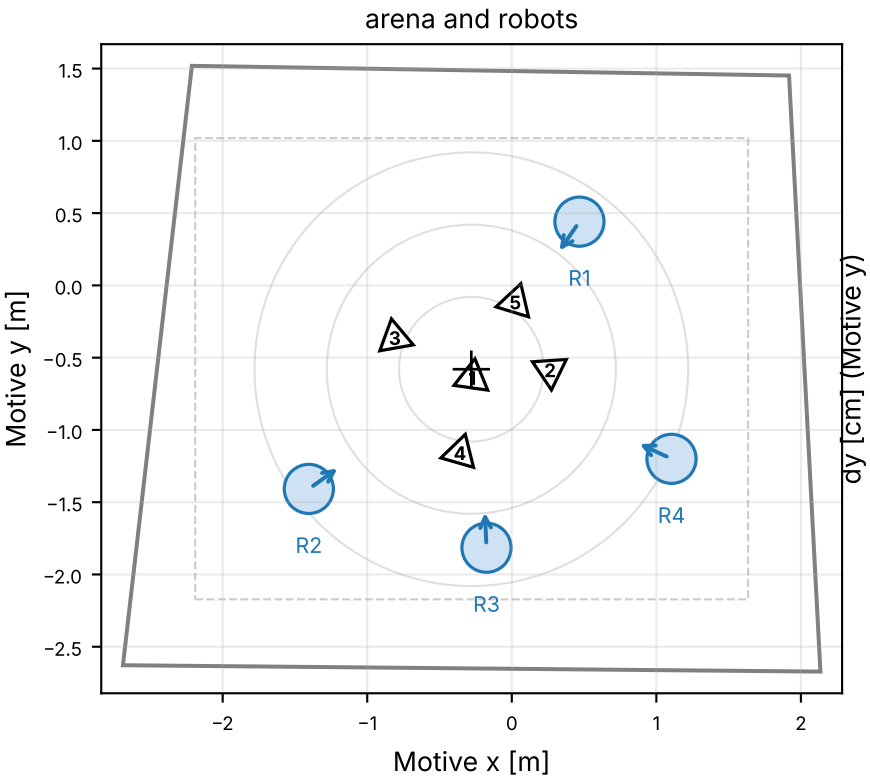
Place and capture: python -m tools.place_check --layout R03

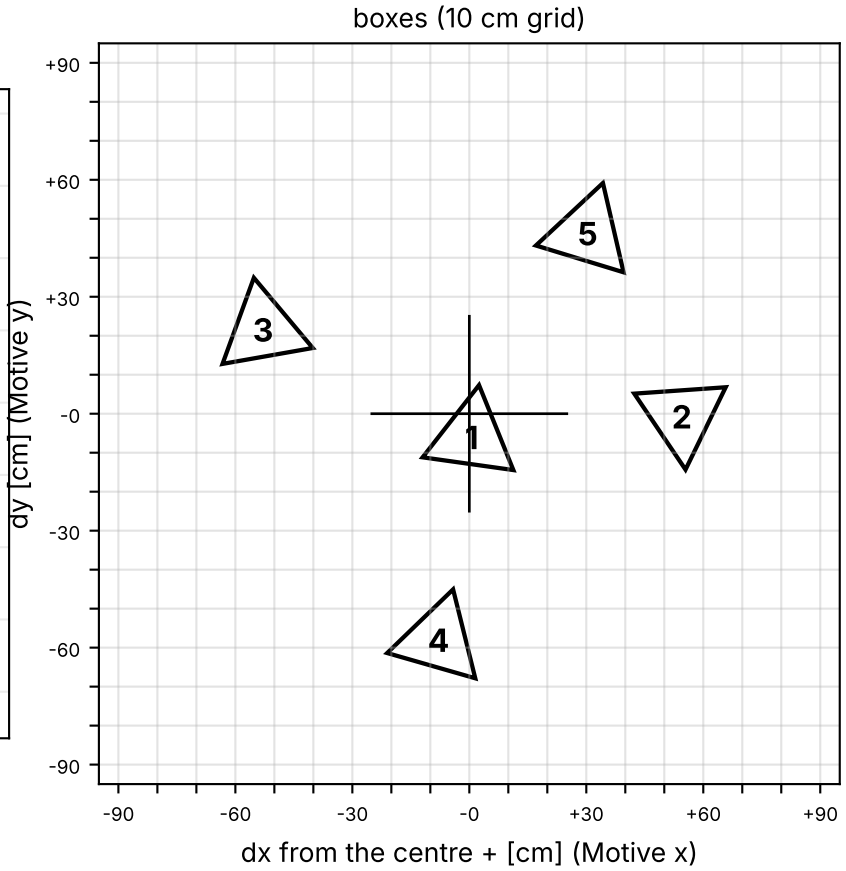
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R03 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R04 day 1 A base (test)

5 boxes, sparsa, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+1	-6
2	+55	-1
3	-53	+21
4	-8	-58
5	+30	+46

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.47	+0.44
R2	-1.40	-1.41
R3	-0.17	-1.81
R4	+1.10	-1.20

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

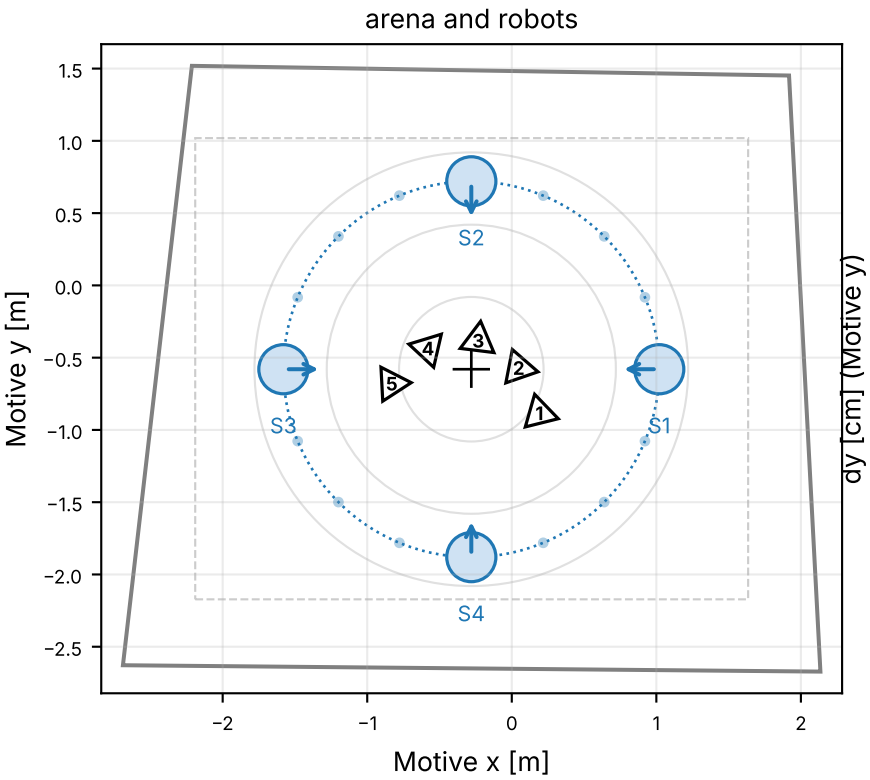
Place and capture: `python -m tools.place_check --layout R04`

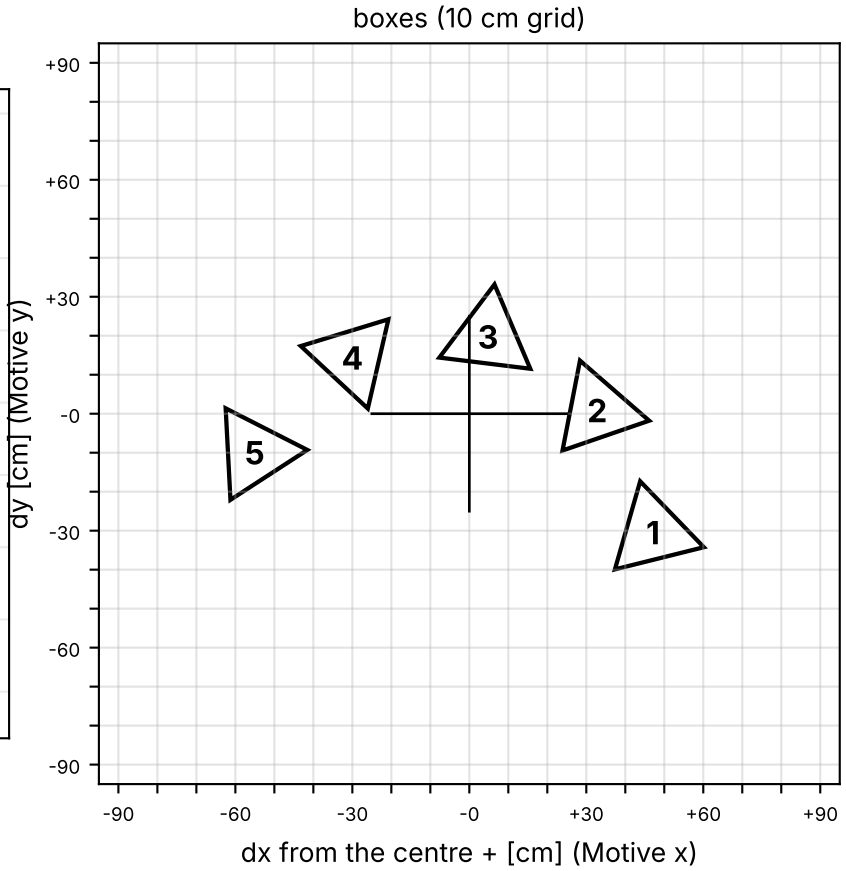
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R04 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R05 day 1 A base (test)

5 boxes, fila/arco, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+47	-31
2	+33	+1
3	+5	+20
4	-30	+14
5	-55	-10

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

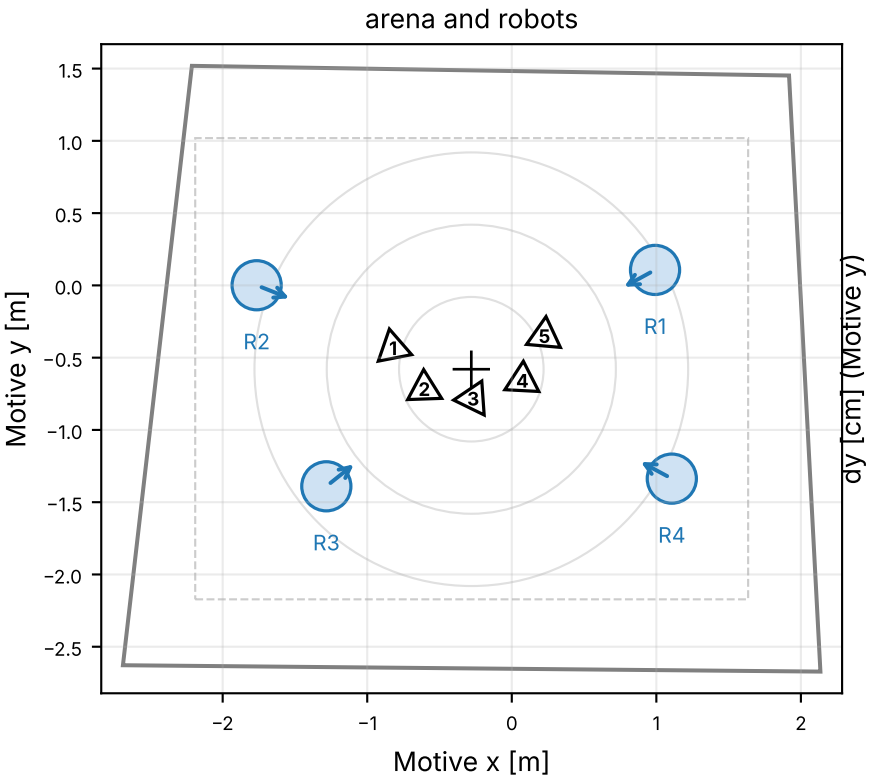
Place and capture: python -m tools.place_check --layout R05

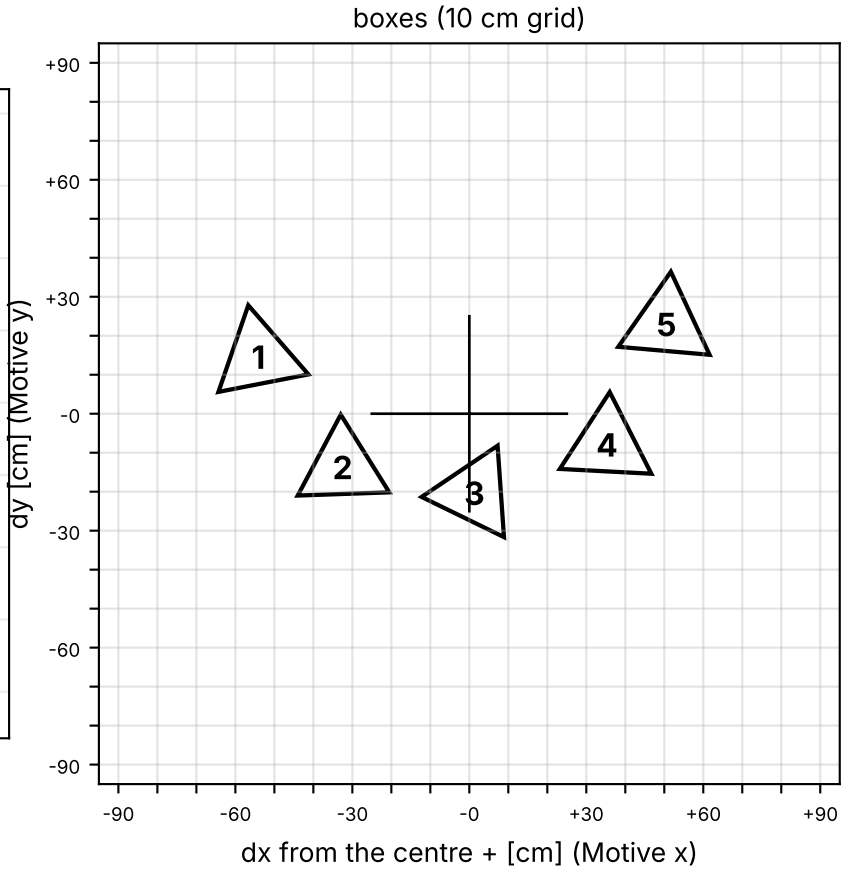
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R05 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R06 day 1 A base (test)

5 boxes, fila/arco, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-54	+14
2	-32	-14
3	+1	-20
4	+35	-8
5	+50	+23

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.99	+0.11
R2	-1.76	+0.00
R3	-1.28	-1.39
R4	+1.11	-1.34

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

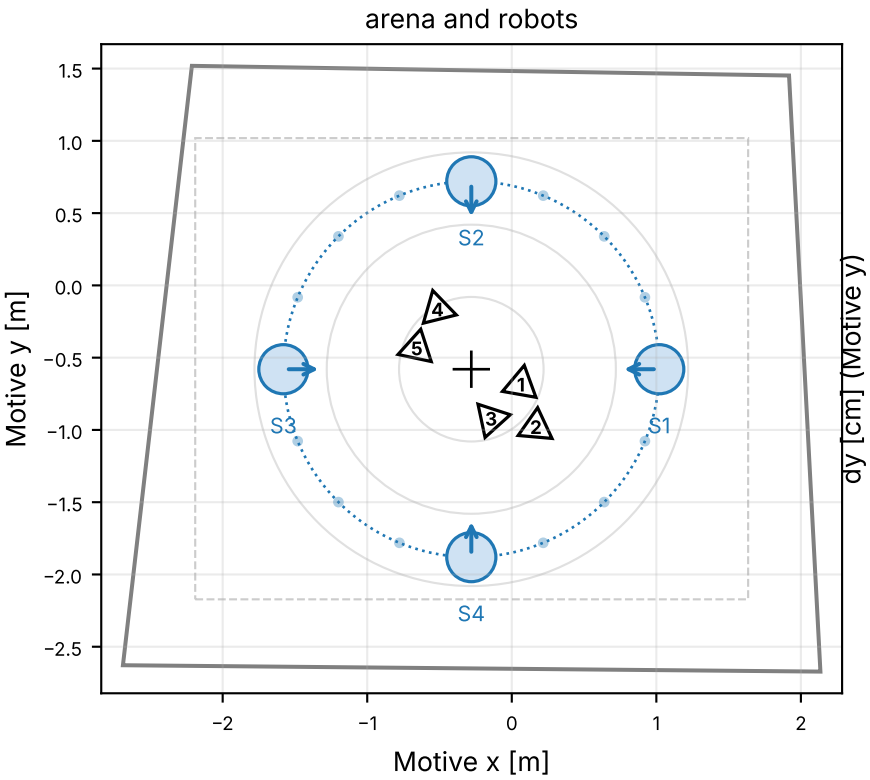
Place and capture: python -m tools.place_check --layout R06

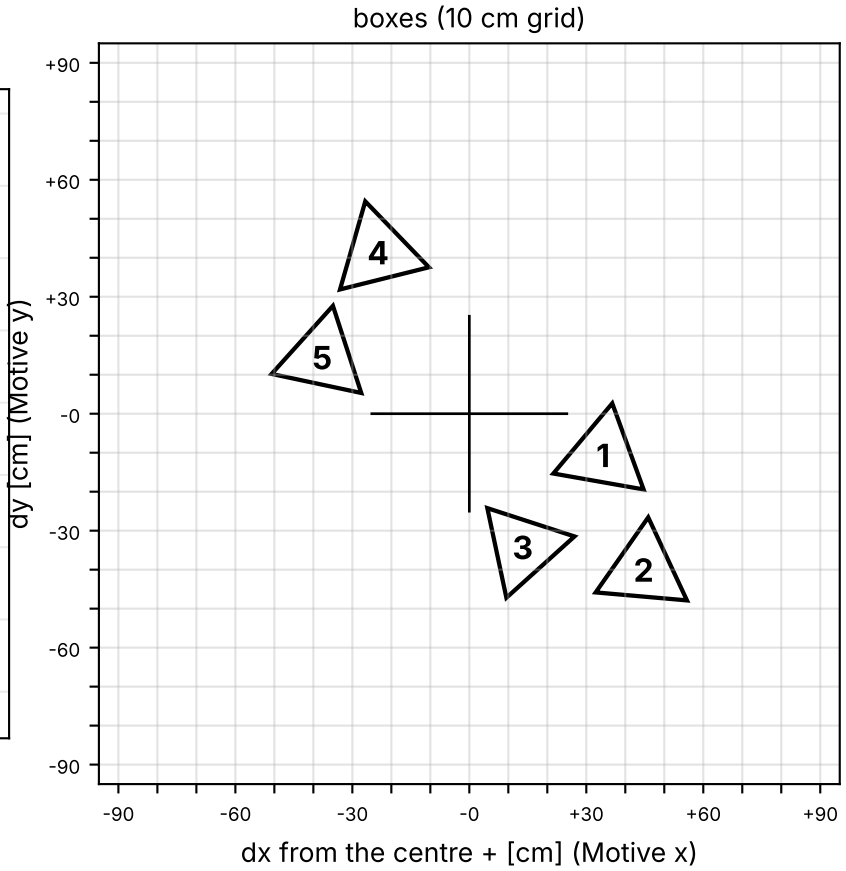
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R06 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R07 day 1 A base (test)

5 boxes, due gruppi, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+34	-11
2	+45	-40
3	+14	-34
4	-23	+41
5	-38	+14

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

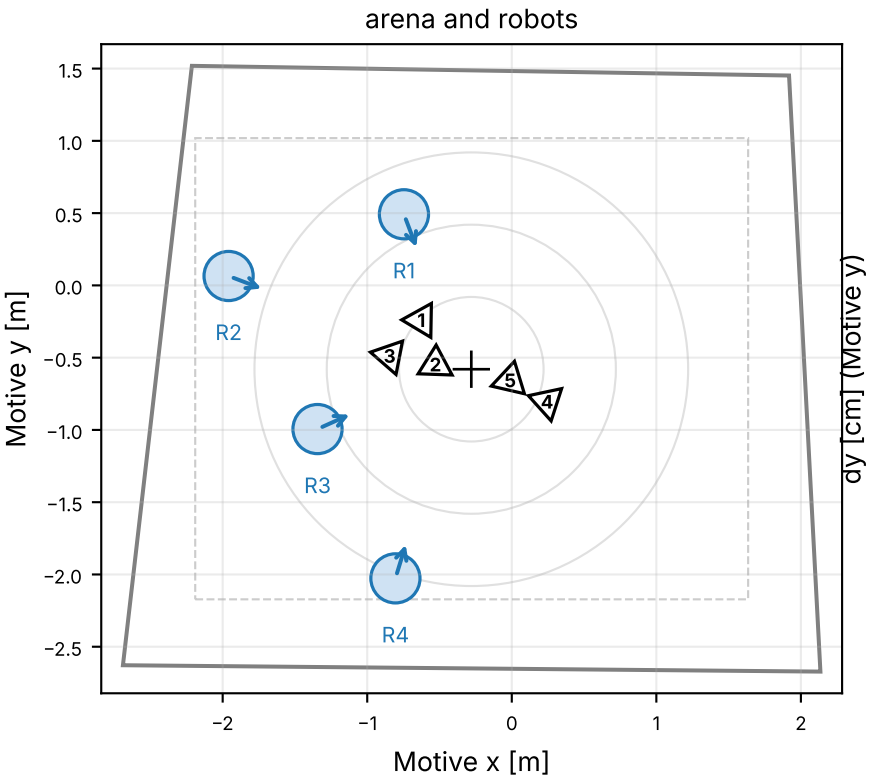
Place and capture: python -m tools.place_check --layout R07

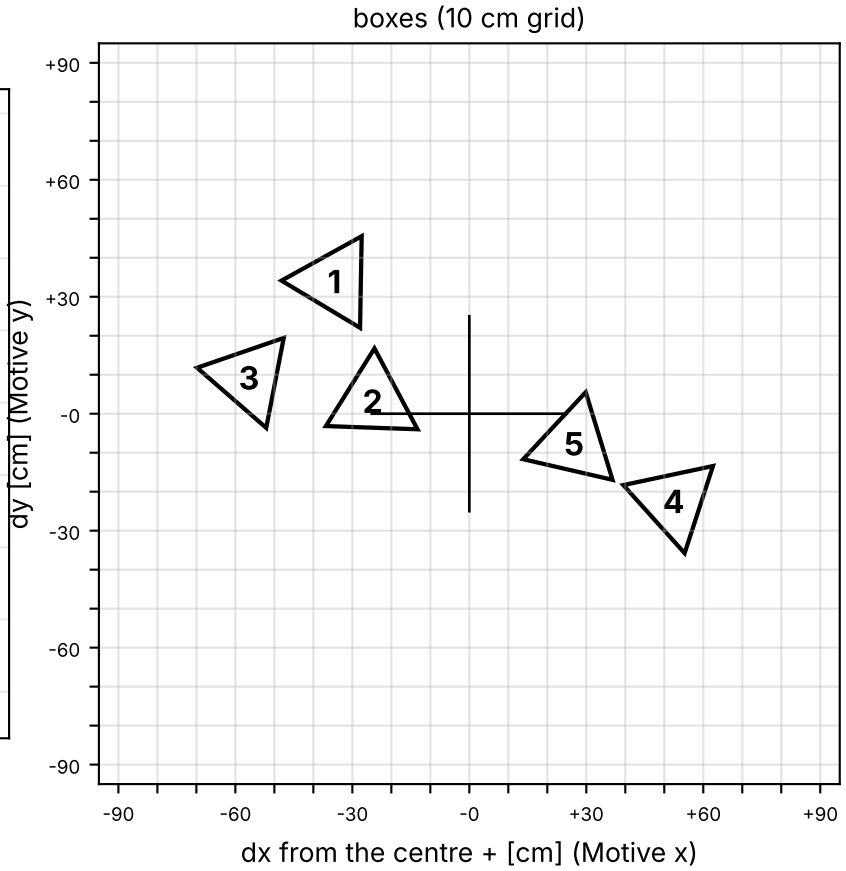
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R07 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R08 day 1 A base (test)

5 boxes, due gruppi, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-35	+34
2	-25	+3
3	-56	+9
4	+52	-23
5	+27	-8

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	-0.75	+0.49
R2	-1.96	+0.07
R3	-1.34	-0.99
R4	-0.81	-2.03

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

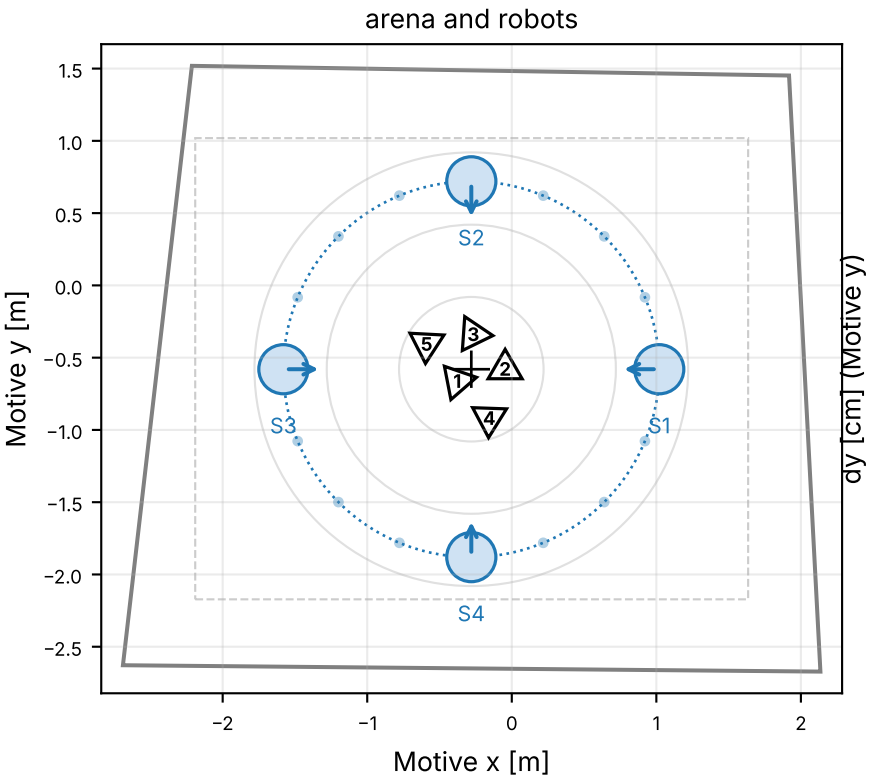
Place and capture: python -m tools.place_check --layout R08

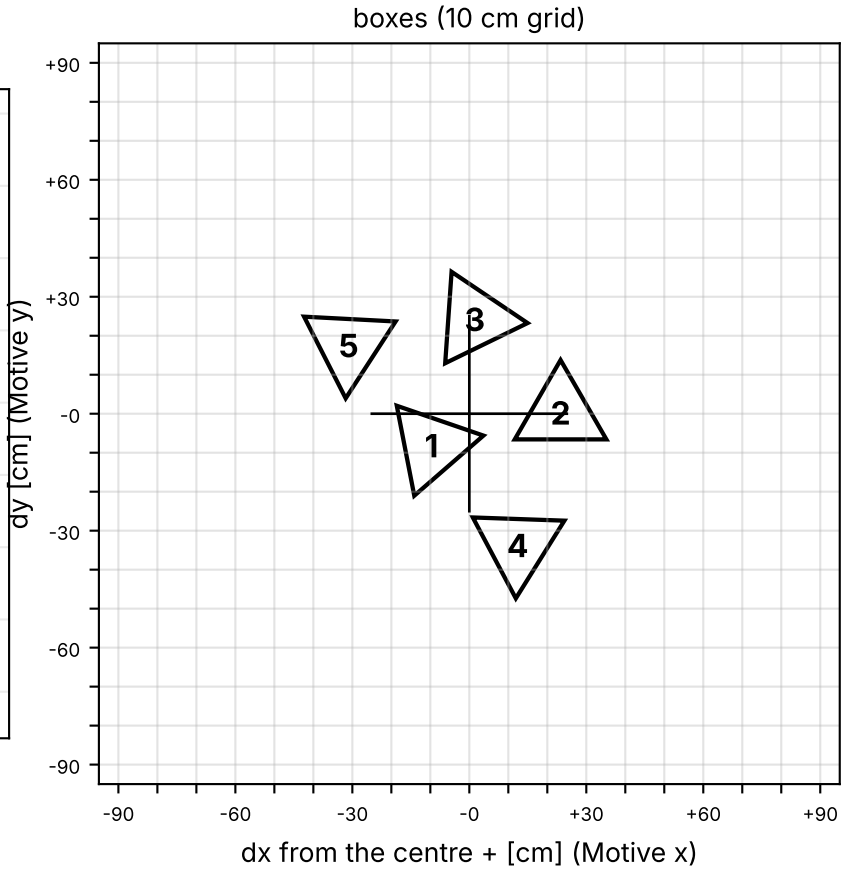
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R08 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R09 day 1 A base (dev)

5 boxes, fitta, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-10	-8
2	+23	+0
3	+1	+24
4	+12	-34
5	-31	+17

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

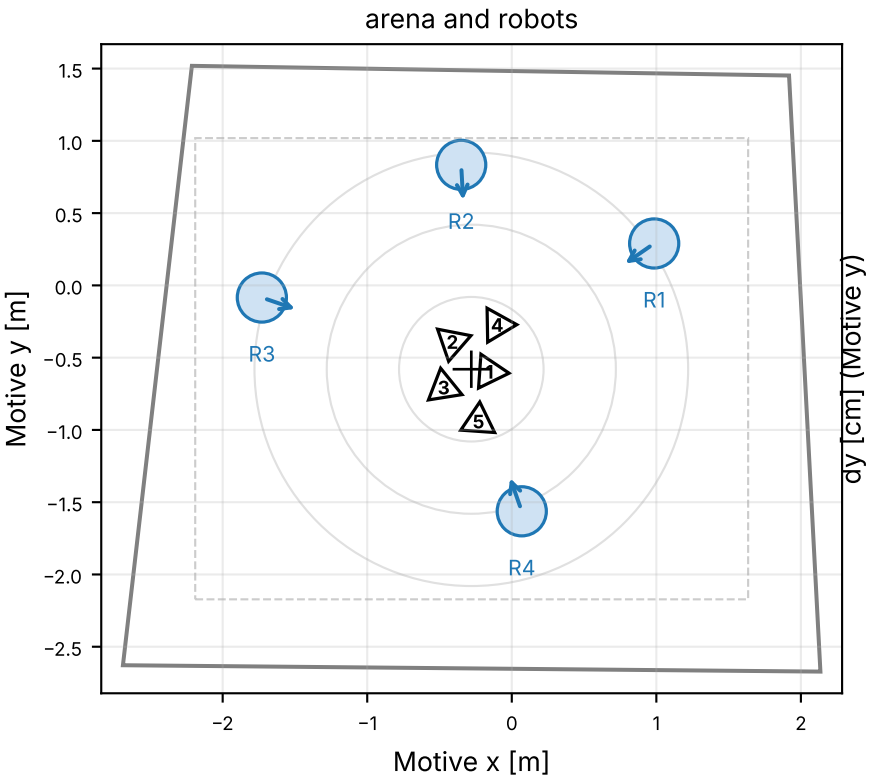
Place and capture: python -m tools.place_check --layout R09

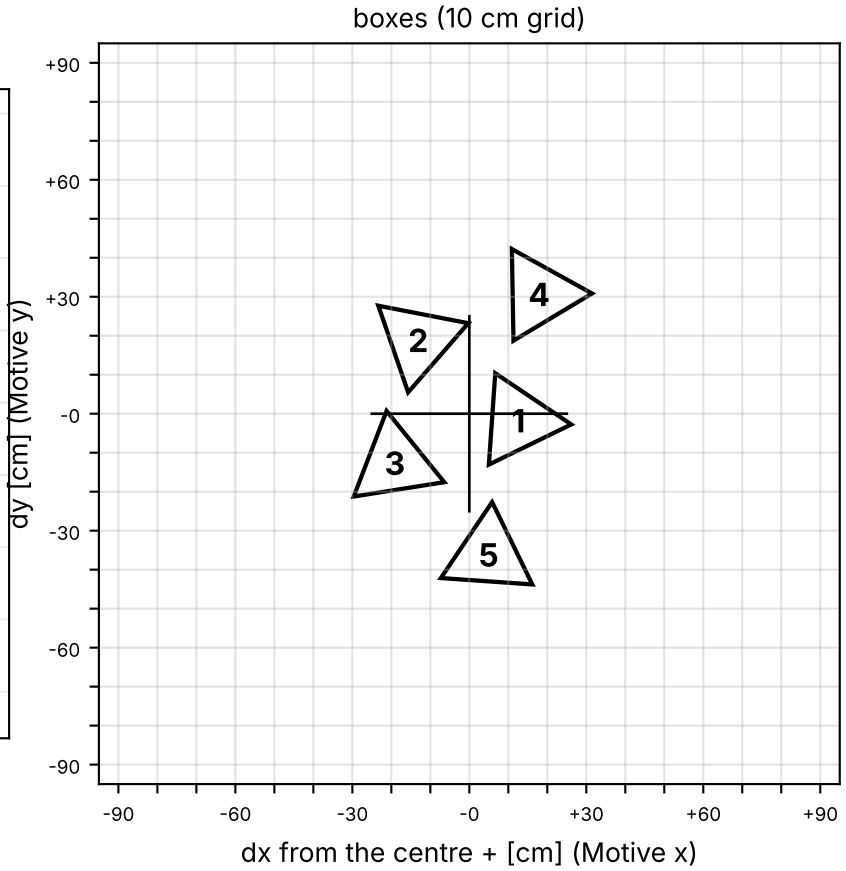
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R09 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R10 day 1 A base (dev)

5 boxes, fitta, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+13	-2
2	-13	+19
3	-19	-13
4	+18	+31
5	+5	-36

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.98	+0.29
R2	-0.35	+0.83
R3	-1.73	-0.08
R4	+0.07	-1.56

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

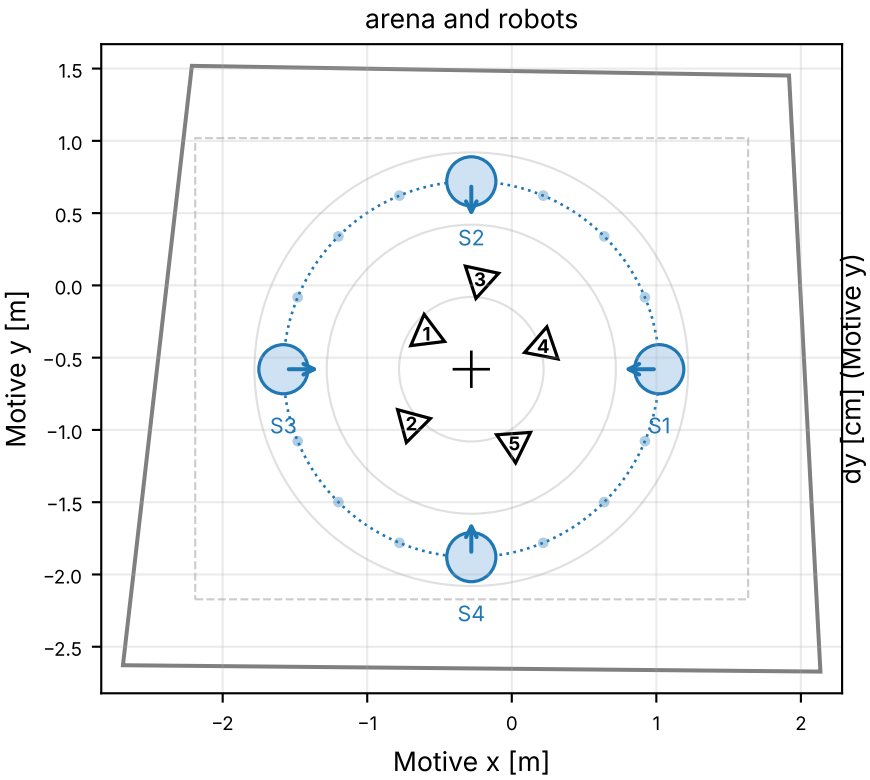
Place and capture: python -m tools.place_check --layout R10

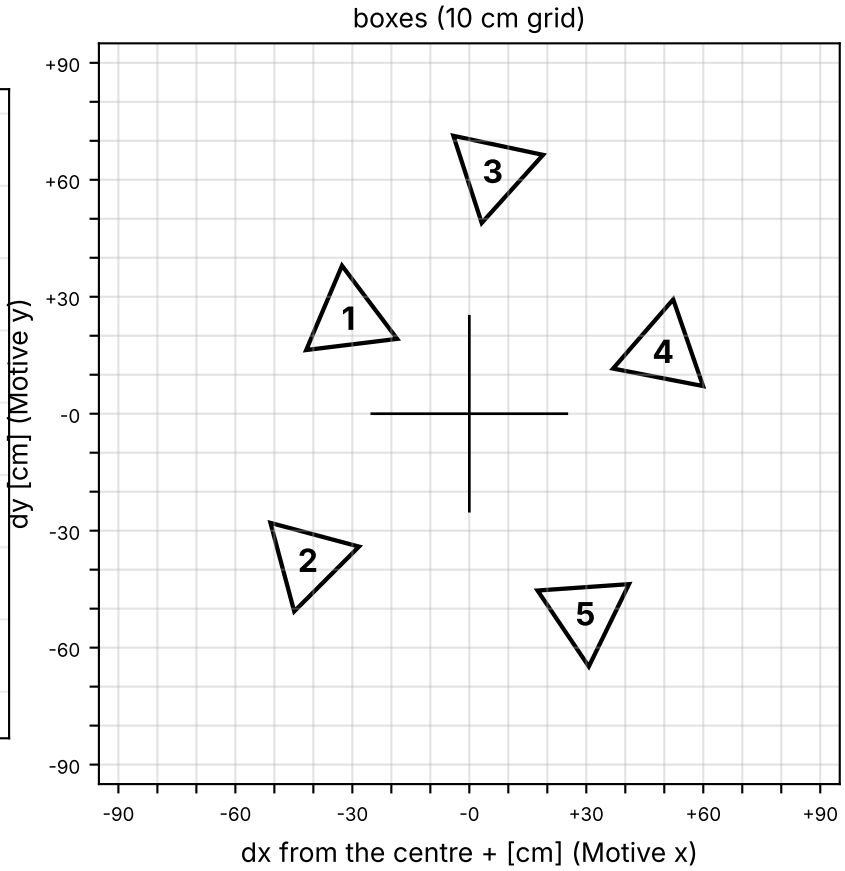
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R10 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R11 day 1 A base (dev)

5 boxes, sparsa, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-31	+24
2	-41	-38
3	+6	+62
4	+50	+16
5	+30	-51

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

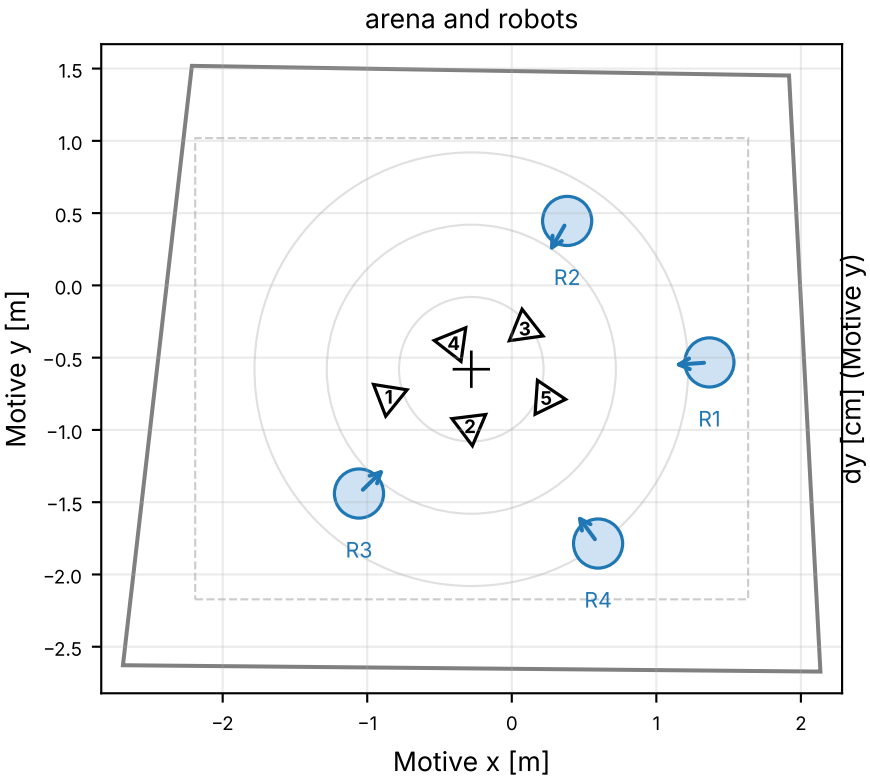
Place and capture: `python -m tools.place_check --layout R11`

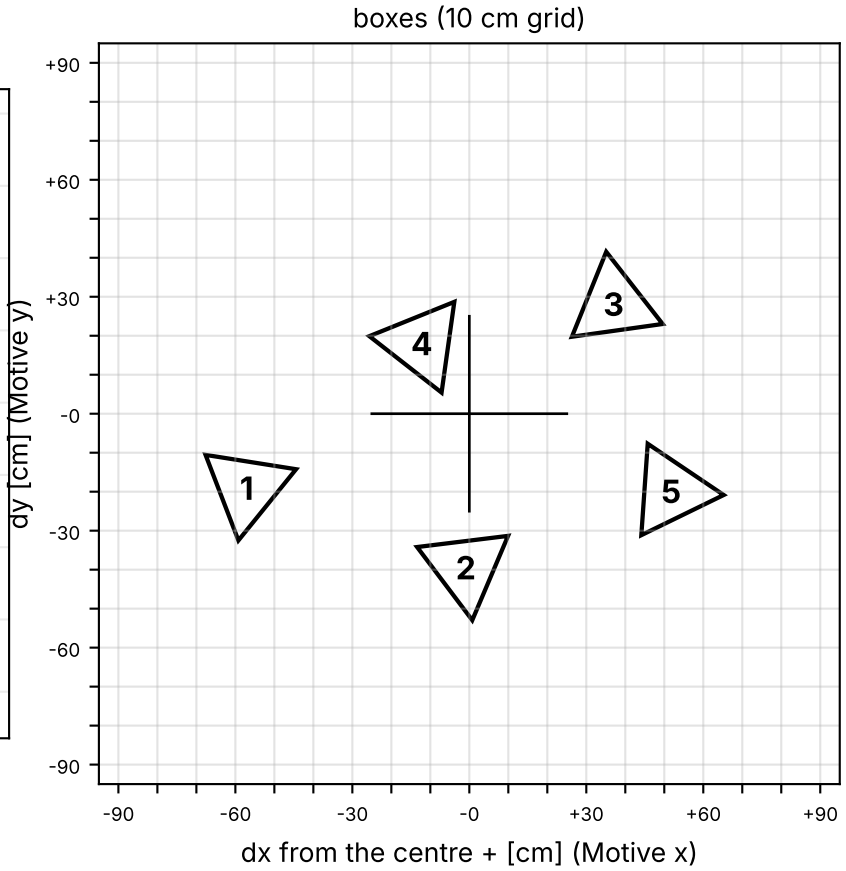
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R11 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius
1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R12 day 1 A base (dev)

5 boxes, sparsa, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-57	-19
2	-1	-40
3	+37	+28
4	-12	+18
5	+52	-20

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+1.37	-0.53
R2	+0.38	+0.45
R3	-1.06	-1.44
R4	+0.60	-1.79

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

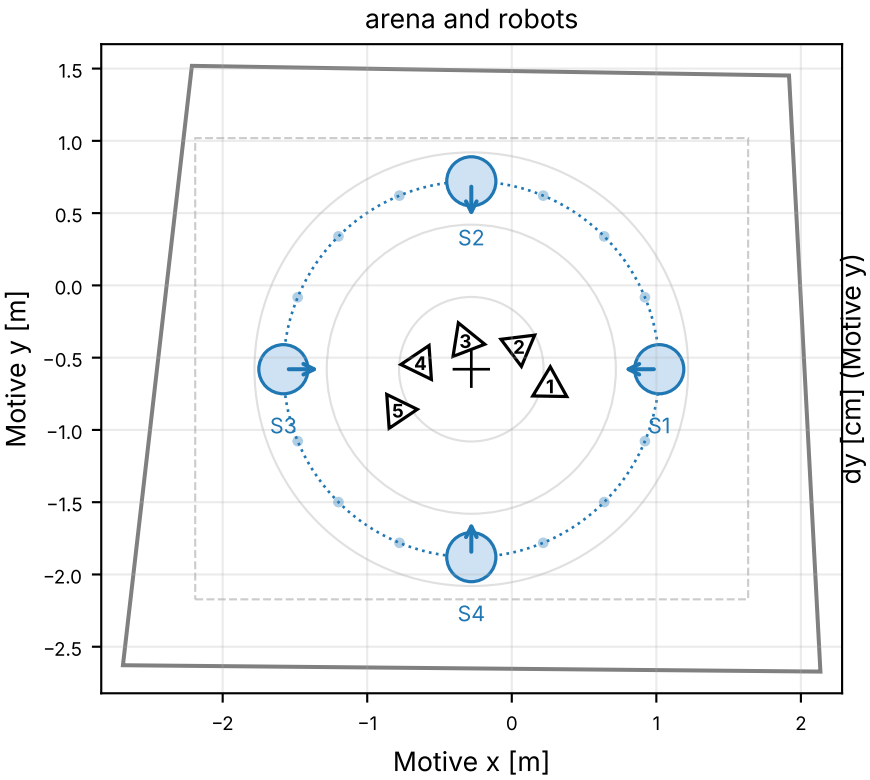
Place and capture: python -m tools.place_check --layout R12

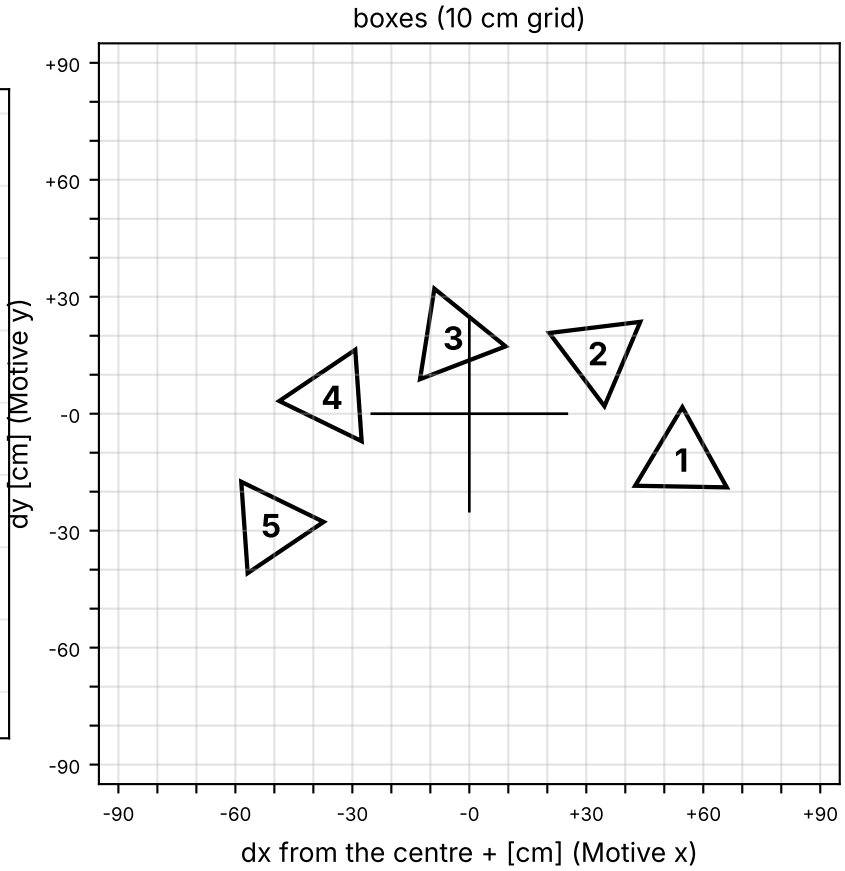
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R12 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R13 day 1 A base (dev)

5 boxes, fila/arco, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+54	-12
2	+33	+15
3	-4	+19
4	-35	+4
5	-51	-29

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

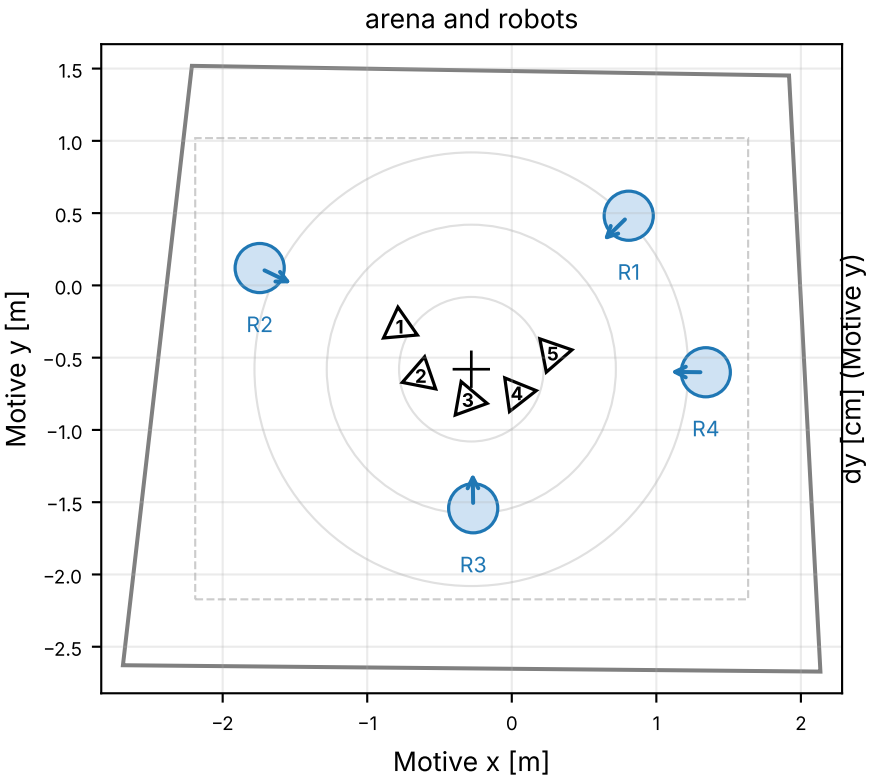
Place and capture: python -m tools.place_check --layout R13

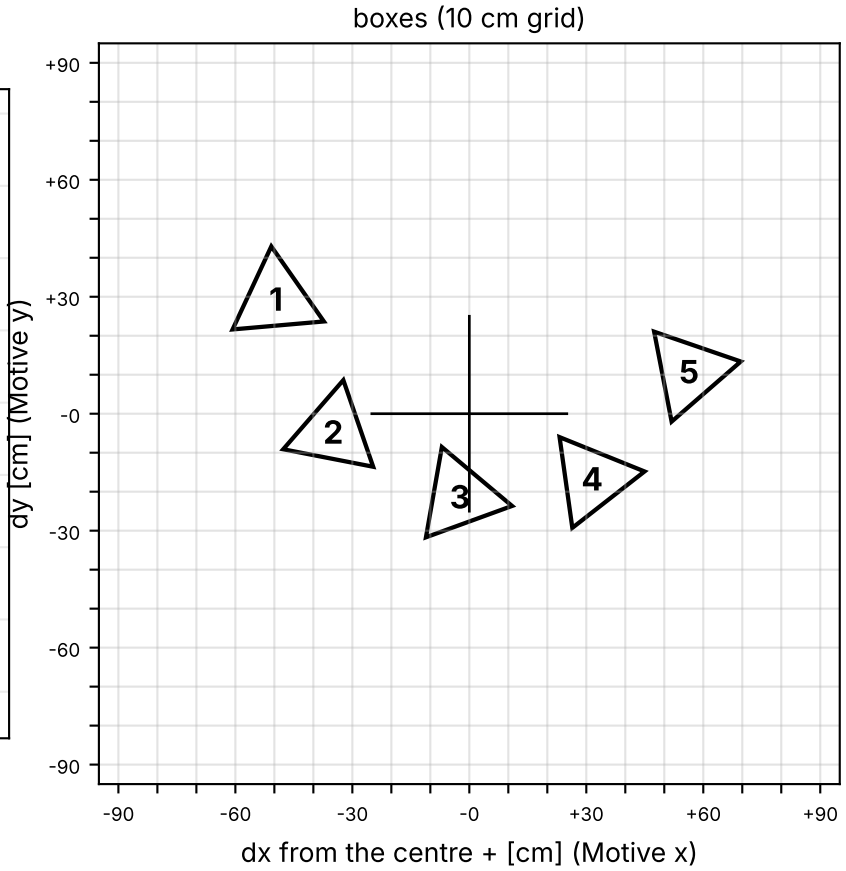
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R13 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R14 day 1 A base (dev)

5 boxes, fila/arco, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-50	+29
2	-35	-5
3	-2	-21
4	+32	-17
5	+56	+11

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.81	+0.48
R2	-1.74	+0.12
R3	-0.27	-1.54
R4	+1.34	-0.60

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

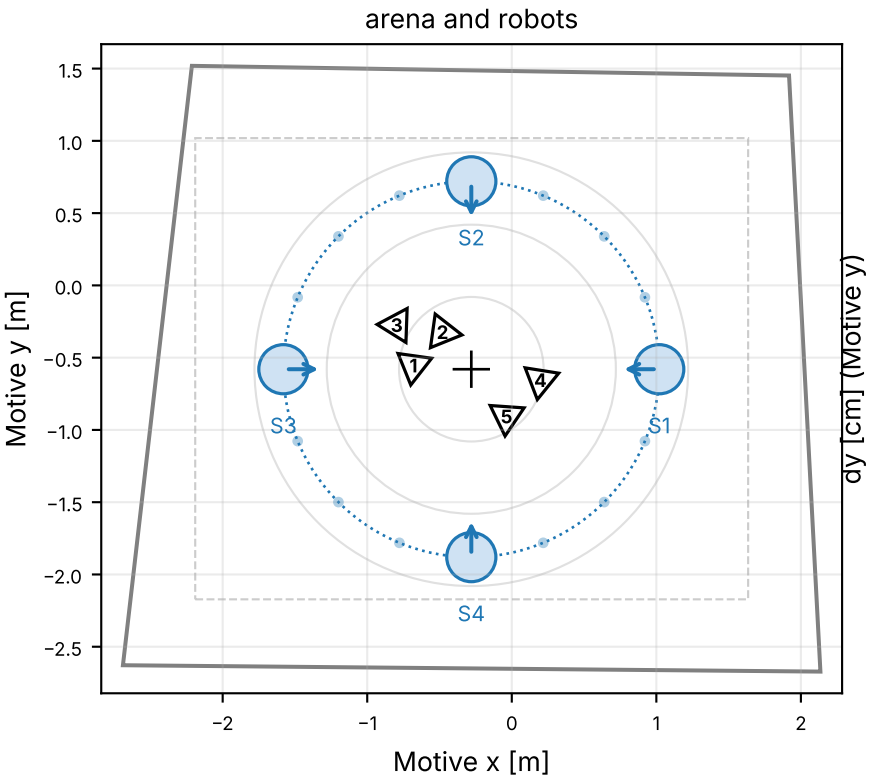
Place and capture: python -m tools.place_check --layout R14

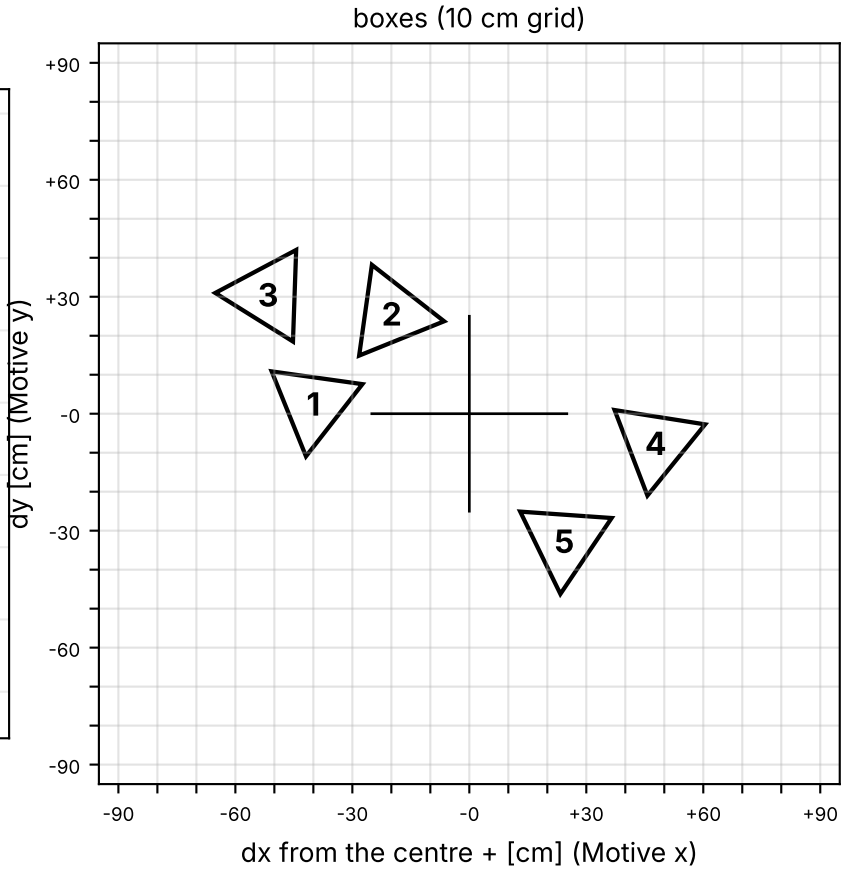
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R14 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R15 day 1 A base (dev)

5 boxes, due gruppi, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-40	+2
2	-20	+26
3	-52	+30
4	+48	-8
5	+24	-33

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

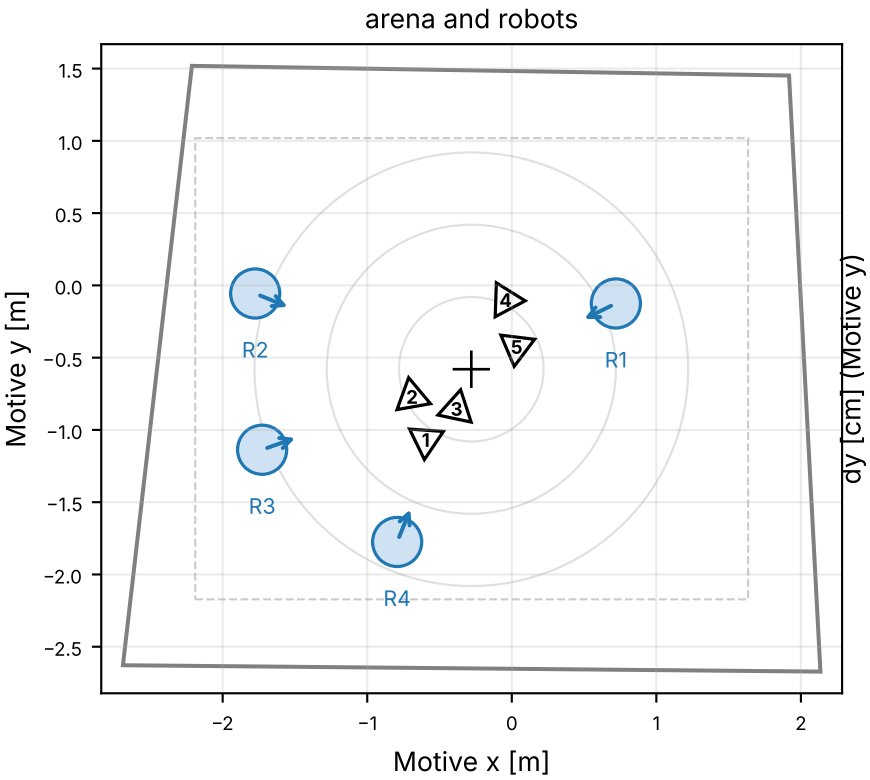
Place and capture: python -m tools.place_check --layout R15

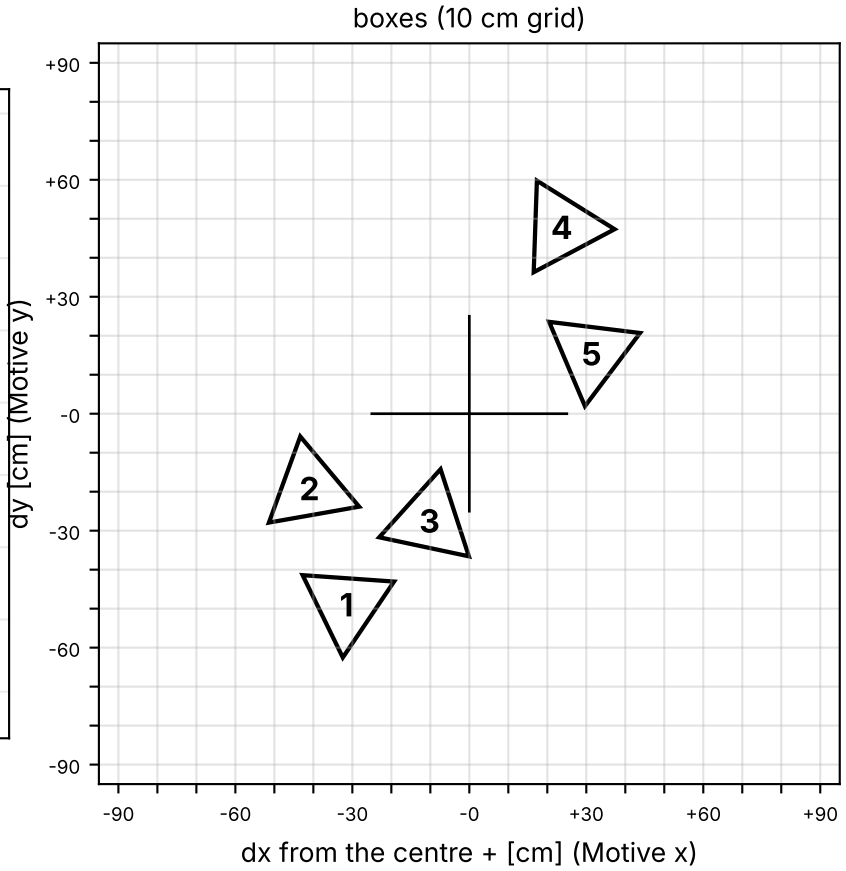
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R15 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R16 day 1 A base (dev)

5 boxes, due gruppi, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-31	-49
2	-41	-19
3	-10	-28
4	+24	+48
5	+31	+15

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.72	-0.12
R2	-1.77	-0.06
R3	-1.73	-1.14
R4	-0.79	-1.77

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

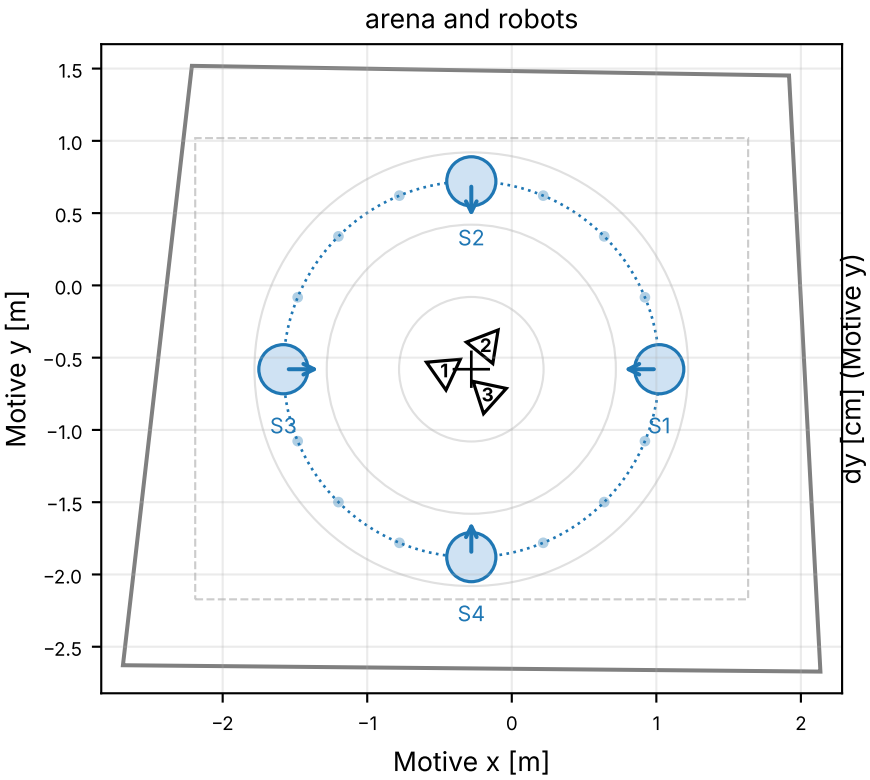
Place and capture: python -m tools.place_check --layout R16

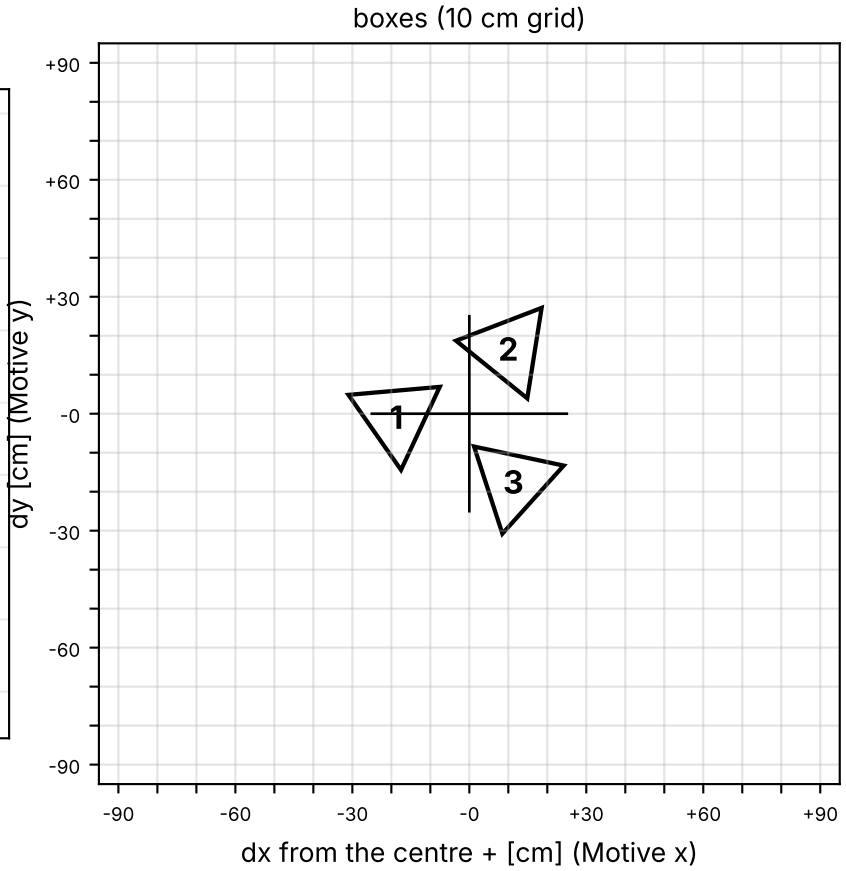
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R16 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R17 day 2 A base (test)

3 boxes, fitta, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-19	-1
2	+10	+17
3	+11	-18

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

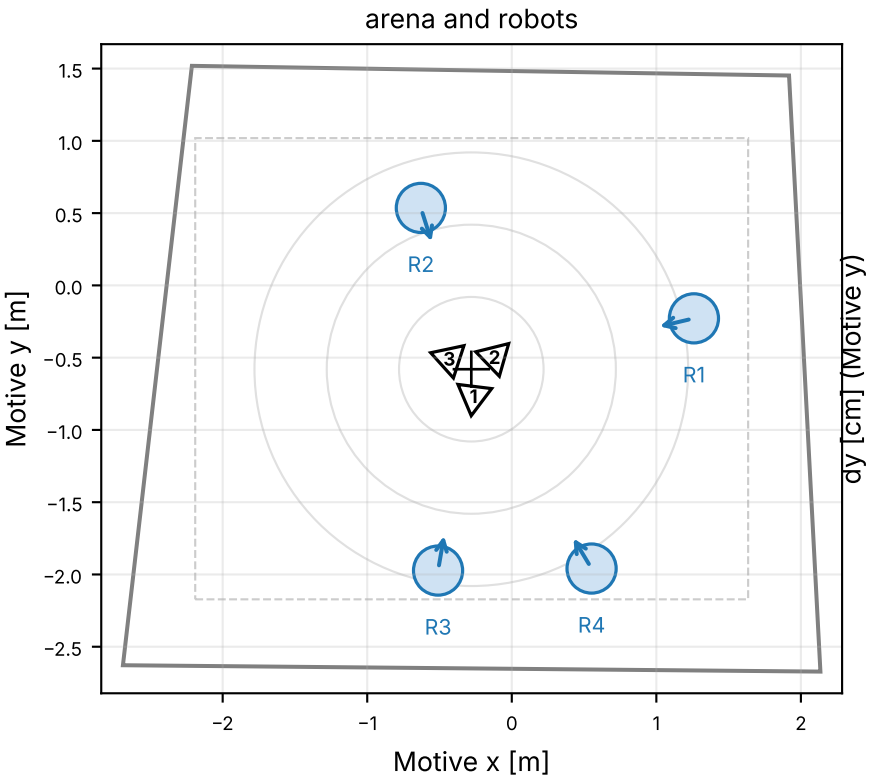
Place and capture: `python -m tools.place_check --layout R17`

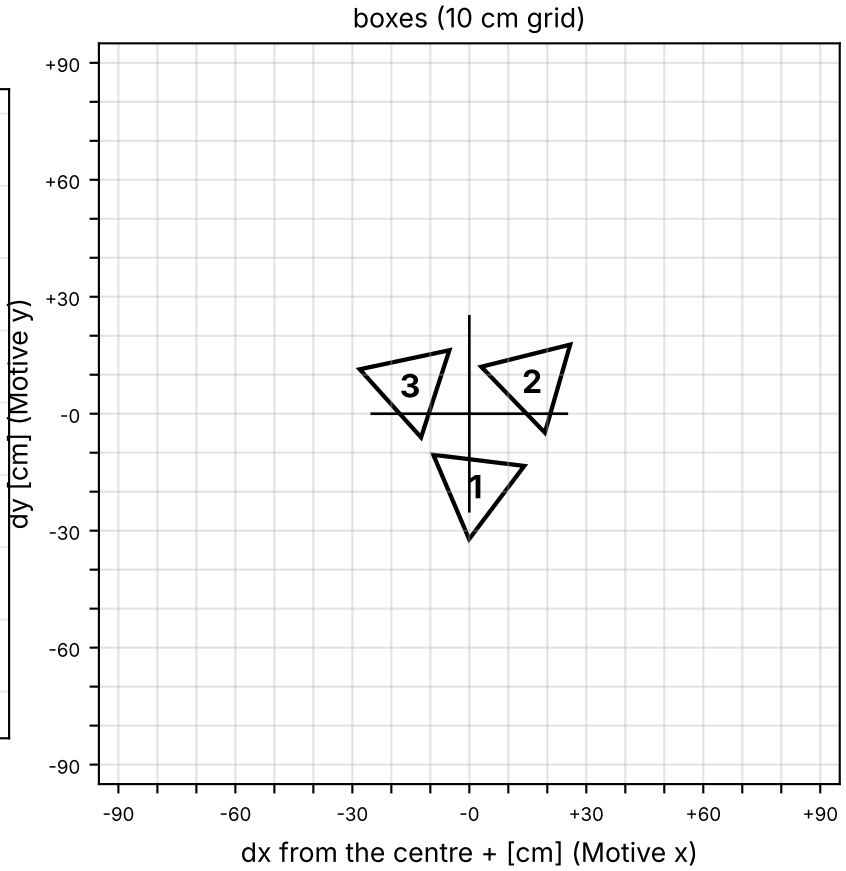
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R17 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R18 day 2 A base (test)

3 boxes, fitta, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+2	-19
2	+16	+8
3	-15	+7

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+1.26	-0.23
R2	-0.63	+0.54
R3	-0.51	-1.97
R4	+0.55	-1.96

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

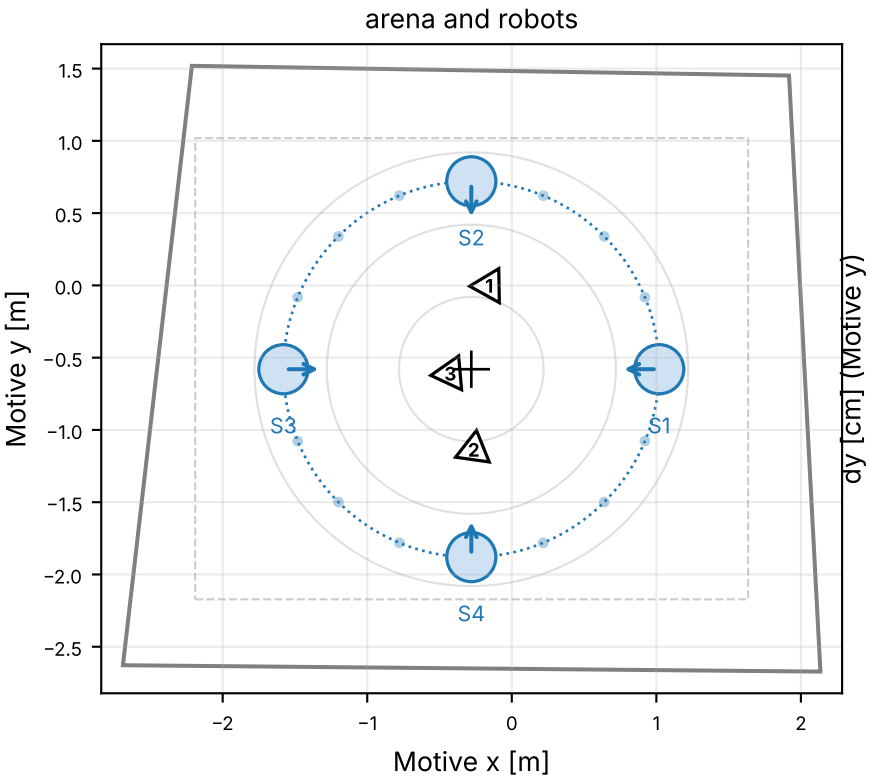
Place and capture: python -m tools.place_check --layout R18

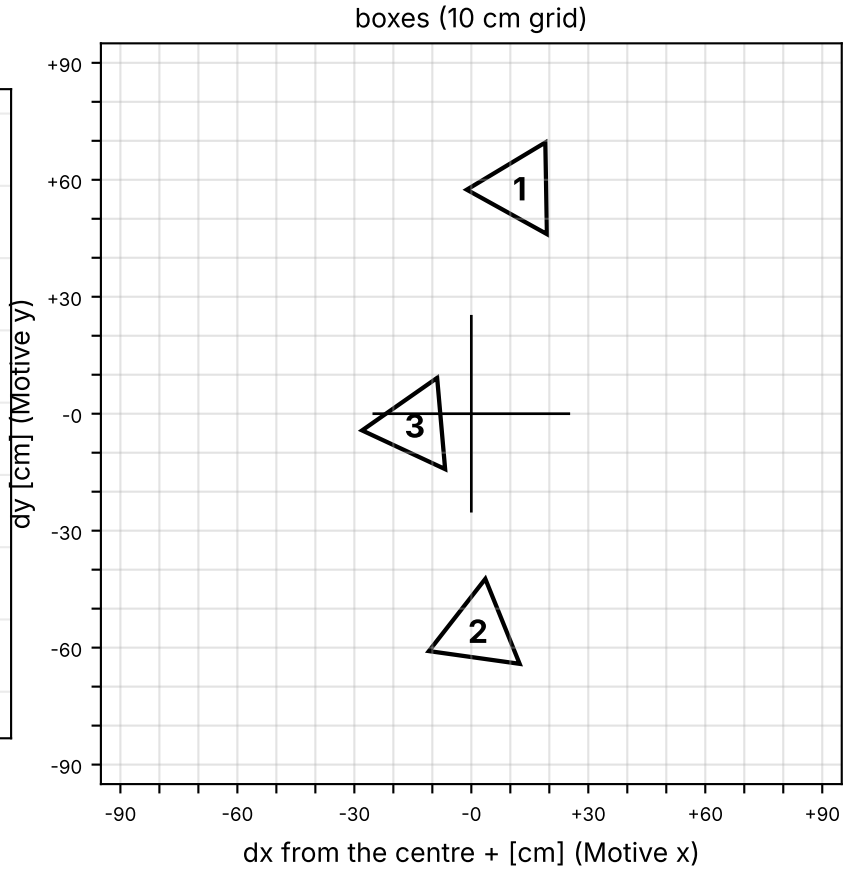
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R18 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R19 day 2 A base (test)

3 boxes, sparsa, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+12	+58
2	+2	-56
3	-14	-3

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

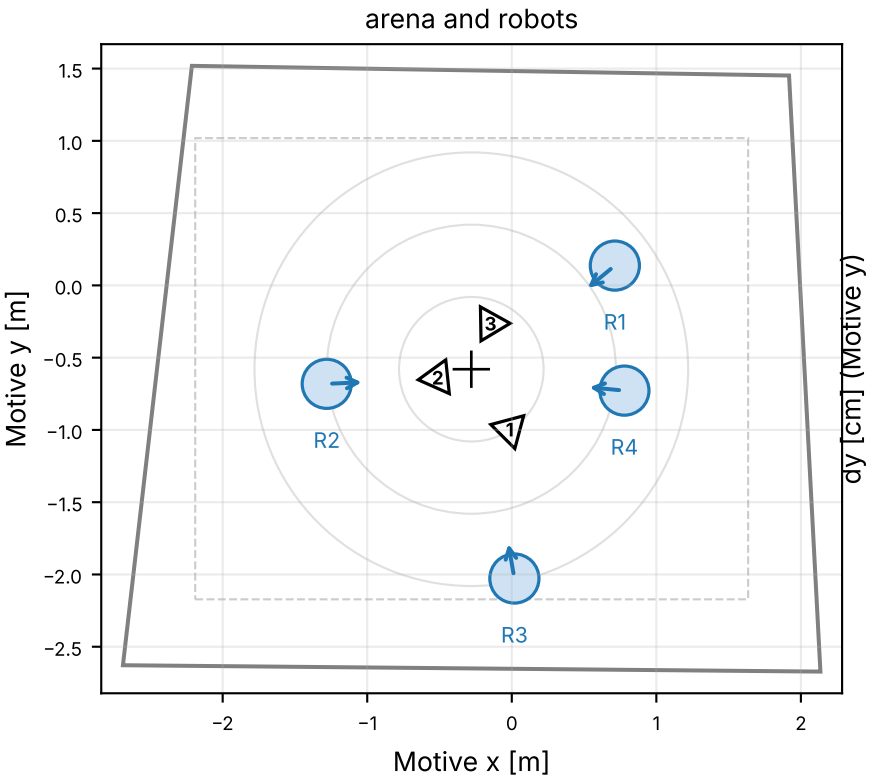
Place and capture: python -m tools.place_check --layout R19

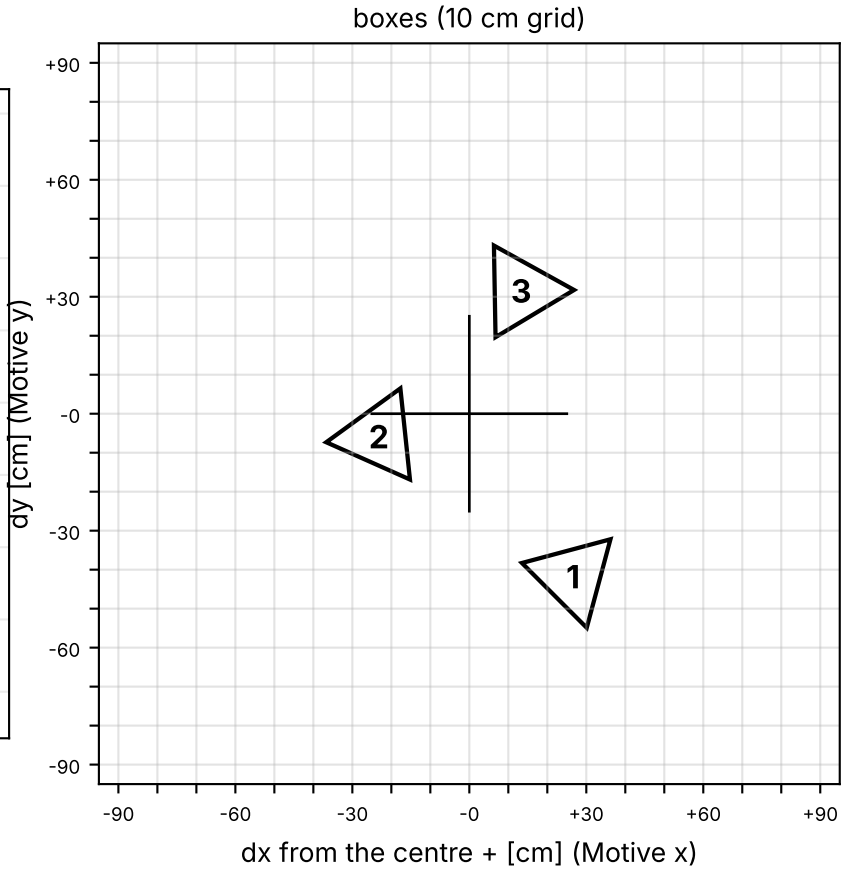
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R19 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R20 day 2 A base (test)

3 boxes, sparsa, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+27	-42
2	-23	-6
3	+13	+31

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.71	+0.14
R2	-1.28	-0.68
R3	+0.02	-2.03
R4	+0.78	-0.73

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

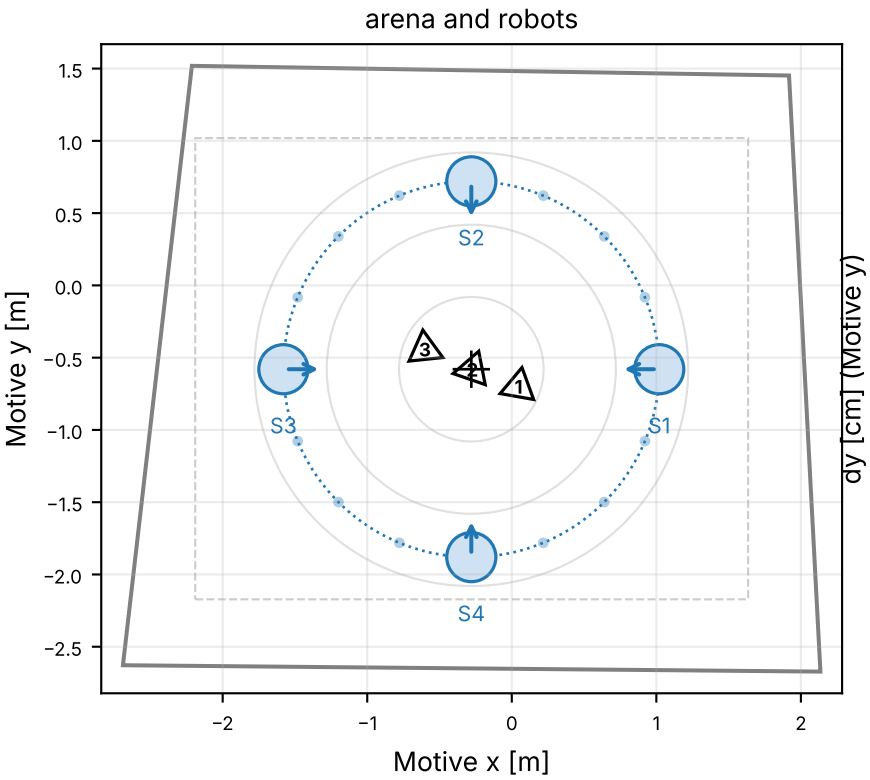
Place and capture: python -m tools.place_check --layout R20

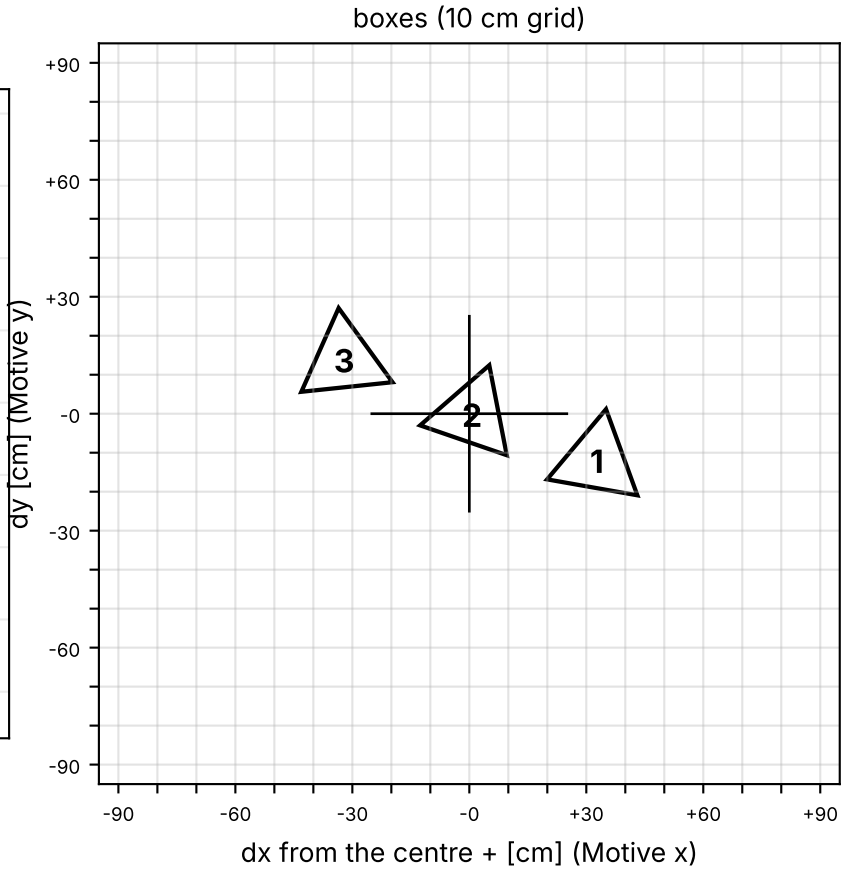
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R20 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R21 day 2 A base (test)

3 boxes, fila/arco, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+33	-12
2	+1	-0
3	-32	+14

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

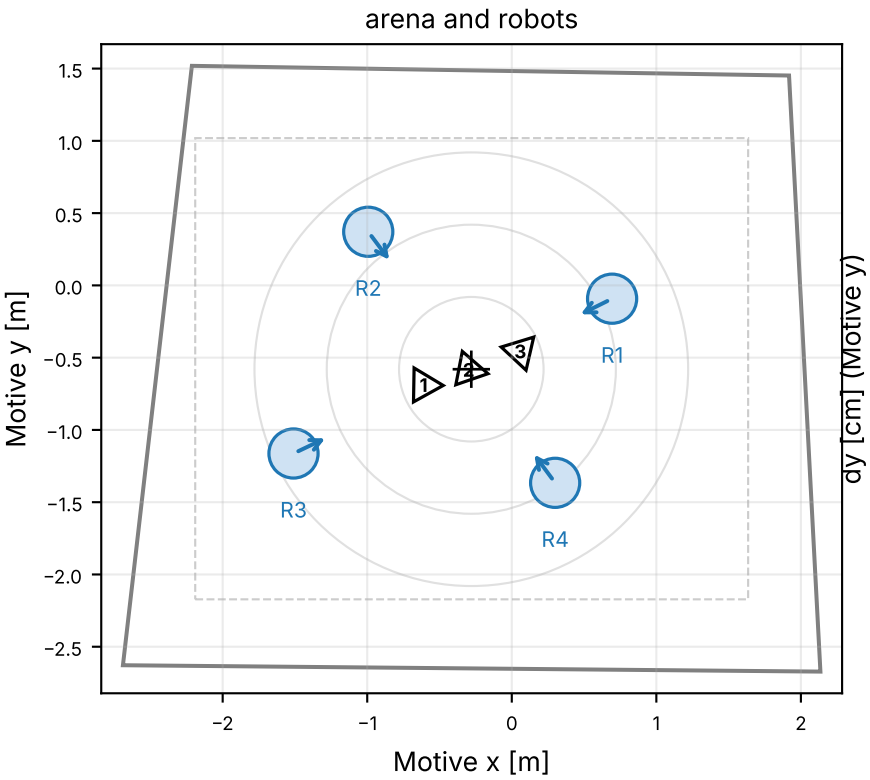
Place and capture: python -m tools.place_check --layout R21

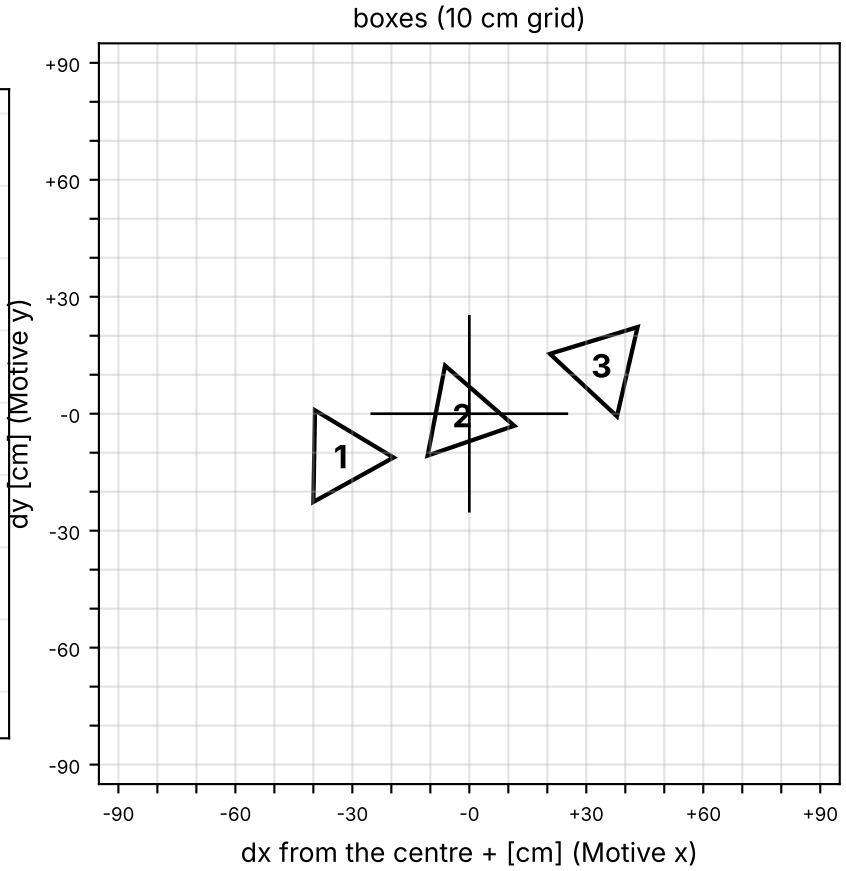
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R21 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R22 day 2 A base (test)

3 boxes, fila/arco, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-33	-11
2	-2	-1
3	+34	+12

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.69	-0.09
R2	-0.99	+0.37
R3	-1.51	-1.16
R4	+0.30	-1.37

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

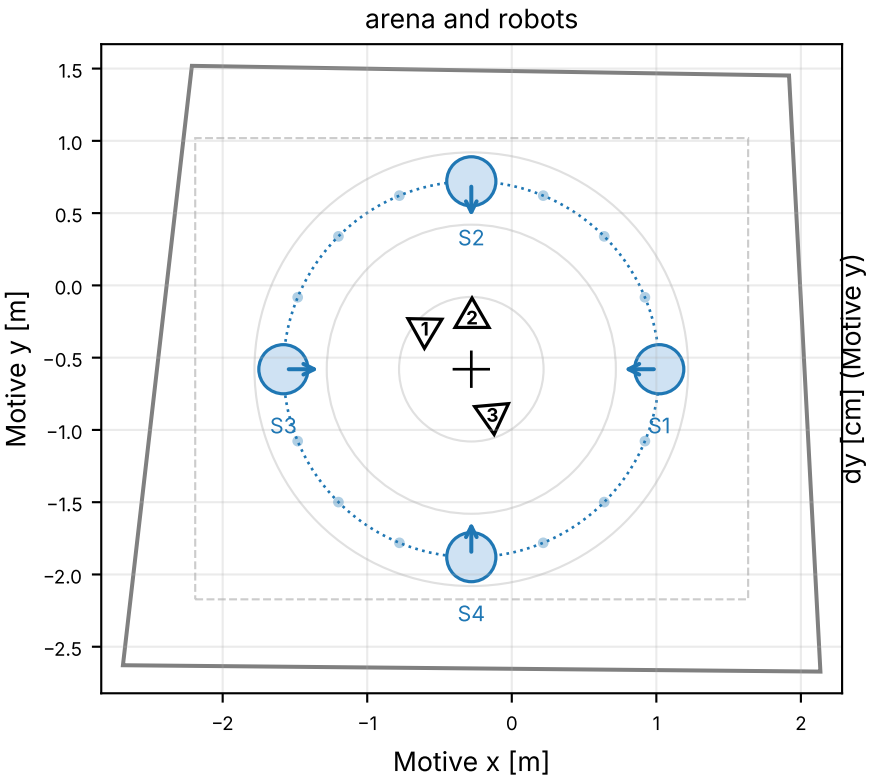
Place and capture: python -m tools.place_check --layout R22

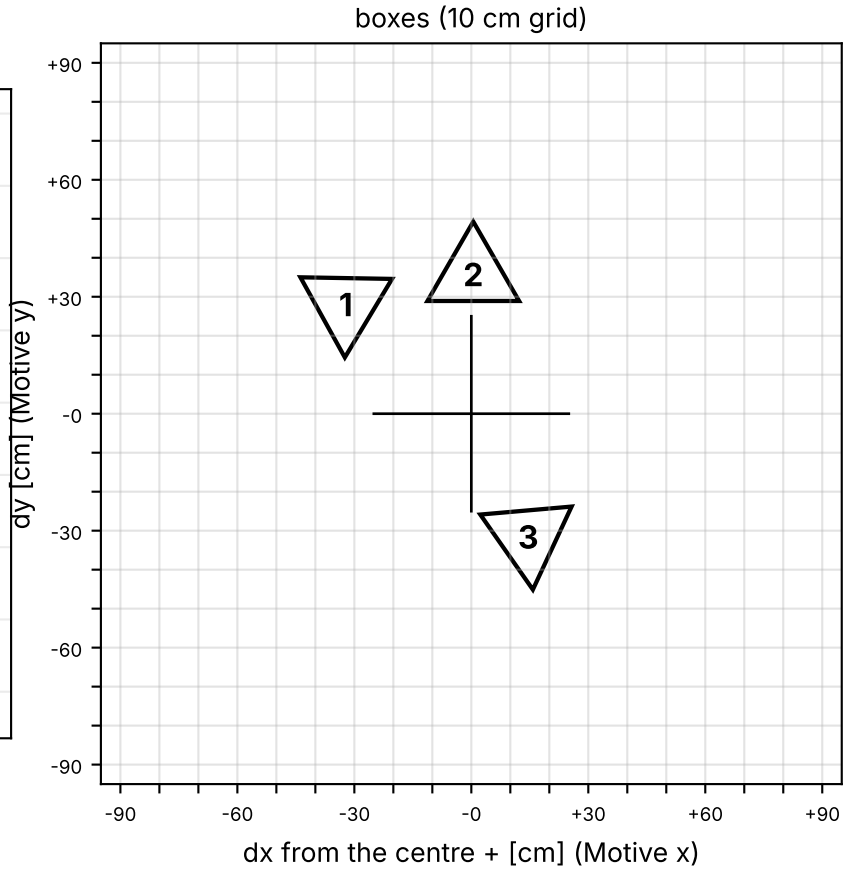
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R22 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R23 day 2 A base (test)

3 boxes, due gruppi (2+1), robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-32	+28
2	+1	+36
3	+15	-32

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

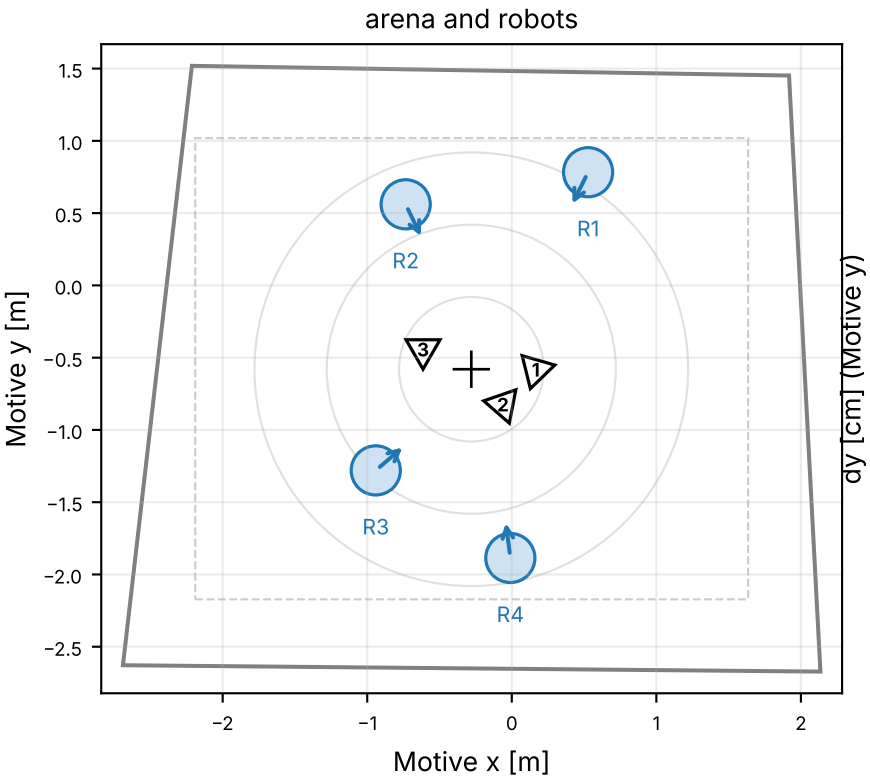
Place and capture: python -m tools.place_check --layout R23

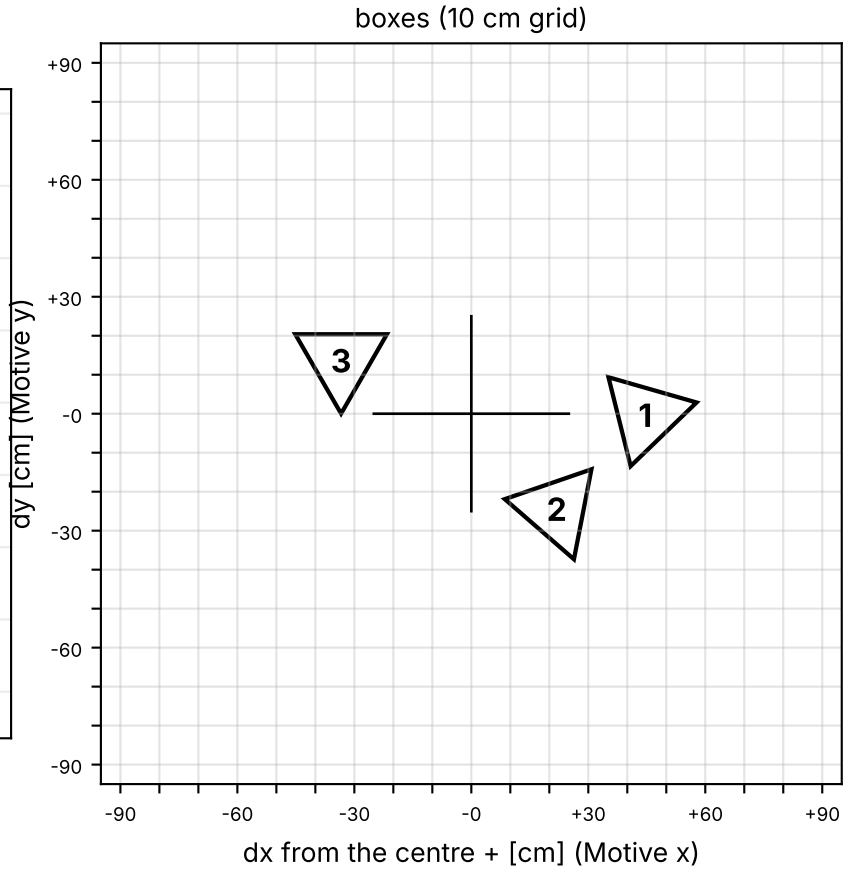
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R23 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R24 day 2 A base (test)

3 boxes, due gruppi (2+1), robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+45	-0
2	+22	-24
3	-33	+14

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.53	+0.78
R2	-0.73	+0.56
R3	-0.94	-1.28
R4	-0.01	-1.89

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

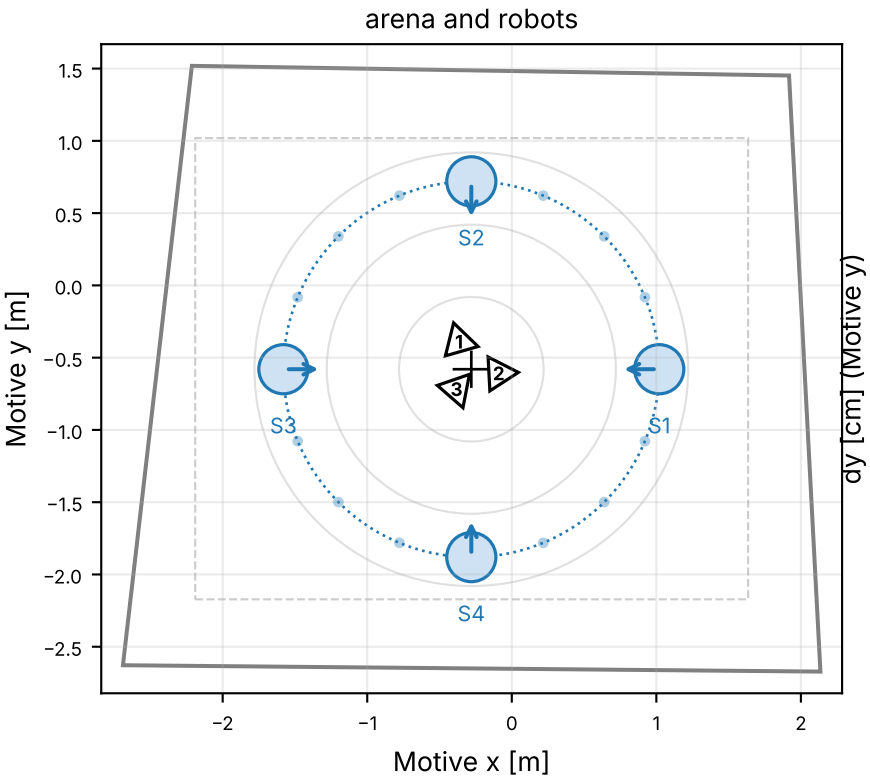
Place and capture: python -m tools.place_check --layout R24

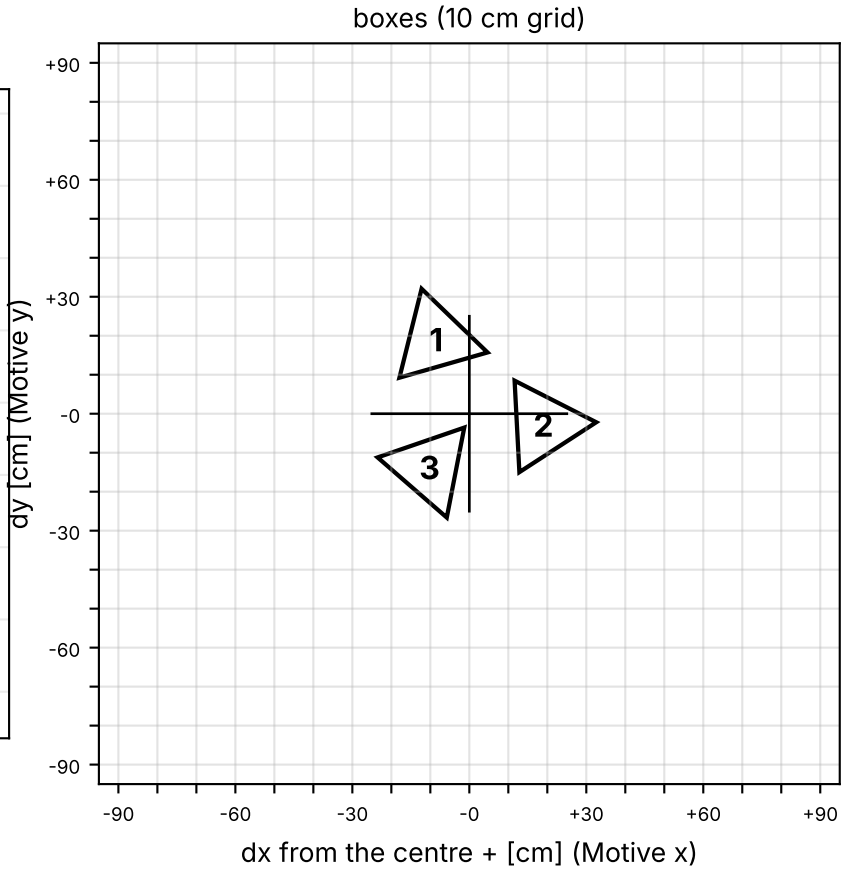
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R24 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R25 day 2 A base (test)

3 boxes, fitta, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-8	+19
2	+19	-3
3	-10	-14

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

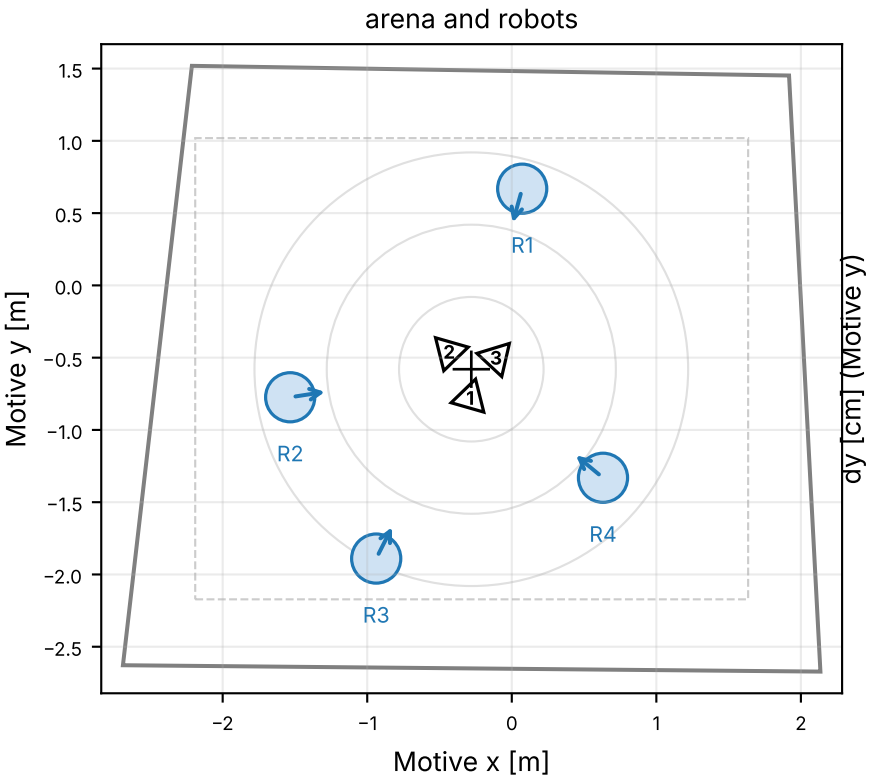
Place and capture: python -m tools.place_check --layout R25

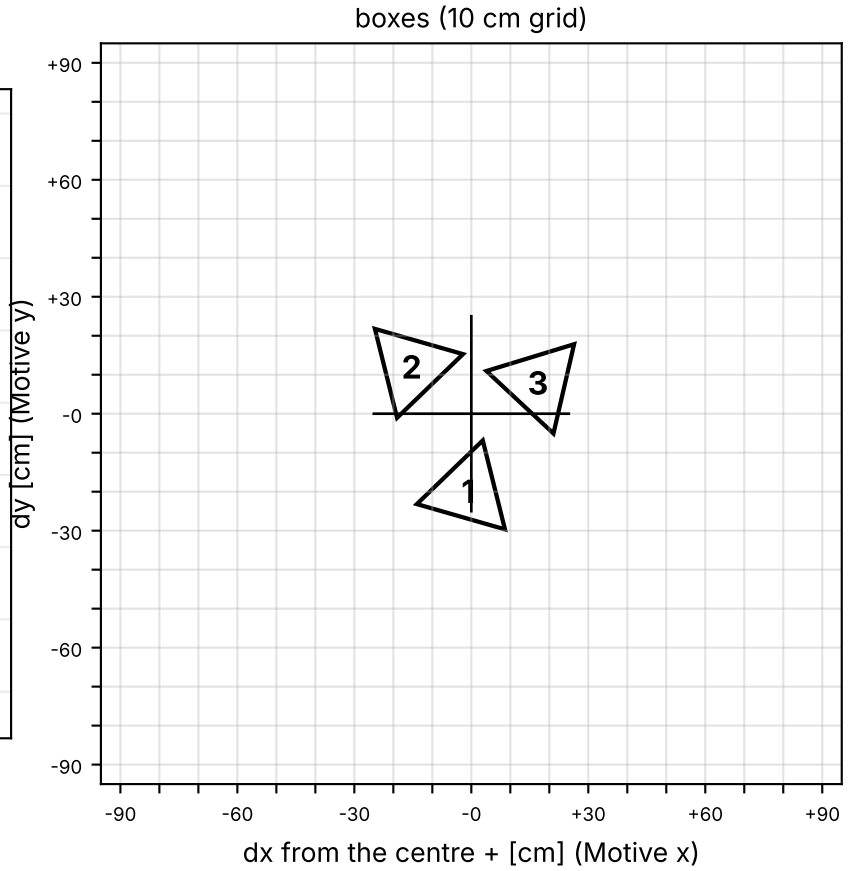
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R25 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R26 day 2 A base (test)

3 boxes, fitta, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-1	-20
2	-15	+12
3	+17	+8

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.07	+0.67
R2	-1.53	-0.77
R3	-0.94	-1.89
R4	+0.63	-1.33

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

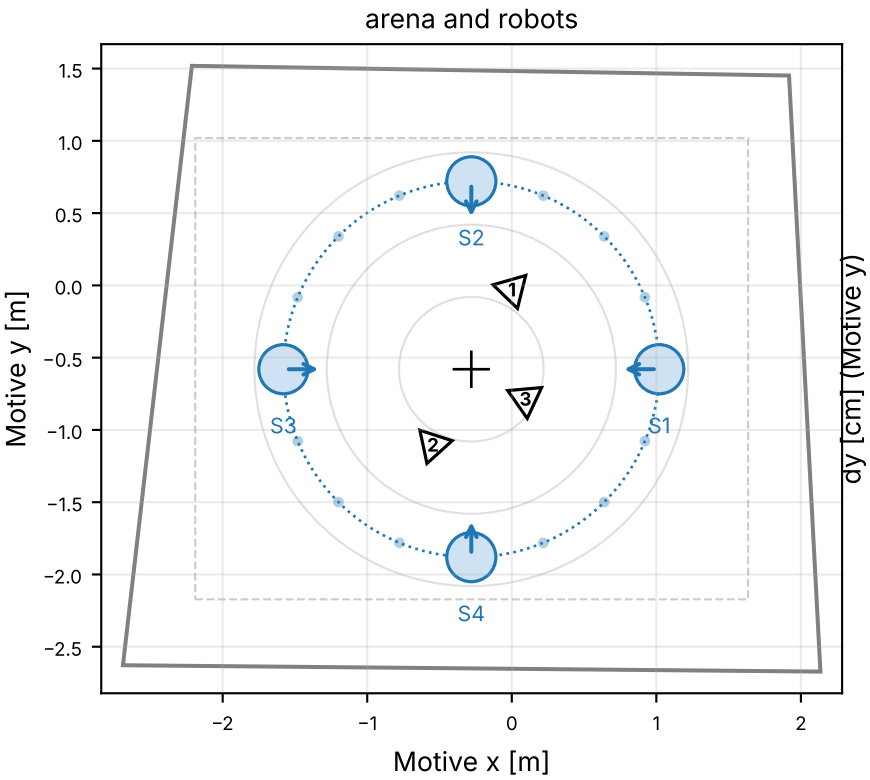
Place and capture: `python -m tools.place_check --layout R26`

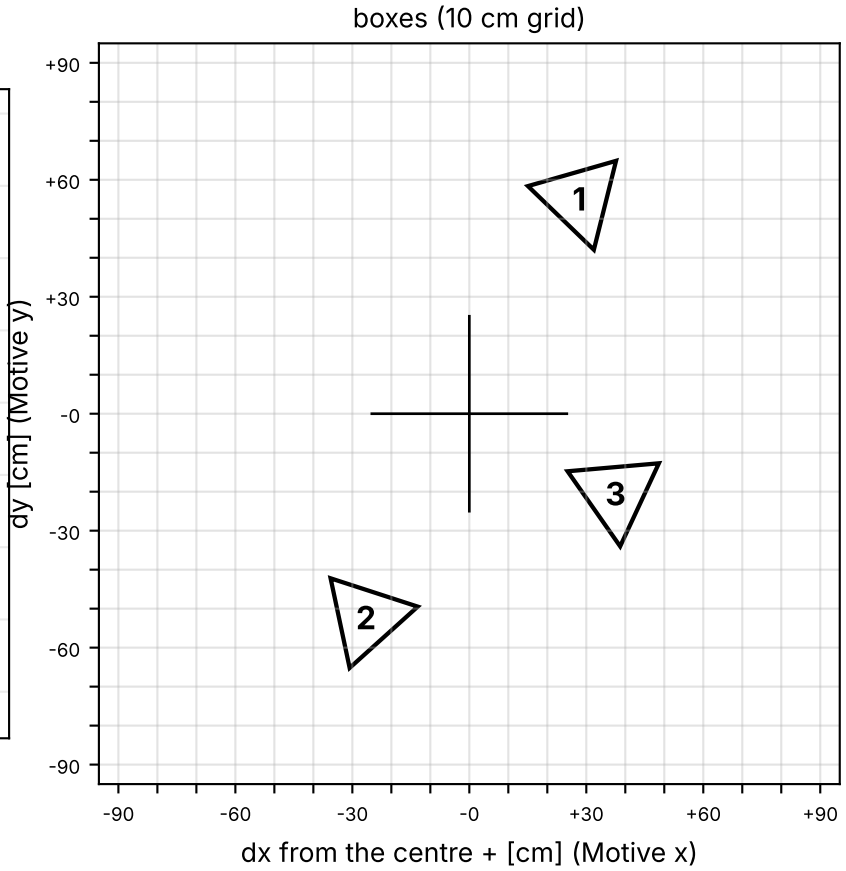
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R26 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R27 day 2 A base (test)

3 boxes, sparsa, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+28	+55
2	-26	-52
3	+38	-21

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

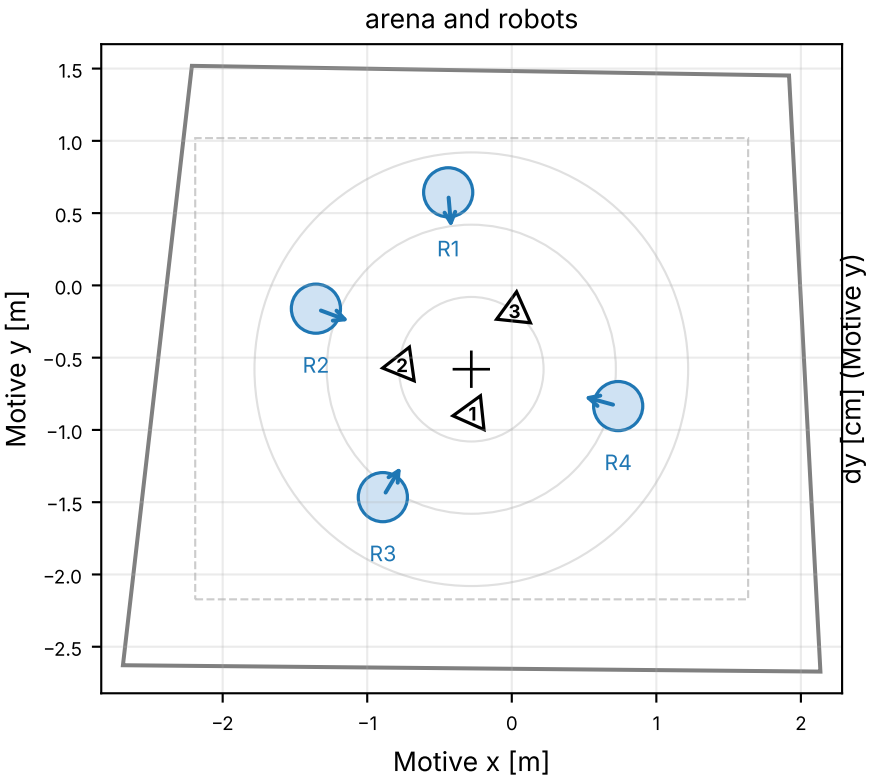
Place and capture: python -m tools.place_check --layout R27

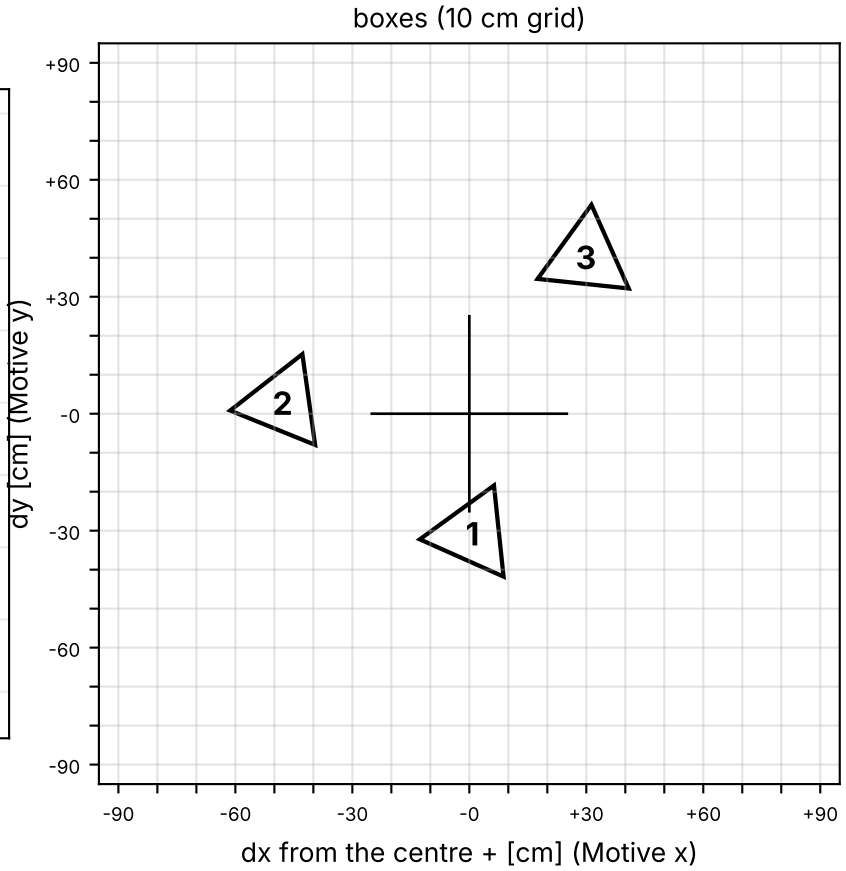
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R27 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R28 day 2 A base (test)

3 boxes, sparsa, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+1	-31
2	-48	+3
3	+30	+40

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	-0.44	+0.64
R2	-1.35	-0.16
R3	-0.89	-1.47
R4	+0.74	-0.83

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

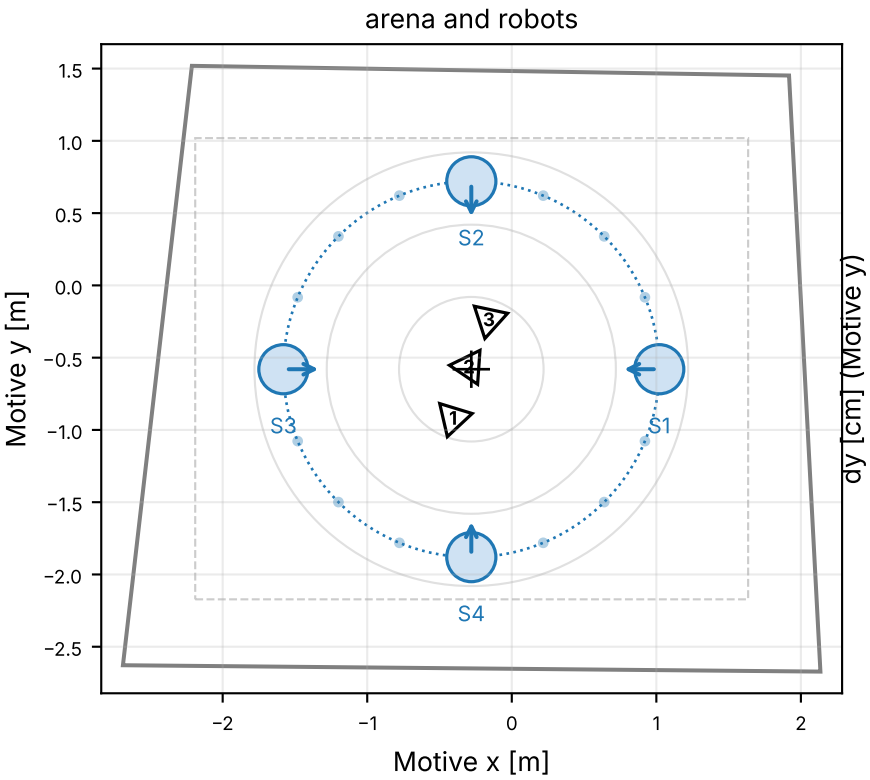
Place and capture: `python -m tools.place_check --layout R28`

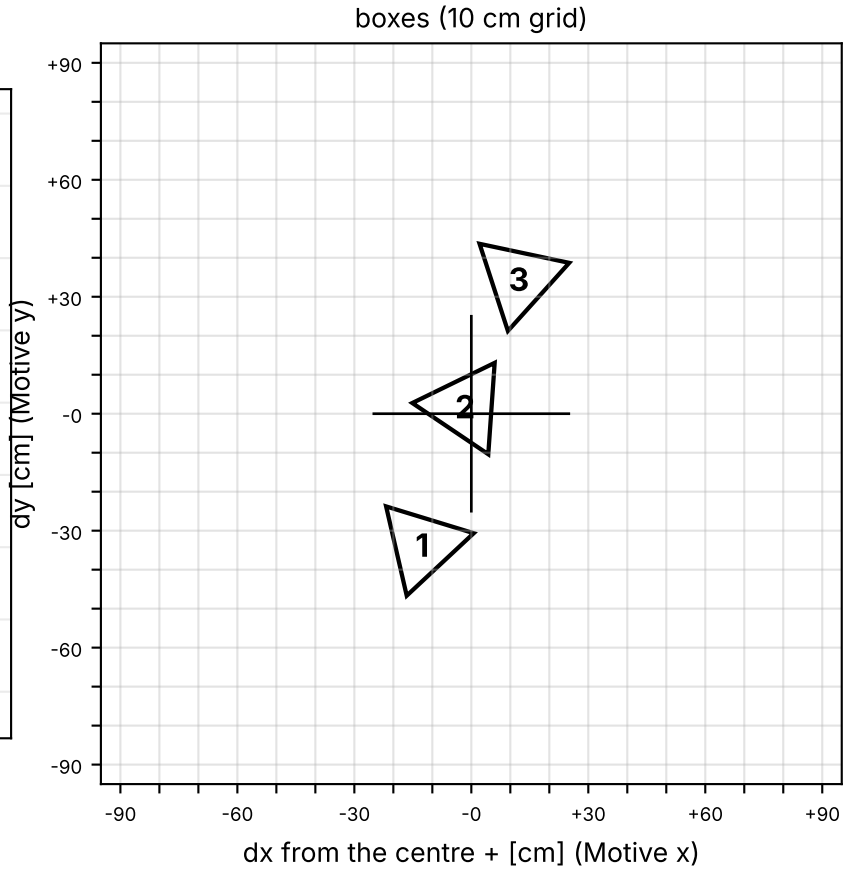
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R28 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R29 day 2 A base (test)

3 boxes, fila/arco, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-13	-34
2	-2	+2
3	+12	+34

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

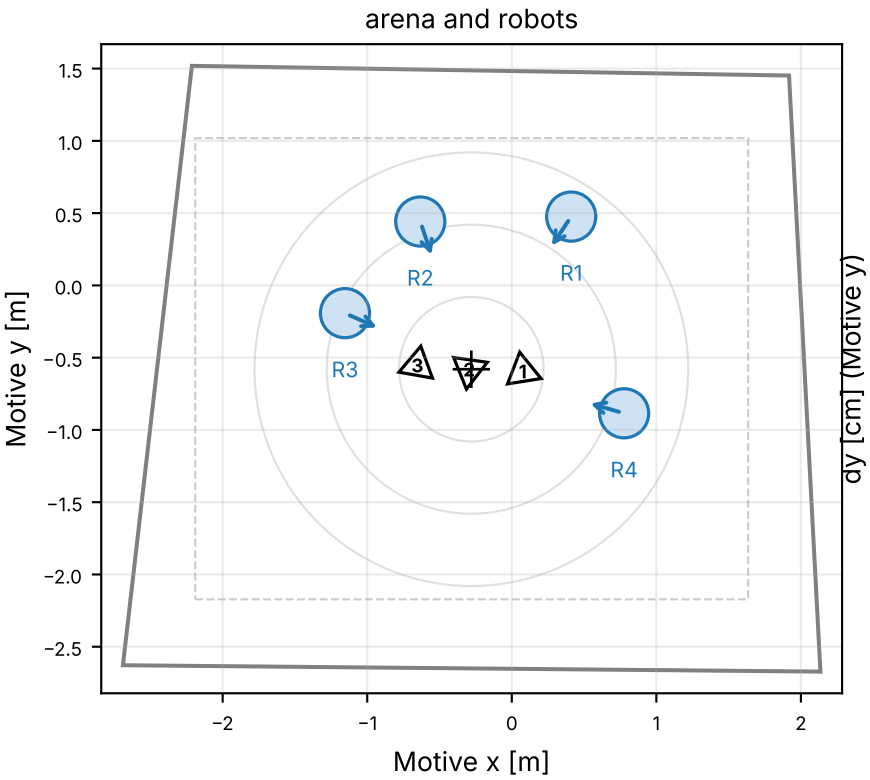
Place and capture: python -m tools.place_check --layout R29

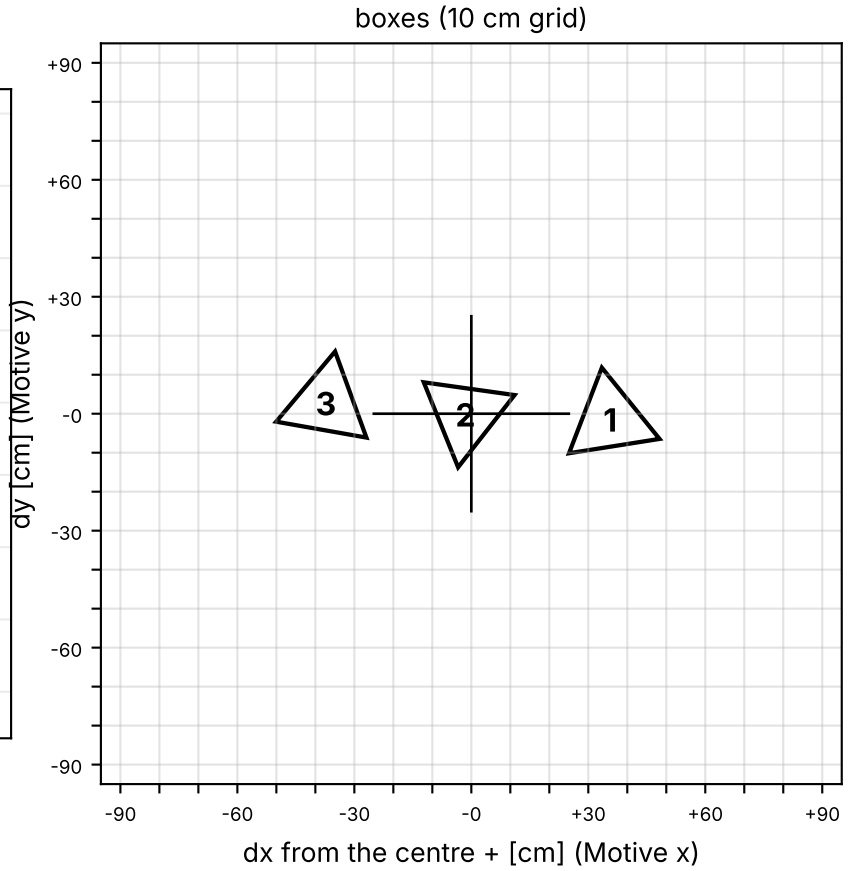
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R29 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R30 day 2 A base (test)

3 boxes, fila/arco, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+36	-2
2	-1	-0
3	-37	+3

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.41	+0.48
R2	-0.63	+0.44
R3	-1.15	-0.19
R4	+0.78	-0.89

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

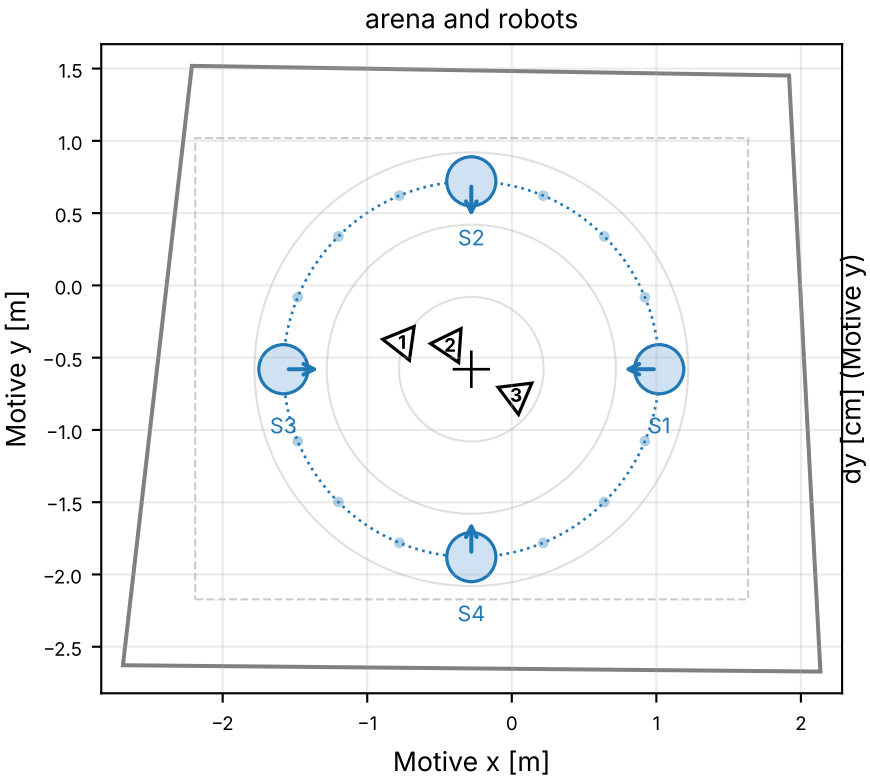
Place and capture: python -m tools.place_check --layout R30

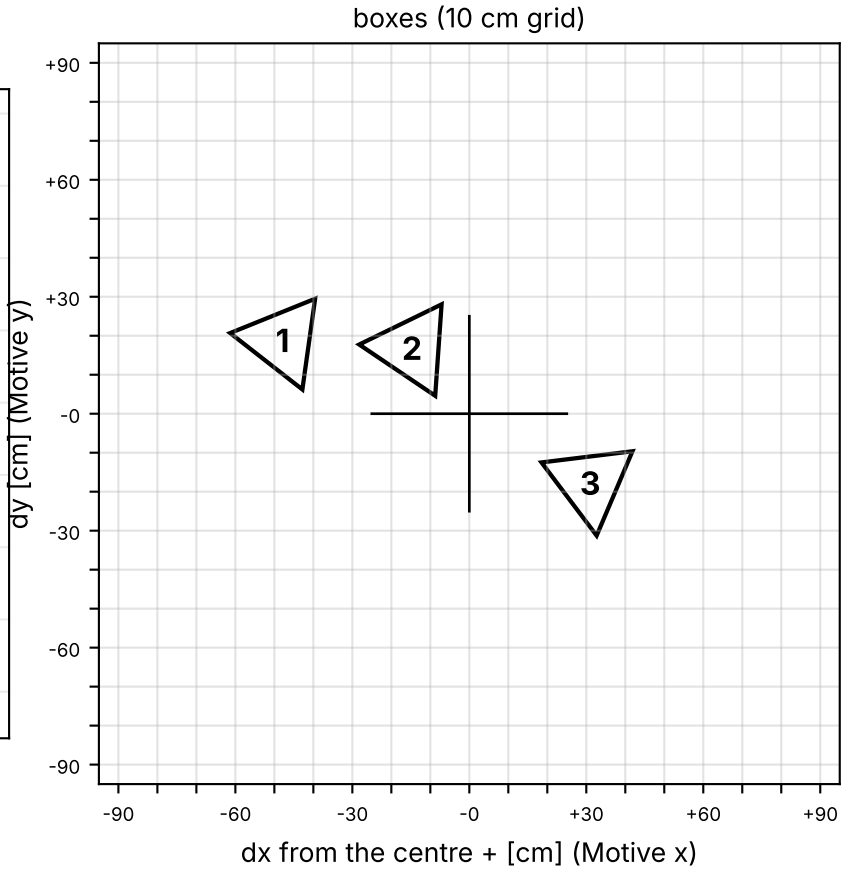
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R30 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R31 day 2 A base (test)

3 boxes, due gruppi (2+1), robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-48	+19
2	-15	+17
3	+31	-18

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

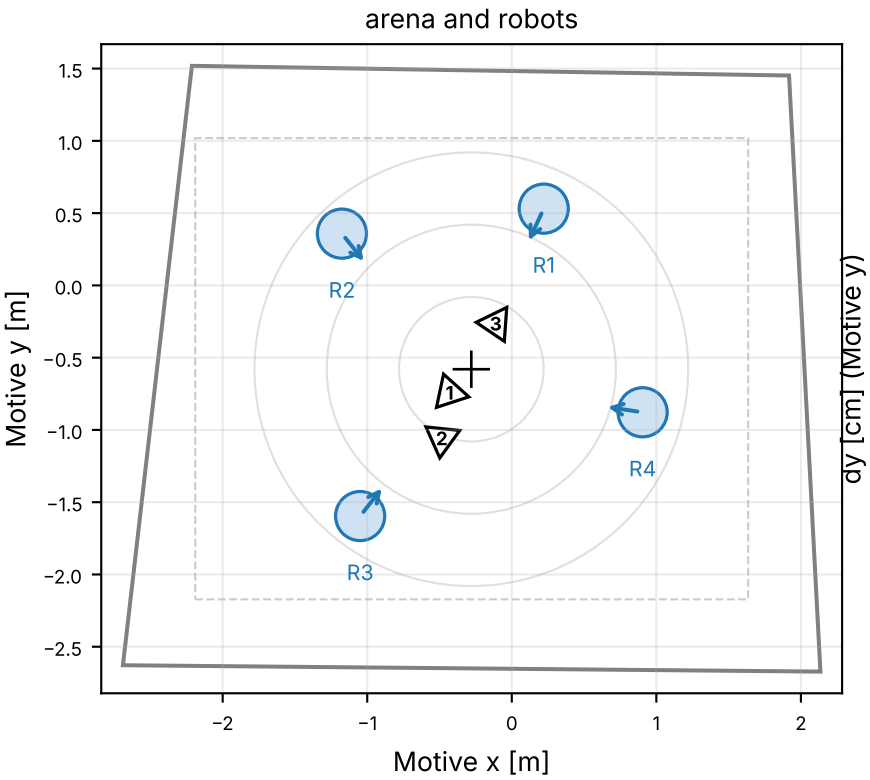
Place and capture: python -m tools.place_check --layout R31

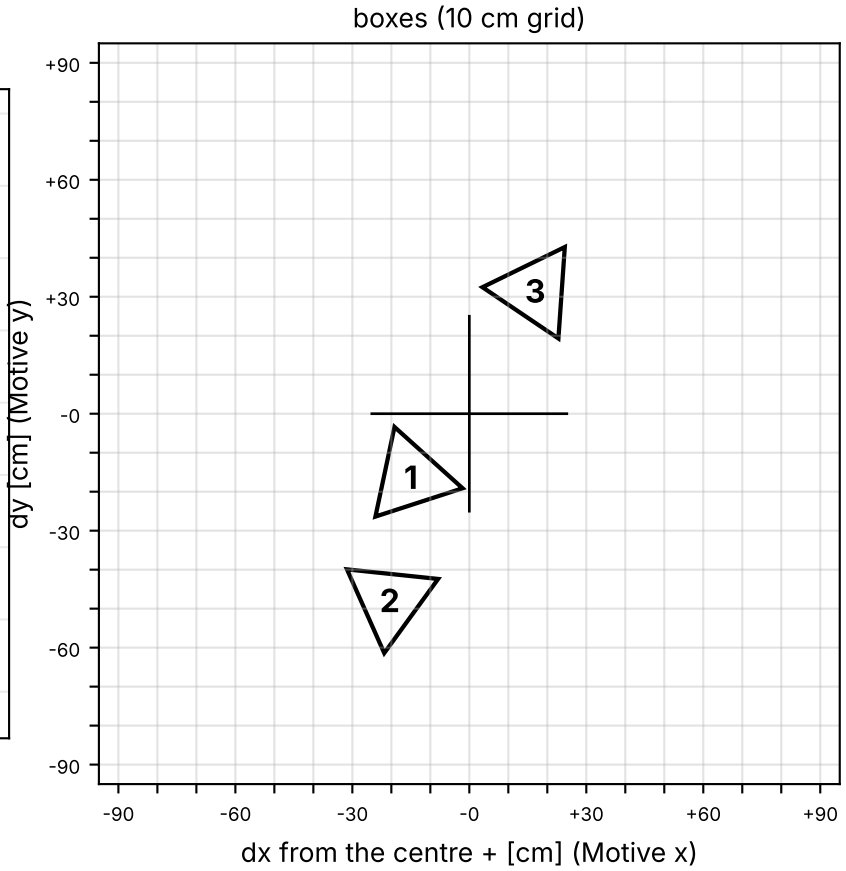
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R31 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R32 day 2 A base (test)

3 boxes, due gruppi (2+1), robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-15	-16
2	-20	-48
3	+17	+31

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.22	+0.53
R2	-1.18	+0.36
R3	-1.05	-1.60
R4	+0.91	-0.88

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

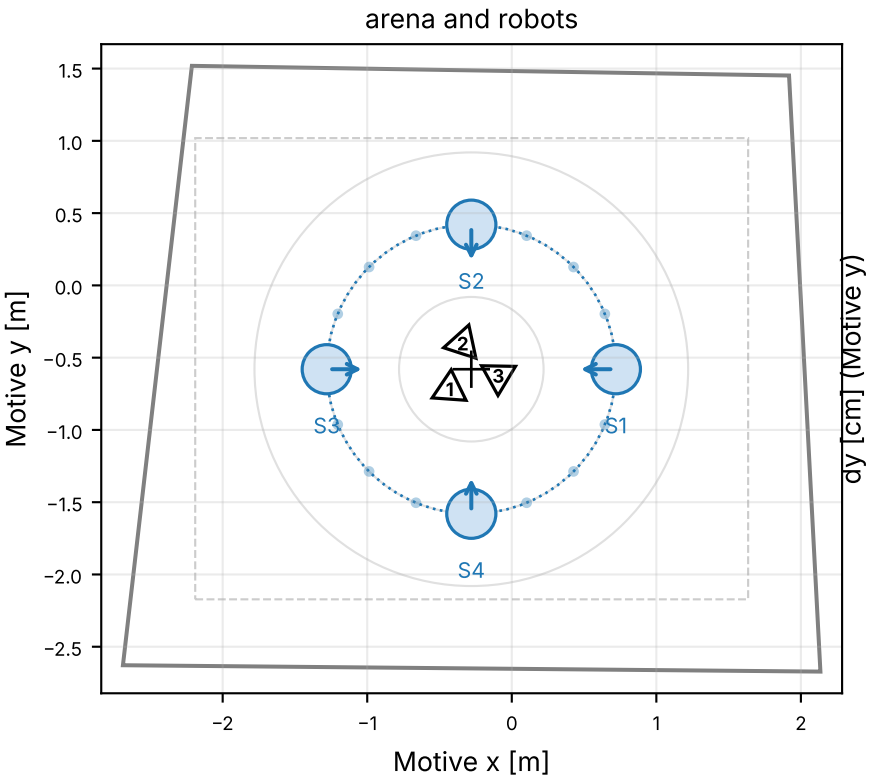
Place and capture: python -m tools.place_check --layout R32

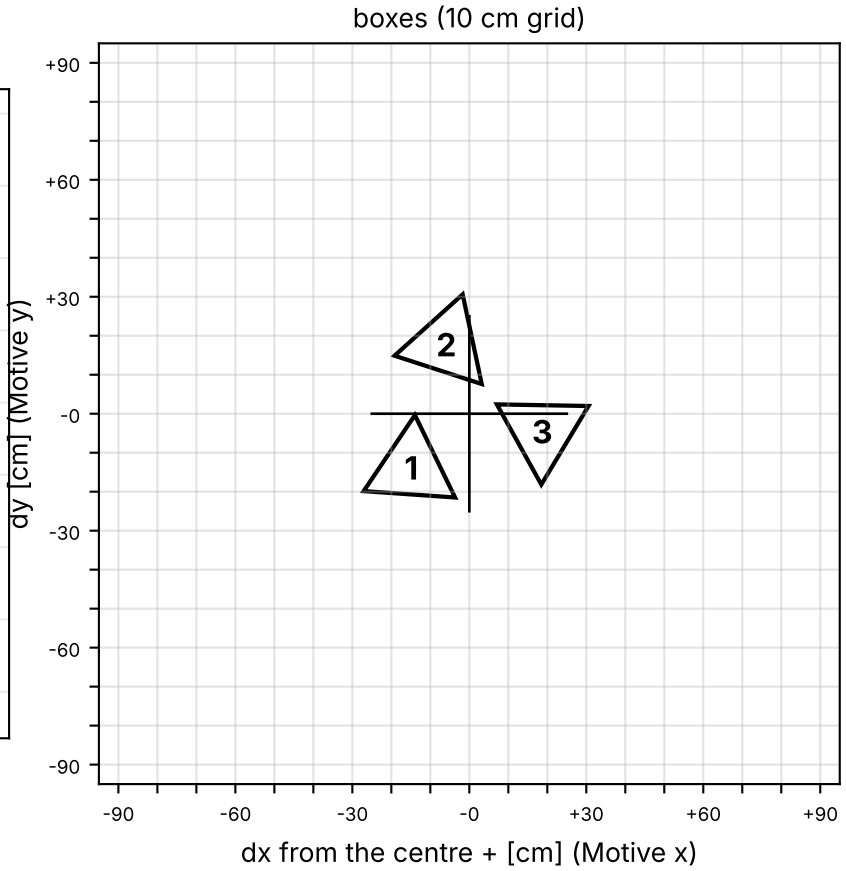
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R32 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R33 day 2 B distanza (test)

3 boxes, fitta, robots: cerchio 1.0 | scatole entro 0.45 m dal centro





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-15	-14
2	-6	+18
3	+19	-5

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

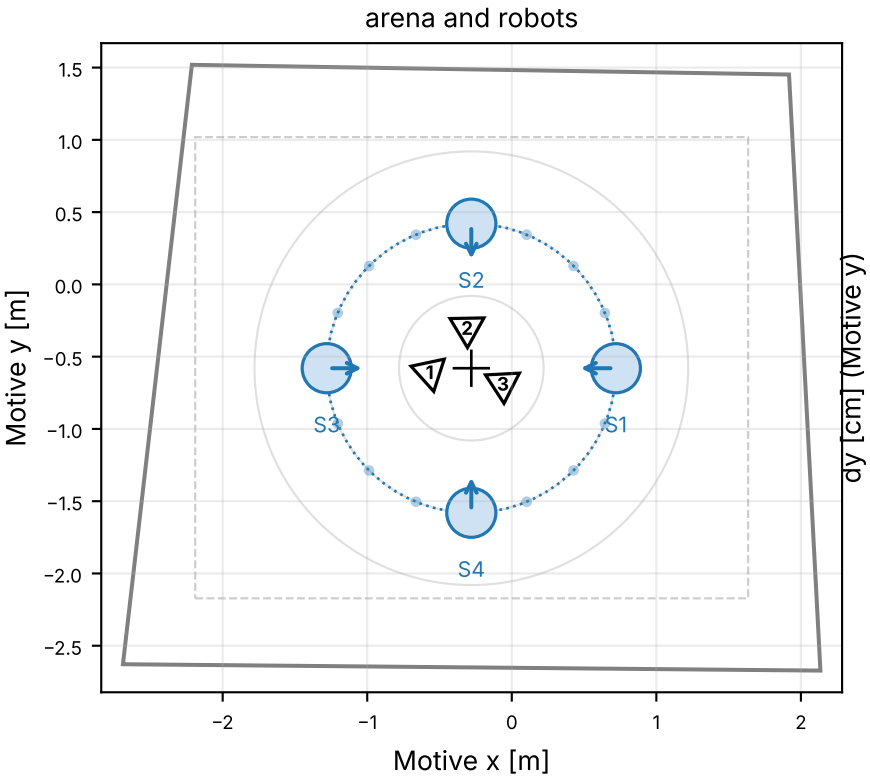
Place and capture: python -m tools.place_check --layout R33

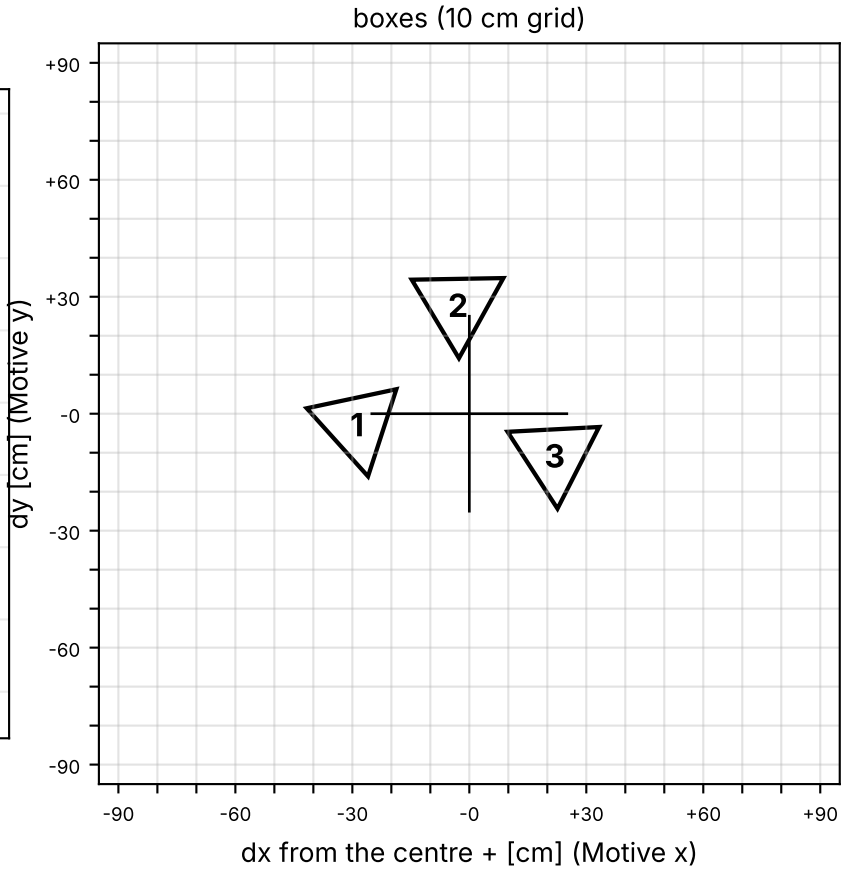
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R33 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R34 day 2 B distanza (test)

3 boxes, sparsa, robots: cerchio 1.0 | scatole entro 0.45 m dal centro





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-29	-3
2	-3	+28
3	+22	-11

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

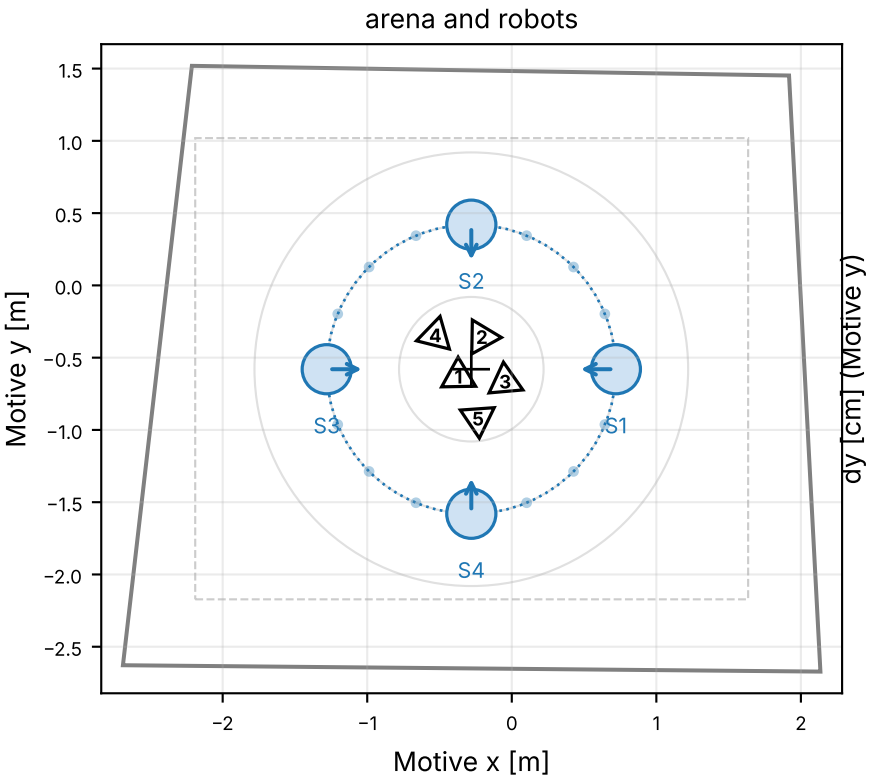
Place and capture: python -m tools.place_check --layout R34

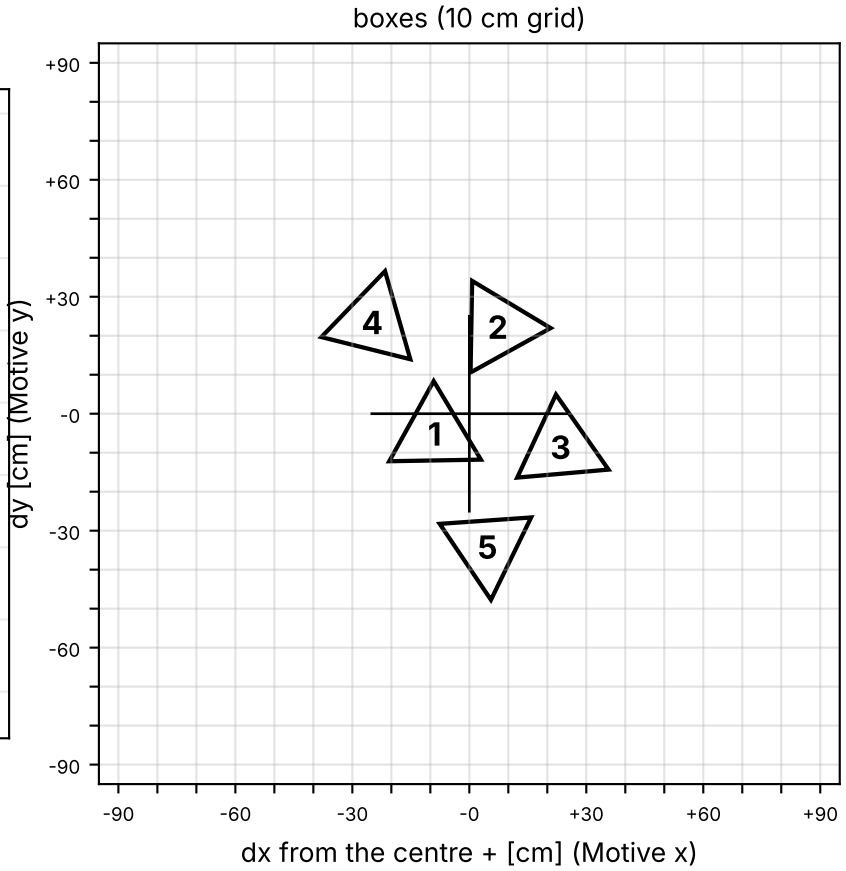
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R34 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R35 day 2 B distanza (test)

5 boxes, fitta, robots: cerchio 1.0 | scatole entro 0.45 m dal centro





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-9	-5
2	+7	+22
3	+23	-9
4	-25	+23
5	+5	-34

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

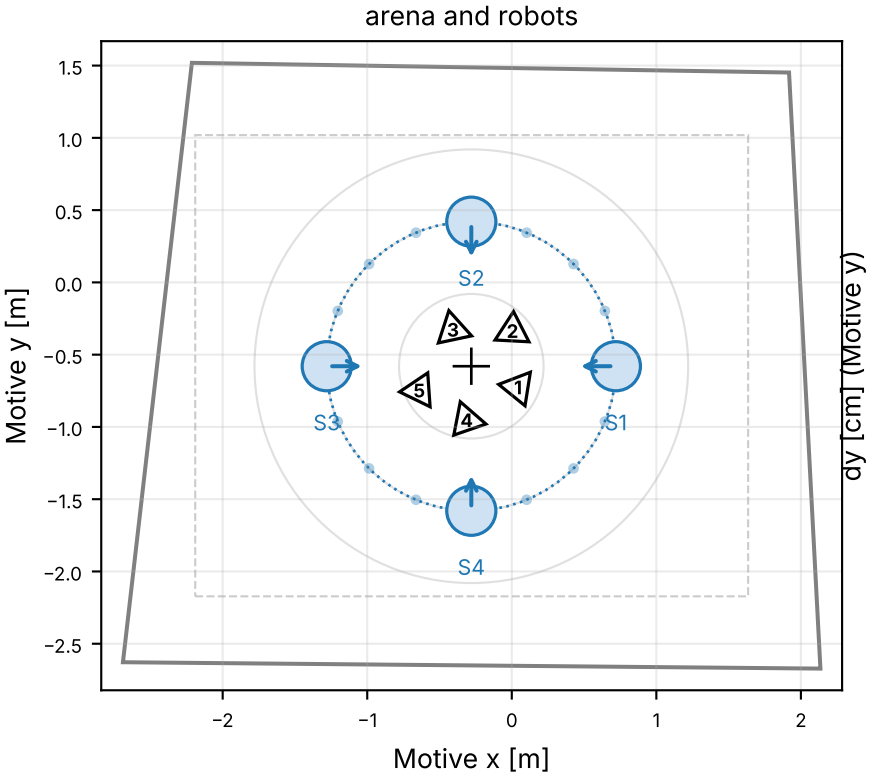
Place and capture: python -m tools.place_check --layout R35

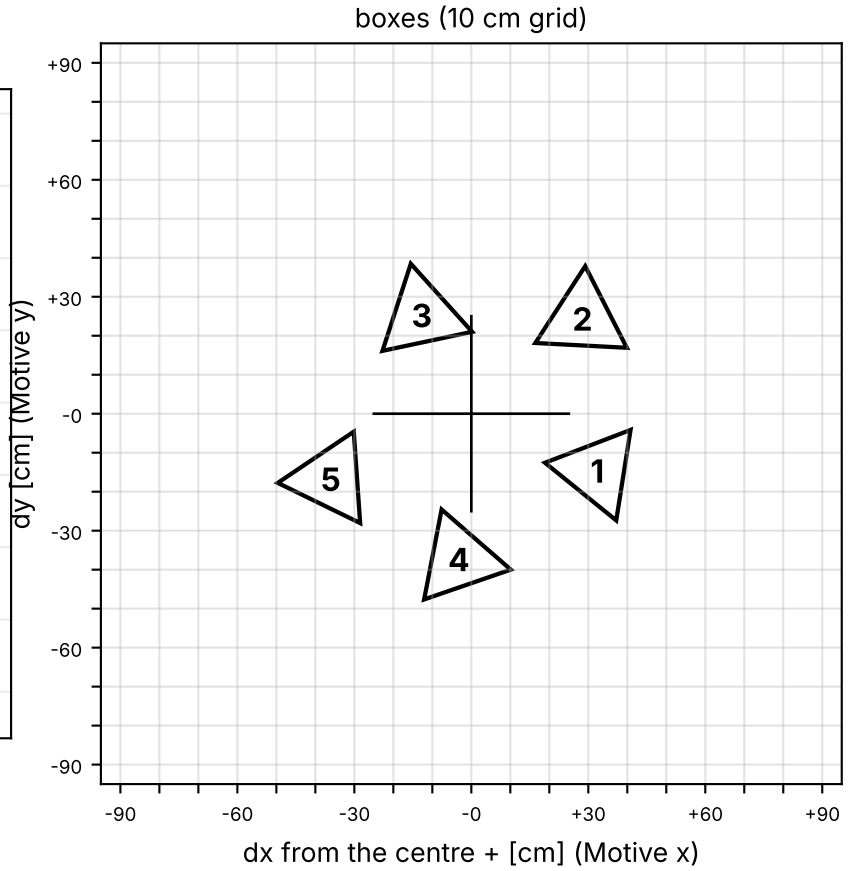
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R35 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R36 day 2 B distanza (test)

5 boxes, sparsa, robots: cerchio 1.0 | scatole entro 0.45 m dal centro





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+32	-15
2	+29	+24
3	-13	+25
4	-3	-37
5	-36	-17

Robots

anywhere near the 1 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

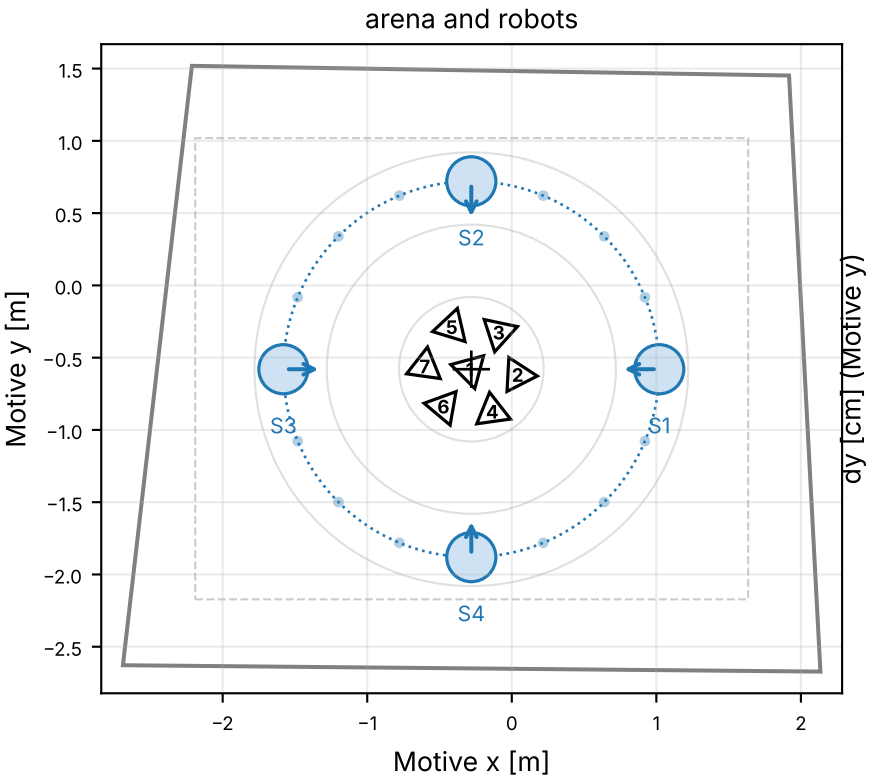
Place and capture: python -m tools.place_check --layout R36

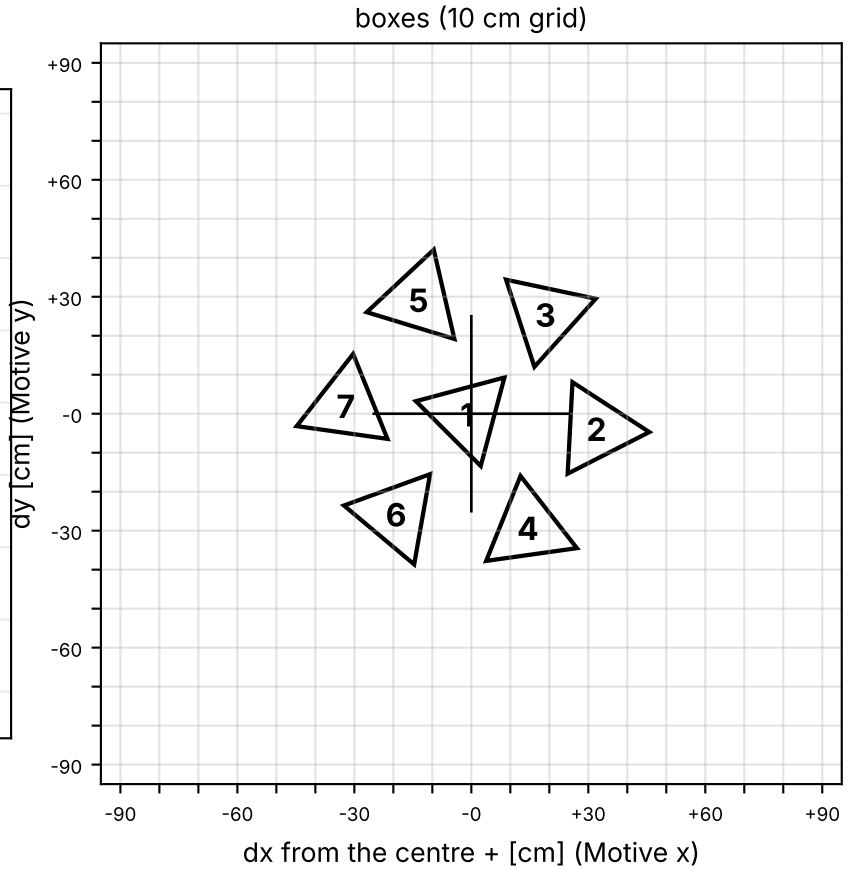
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R36 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R37 day 3 A base (test)

7 boxes, fitta, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-1	-0
2	+32	-4
3	+19	+25
4	+15	-29
5	-14	+29
6	-19	-26
7	-32	+2

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

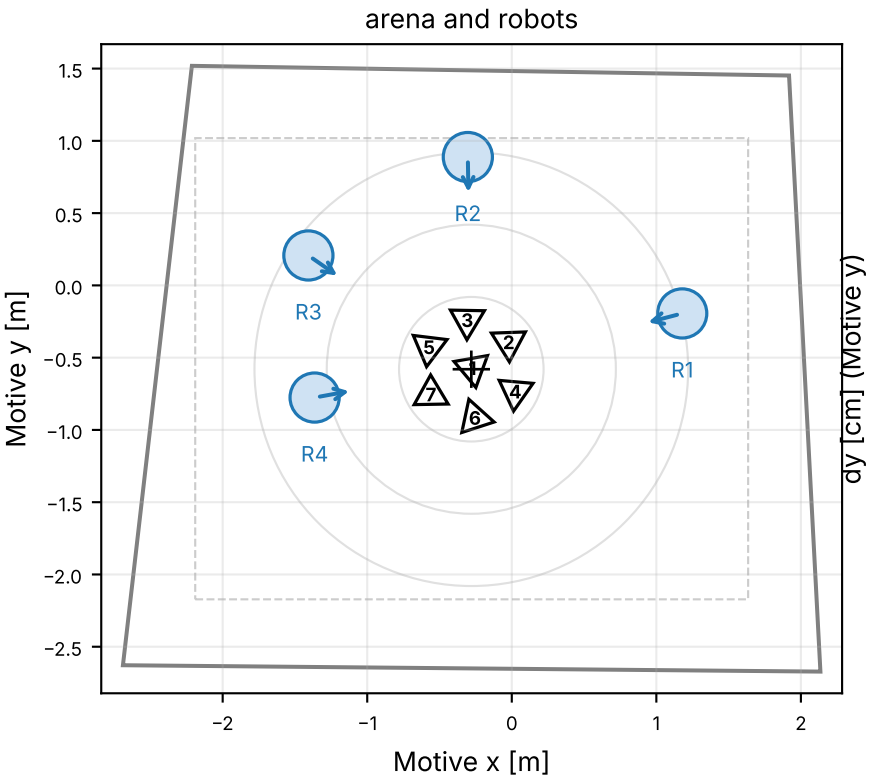
Place and capture: python -m tools.place_check --layout R37

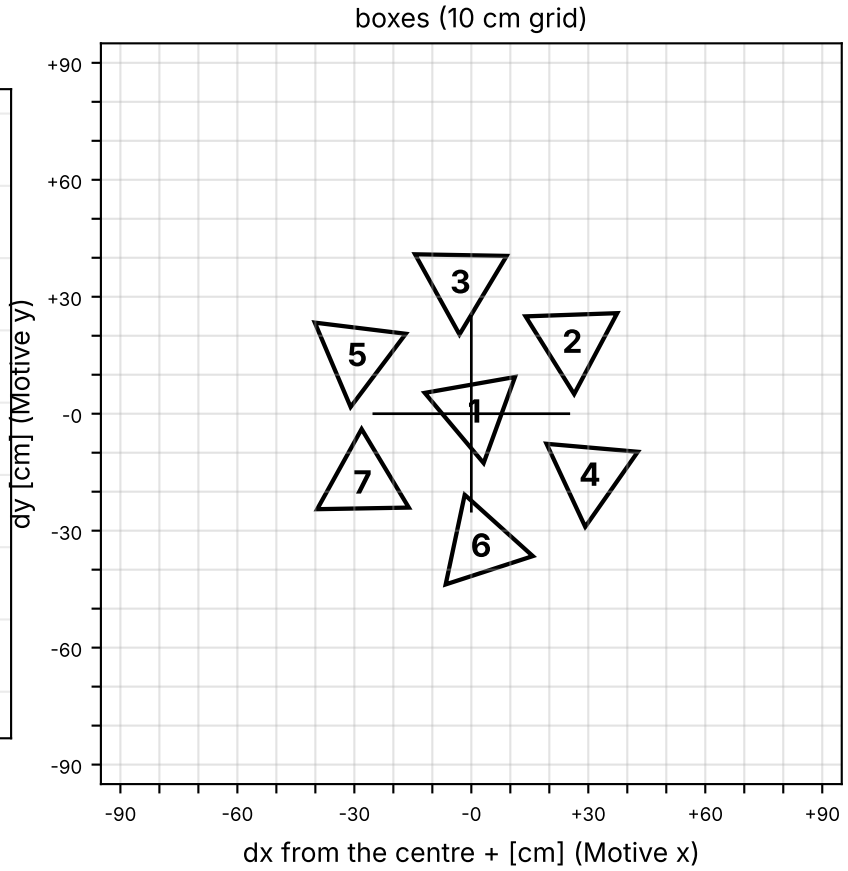
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R37 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R38 day 3 A base (test)

7 boxes, fitta, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+1	+1
2	+26	+19
3	-3	+34
4	+30	-16
5	-29	+15
6	+3	-34
7	-28	-18

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+1.18	-0.19
R2	-0.30	+0.89
R3	-1.41	+0.21
R4	-1.36	-0.78

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

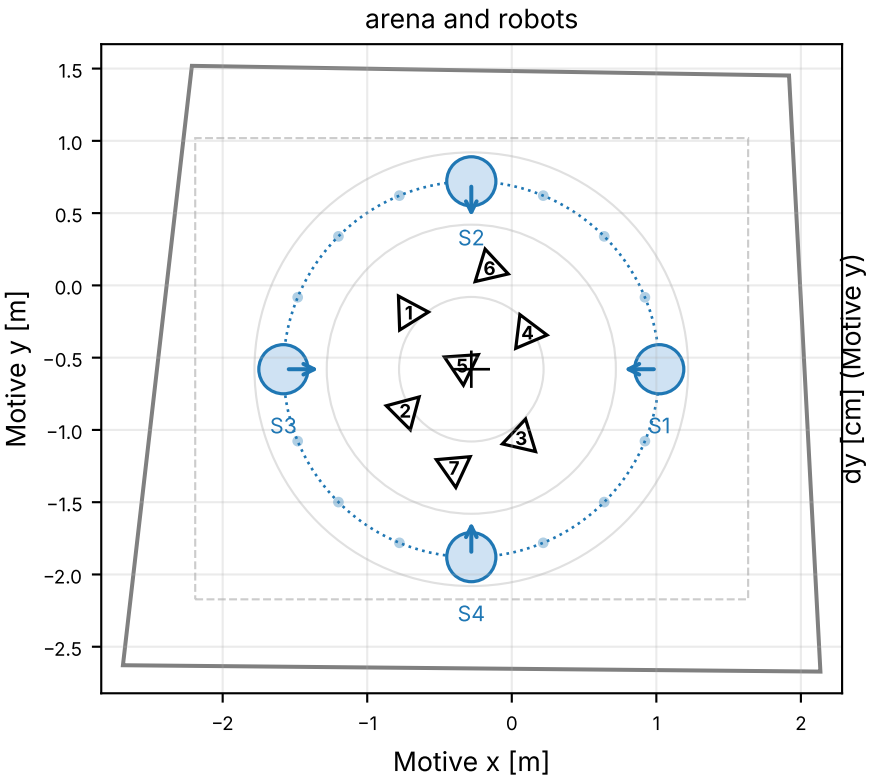
Place and capture: python -m tools.place_check --layout R38

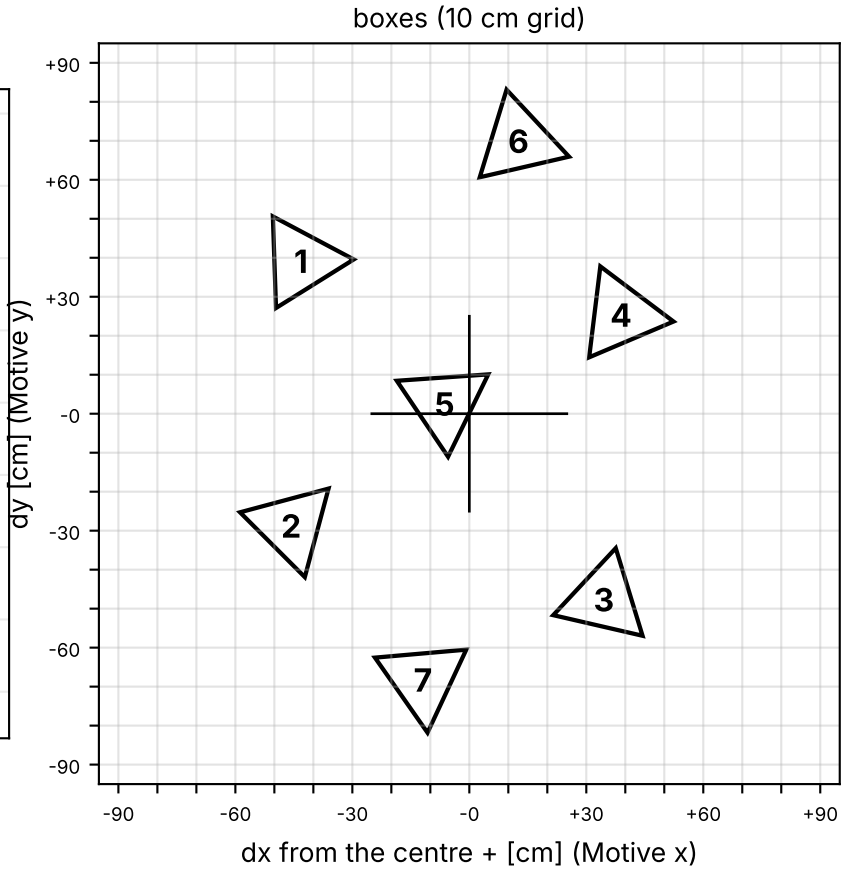
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R38 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R39 day 3 A base (test)

7 boxes, sparsa, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-43	+39
2	-46	-29
3	+34	-48
4	+39	+25
5	-6	+2
6	+13	+70
7	-12	-68

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

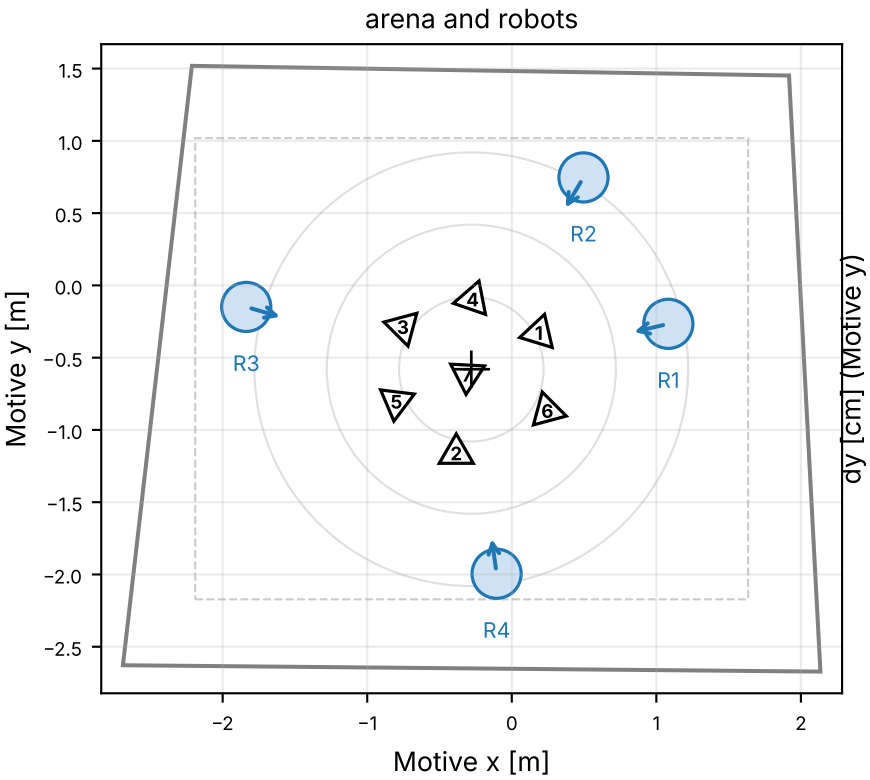
Place and capture: python -m tools.place_check --layout R39

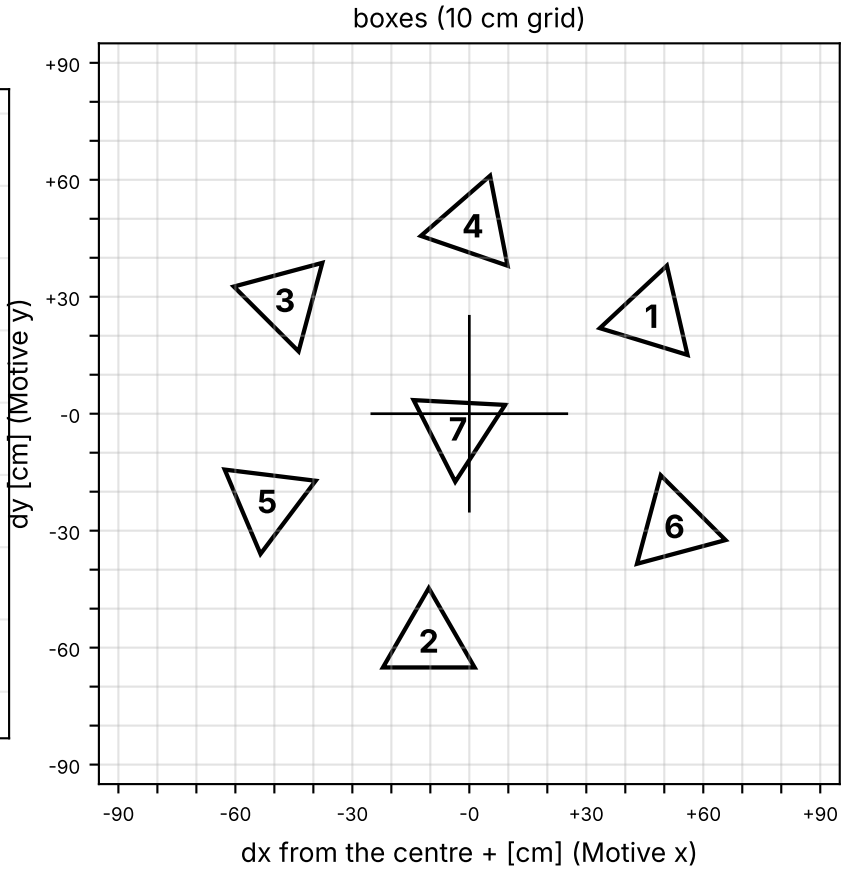
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R39 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R40 day 3 A base (test)

7 boxes, sparsa, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+47	+25
2	-10	-58
3	-47	+29
4	+1	+48
5	-52	-23
6	+53	-29
7	-3	-4

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+1.08	-0.27
R2	+0.50	+0.75
R3	-1.84	-0.15
R4	-0.10	-2.00

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

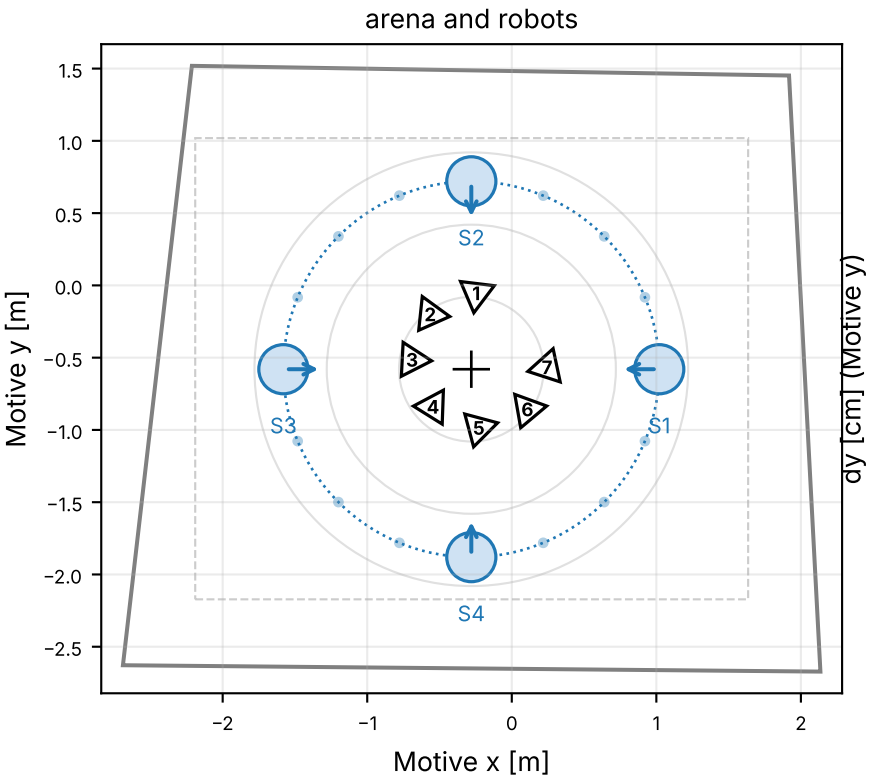
Place and capture: python -m tools.place_check --layout R40

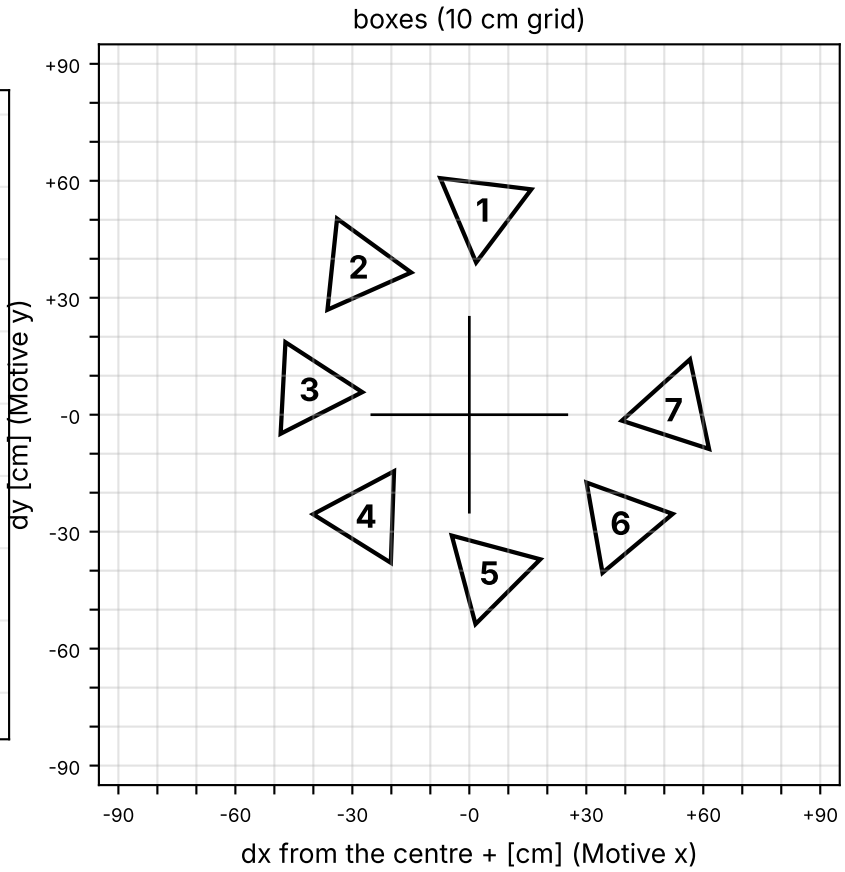
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R40 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R41 day 3 A base (test)

7 boxes, fila/arco, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+3	+52
2	-28	+38
3	-41	+6
4	-26	-26
5	+5	-41
6	+39	-28
7	+52	+1

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

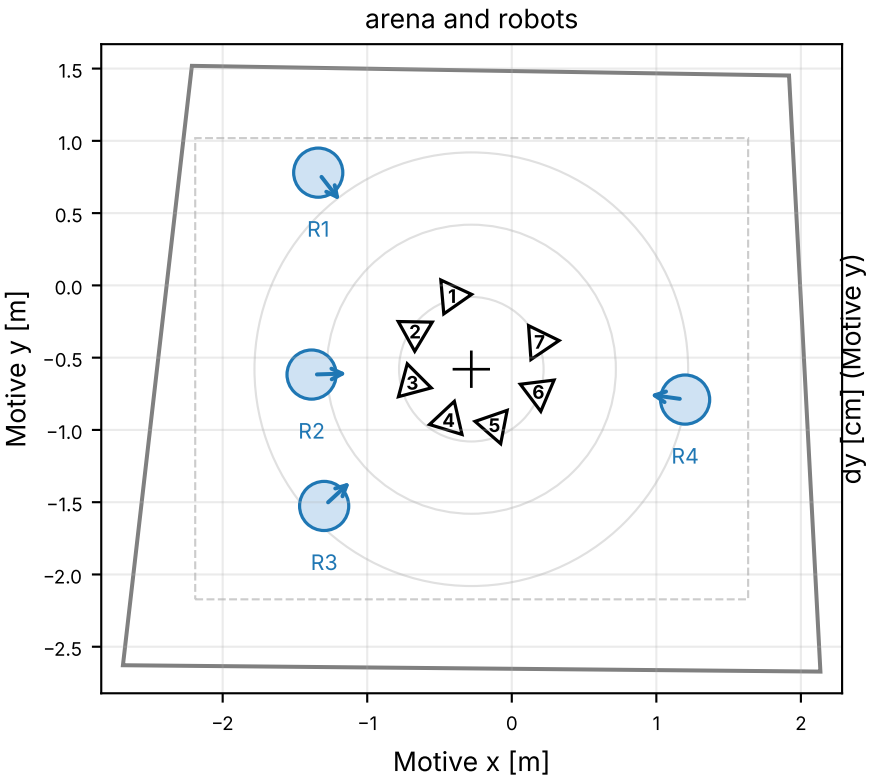
Place and capture: python -m tools.place_check --layout R41

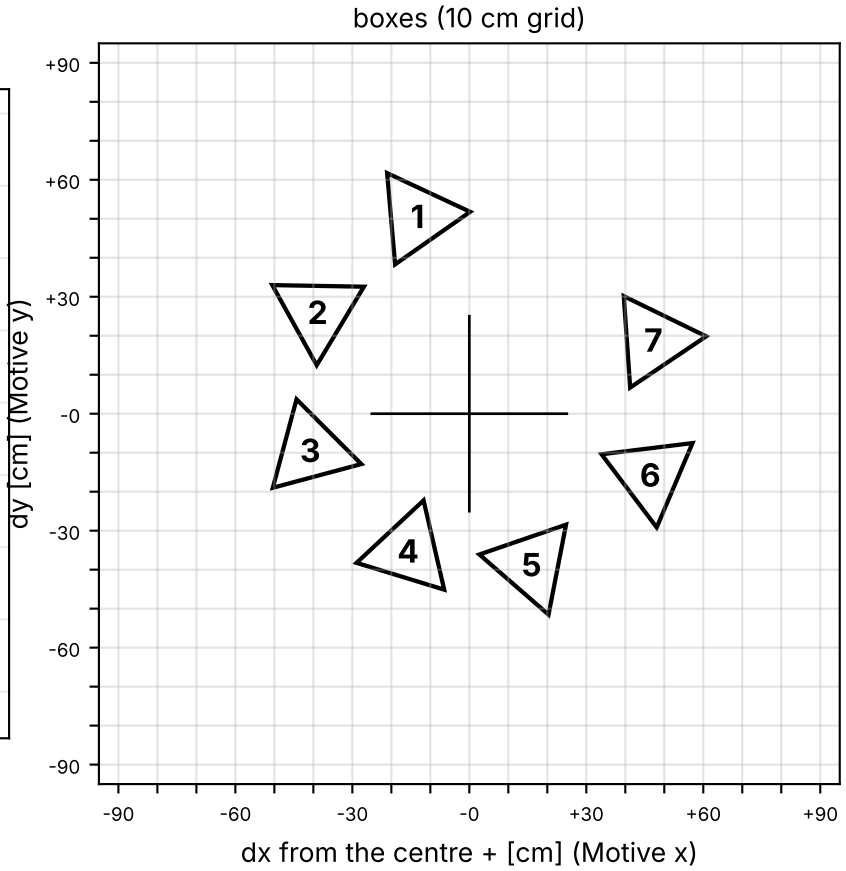
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R41 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R42 day 3 A base (test)

7 boxes, fila/arco, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-13	+51
2	-39	+26
3	-41	-9
4	-16	-35
5	+16	-39
6	+46	-16
7	+47	+19

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	-1.34	+0.78
R2	-1.39	-0.62
R3	-1.30	-1.53
R4	+1.20	-0.79

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

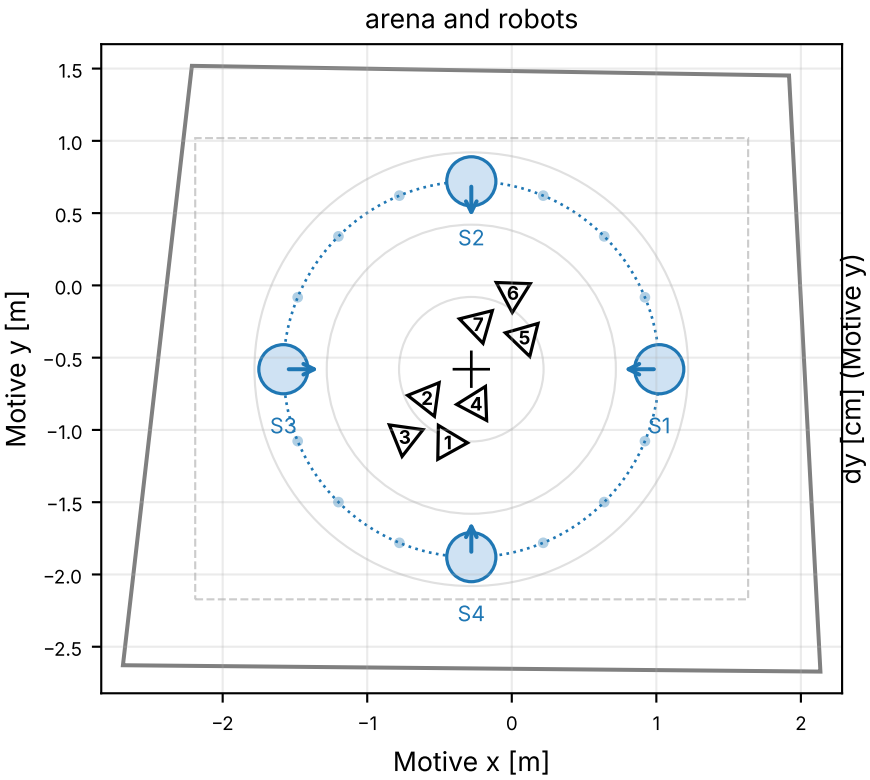
Place and capture: python -m tools.place_check --layout R42

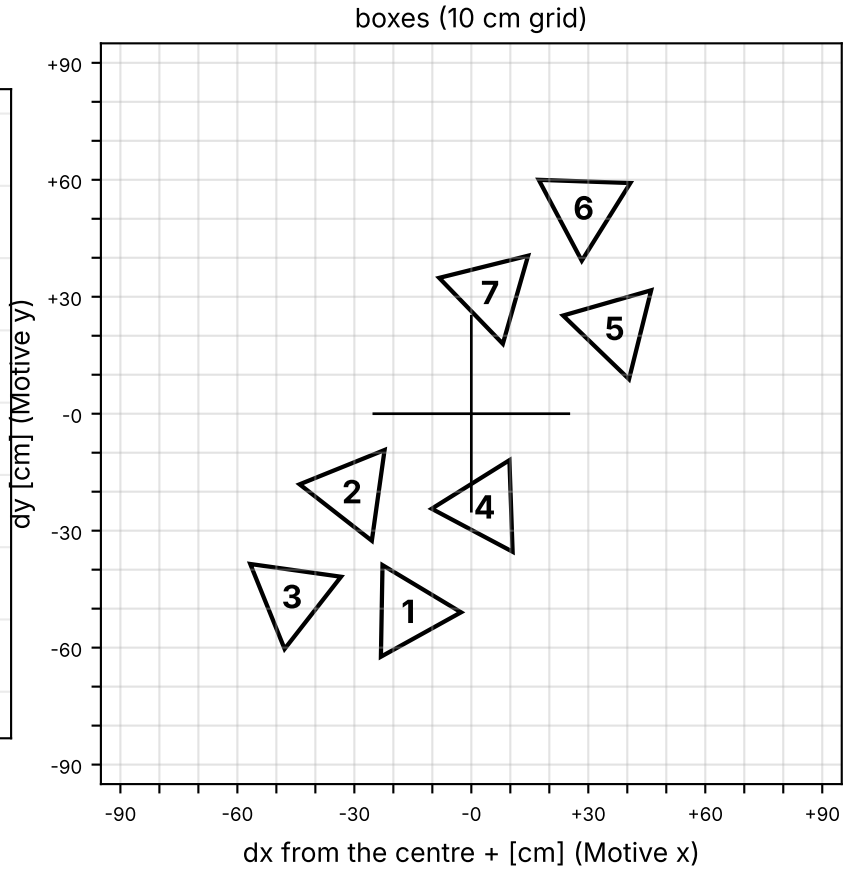
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R42 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R43 day 3 A base (test)

7 boxes, due gruppi, robots: cerchio 1.3 | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-16	-51
2	-31	-20
3	-46	-47
4	+3	-24
5	+37	+22
6	+29	+53
7	+5	+31

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

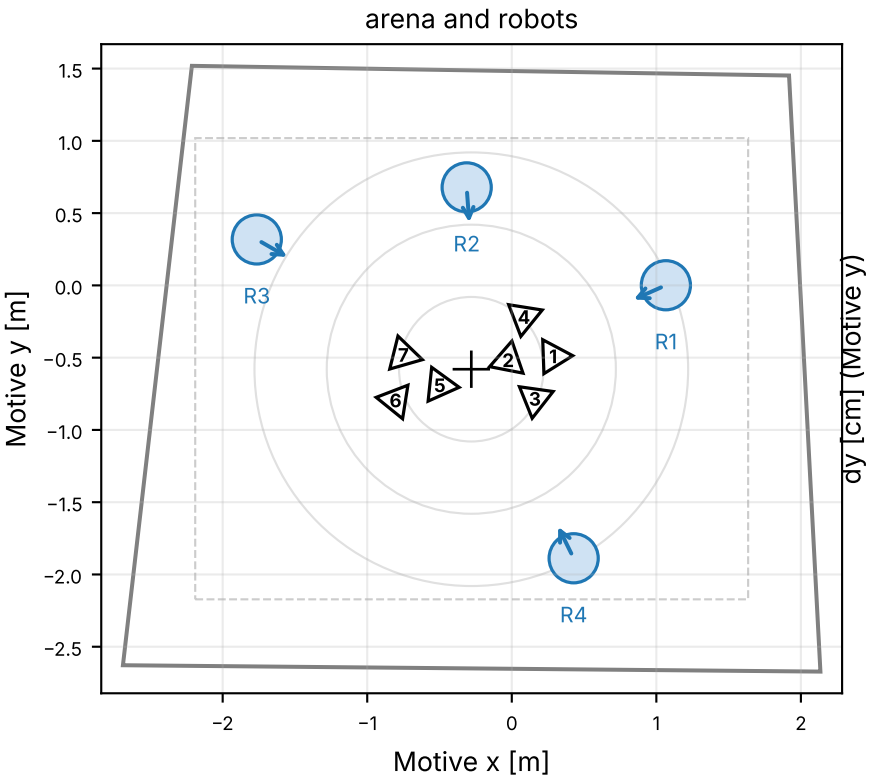
Place and capture: python -m tools.place_check --layout R43

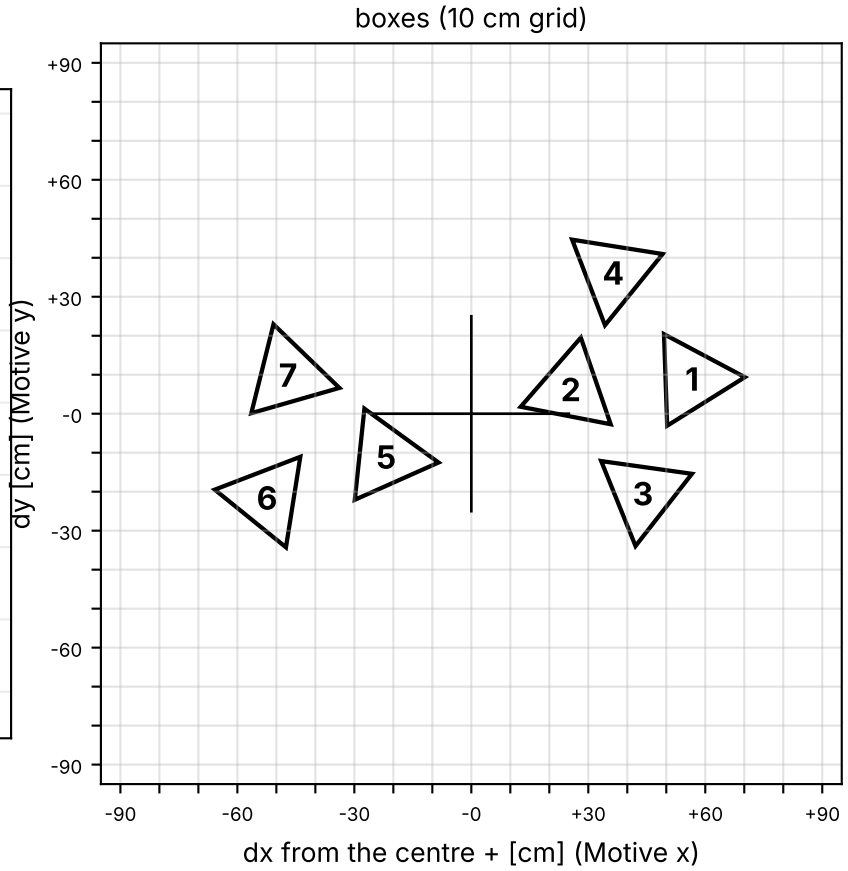
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R43 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R44 day 3 A base (test)

7 boxes, due gruppi, robots: casuale | replica a





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+57	+9
2	+26	+6
3	+44	-21
4	+36	+36
5	-22	-11
6	-52	-22
7	-47	+10

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+1.07	+0.00
R2	-0.31	+0.68
R3	-1.76	+0.32
R4	+0.43	-1.89

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

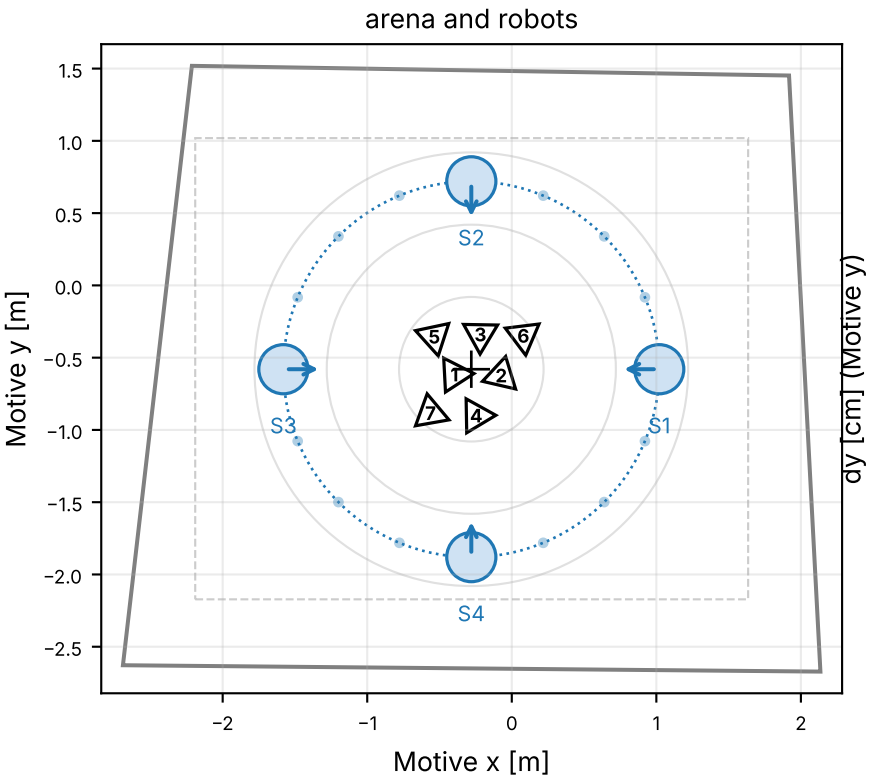
Place and capture: python -m tools.place_check --layout R44

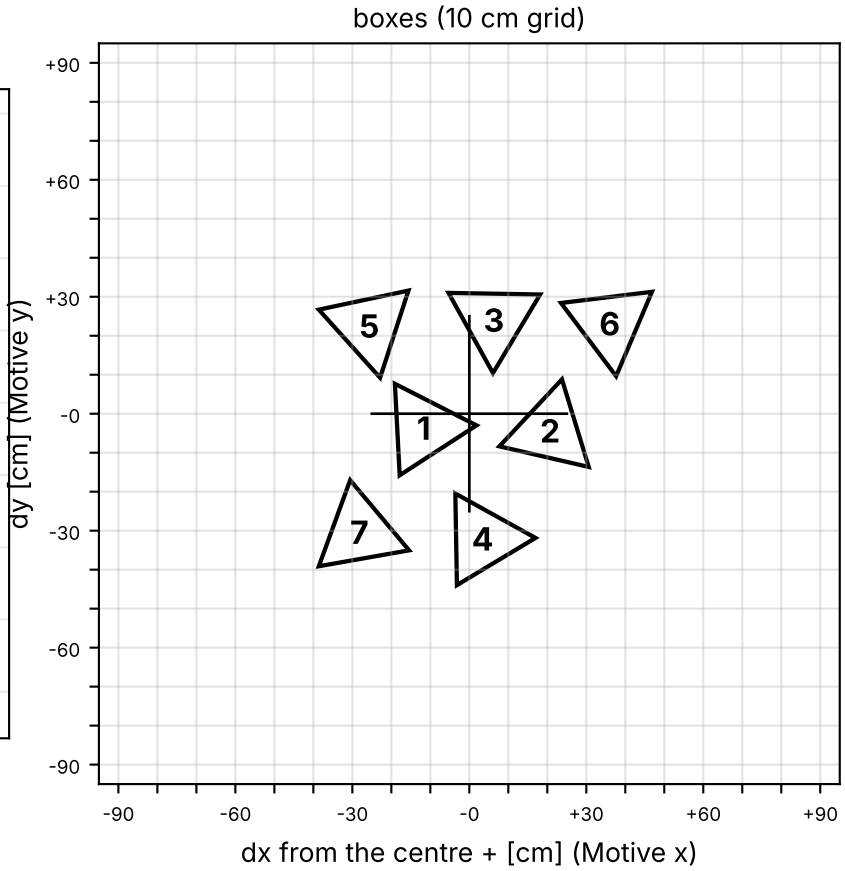
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R44 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R45 day 3 A base (test)

7 boxes, fitta, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-12	-4
2	+21	-4
3	+6	+24
4	+3	-32
5	-26	+22
6	+36	+23
7	-28	-30

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

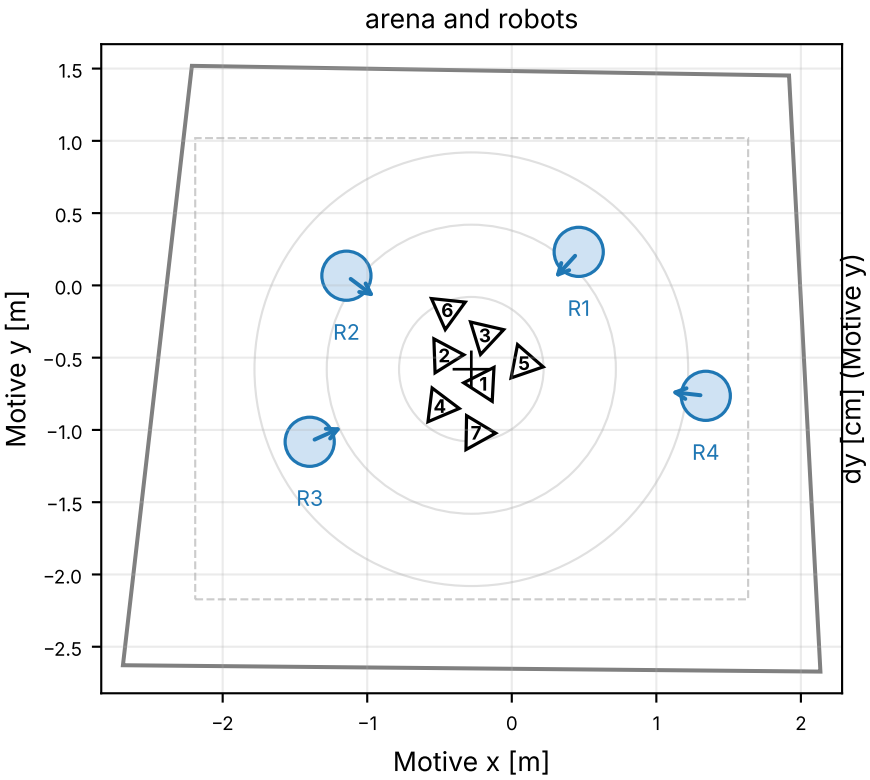
Place and capture: python -m tools.place_check --layout R45

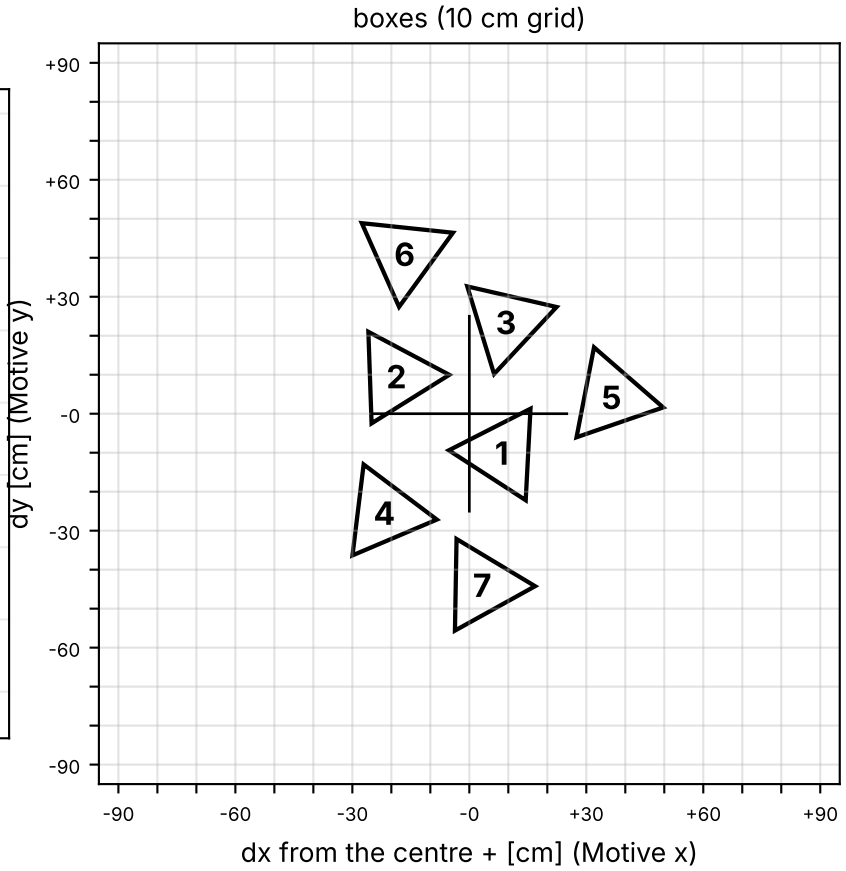
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R45 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R46 day 3 A base (test)

7 boxes, fitta, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+8	-10
2	-19	+9
3	+9	+23
4	-22	-26
5	+36	+4
6	-17	+41
7	+3	-44

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.46	+0.23
R2	-1.14	+0.07
R3	-1.40	-1.08
R4	+1.34	-0.76

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

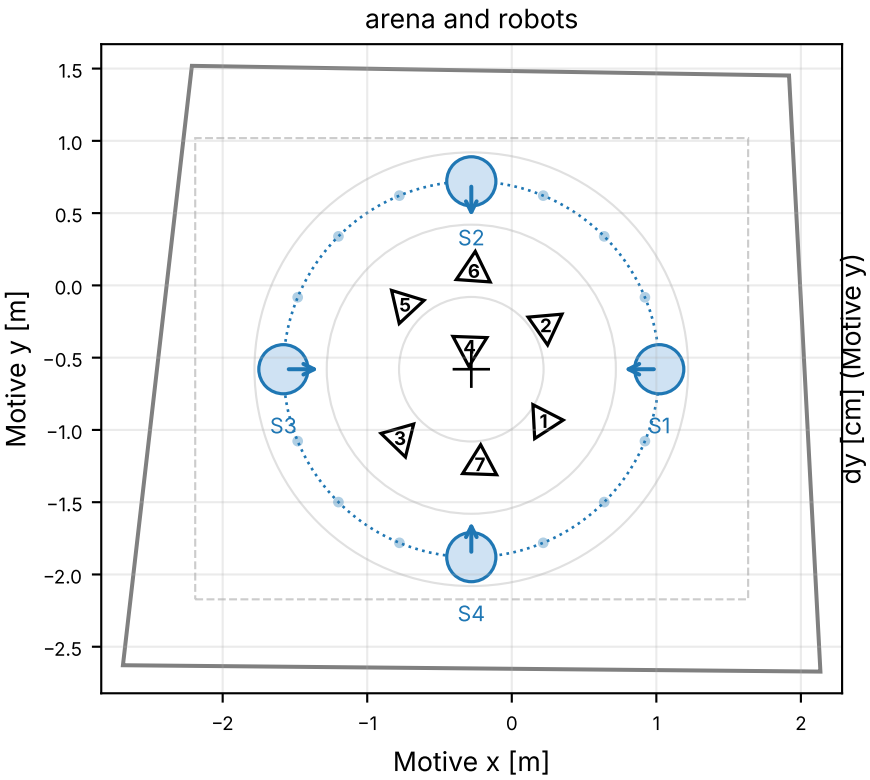
Place and capture: python -m tools.place_check --layout R46

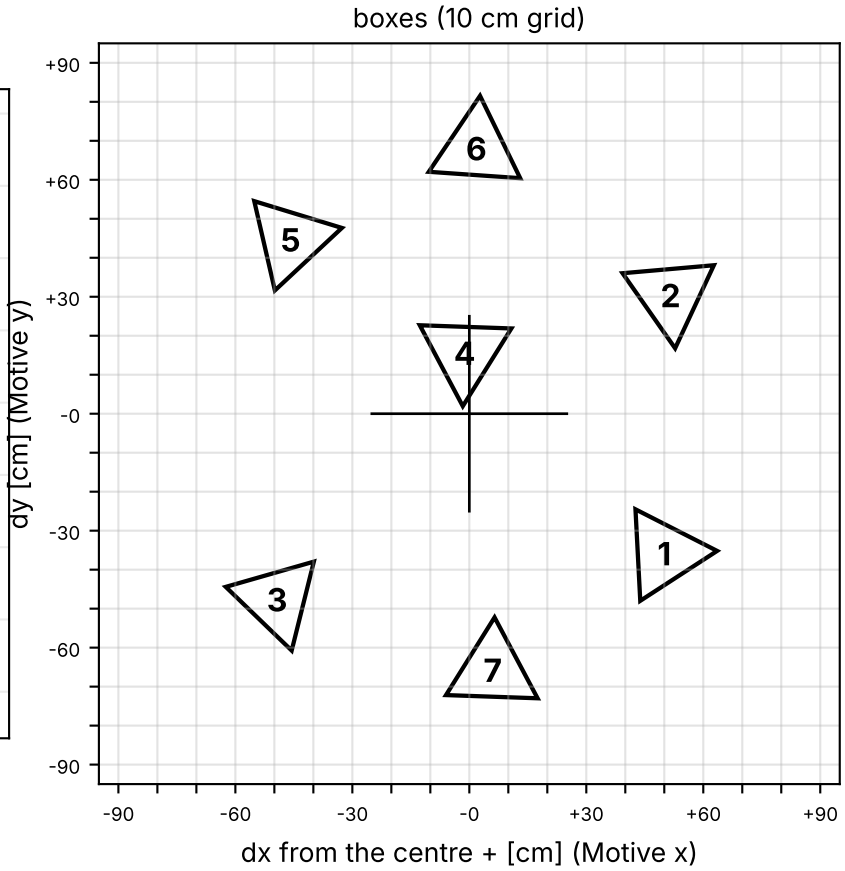
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R46 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R47 day 3 A base (test)

7 boxes, sparsa, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+50	-36
2	+52	+30
3	-49	-48
4	-1	+15
5	-46	+45
6	+2	+68
7	+6	-66

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

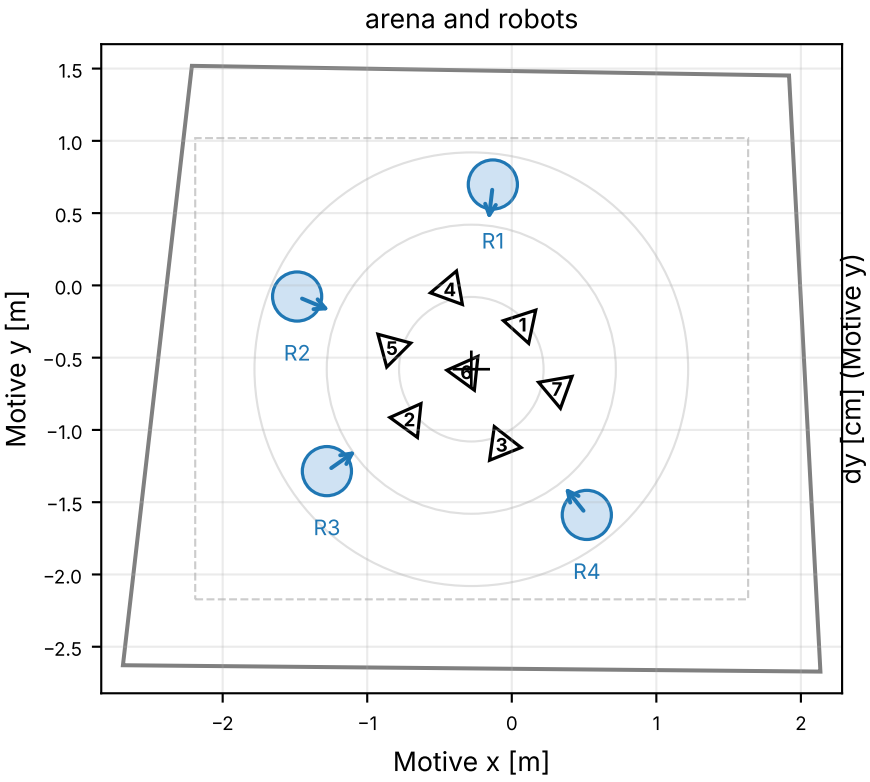
Place and capture: python -m tools.place_check --layout R47

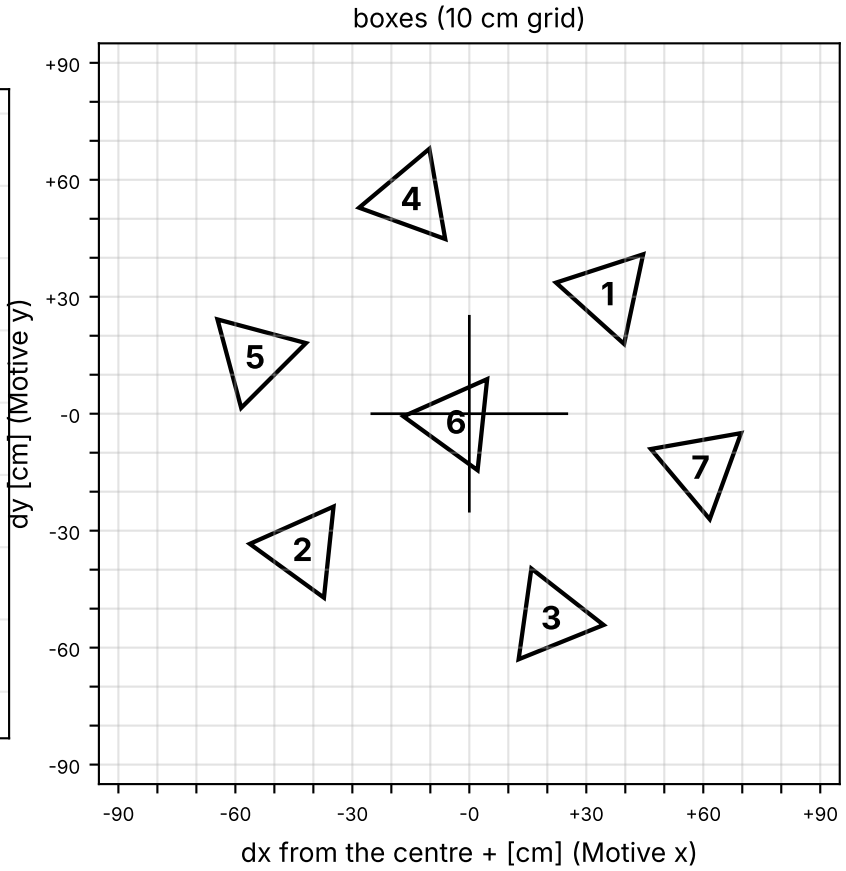
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R47 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R48 day 3 A base (test)

7 boxes, sparsa, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+36	+31
2	-43	-35
3	+21	-52
4	-15	+55
5	-55	+15
6	-3	-2
7	+59	-14

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	-0.13	+0.70
R2	-1.49	-0.08
R3	-1.28	-1.28
R4	+0.52	-1.59

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

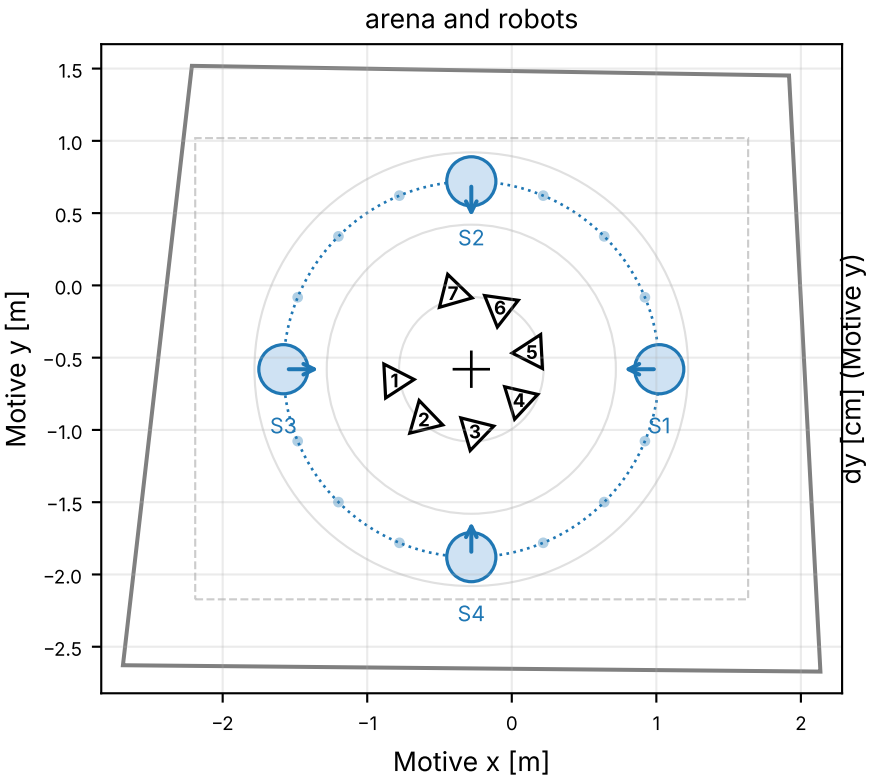
Place and capture: python -m tools.place_check --layout R48

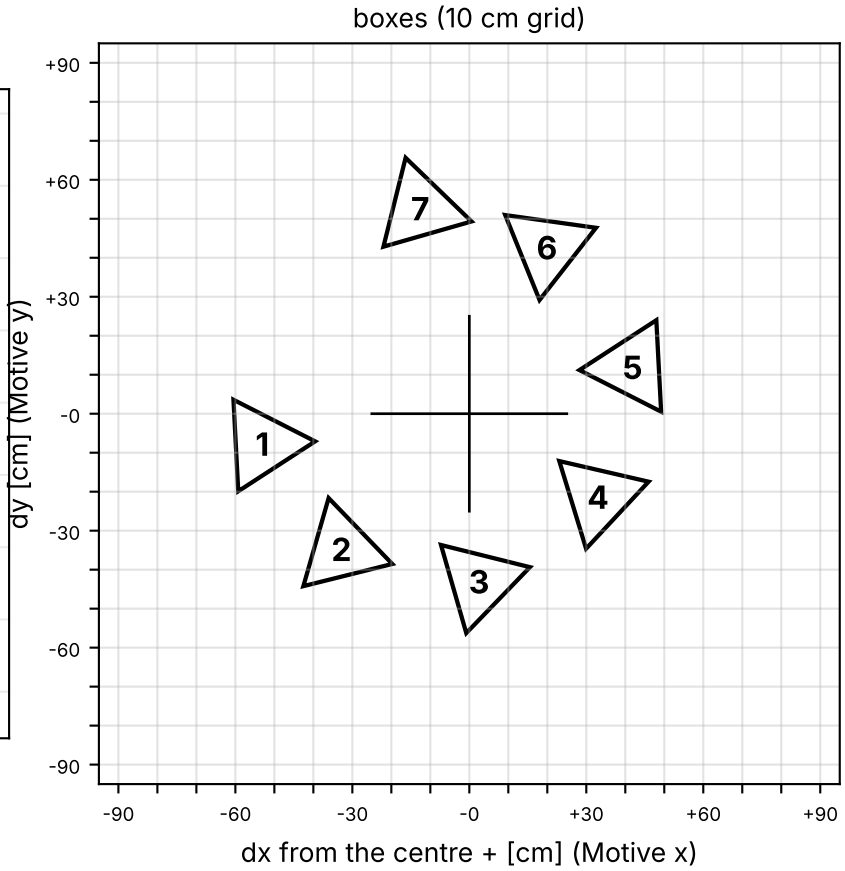
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R48 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R49 day 3 A base (test)

7 boxes, fila/arco, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-53	-8
2	-33	-35
3	+3	-43
4	+33	-21
5	+42	+12
6	+20	+43
7	-13	+53

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

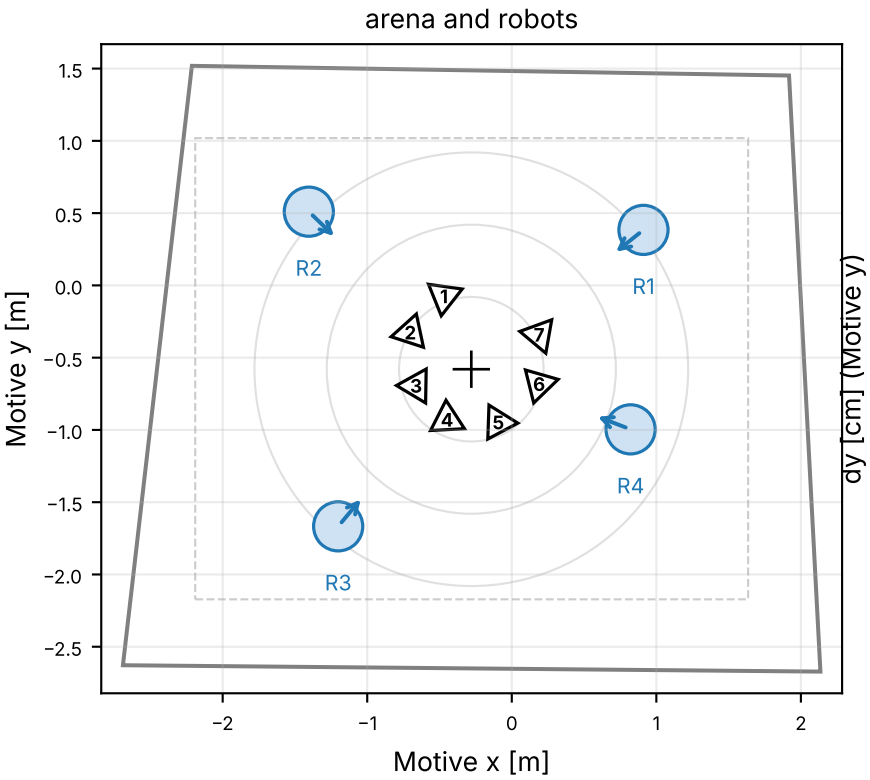
Place and capture: python -m tools.place_check --layout R49

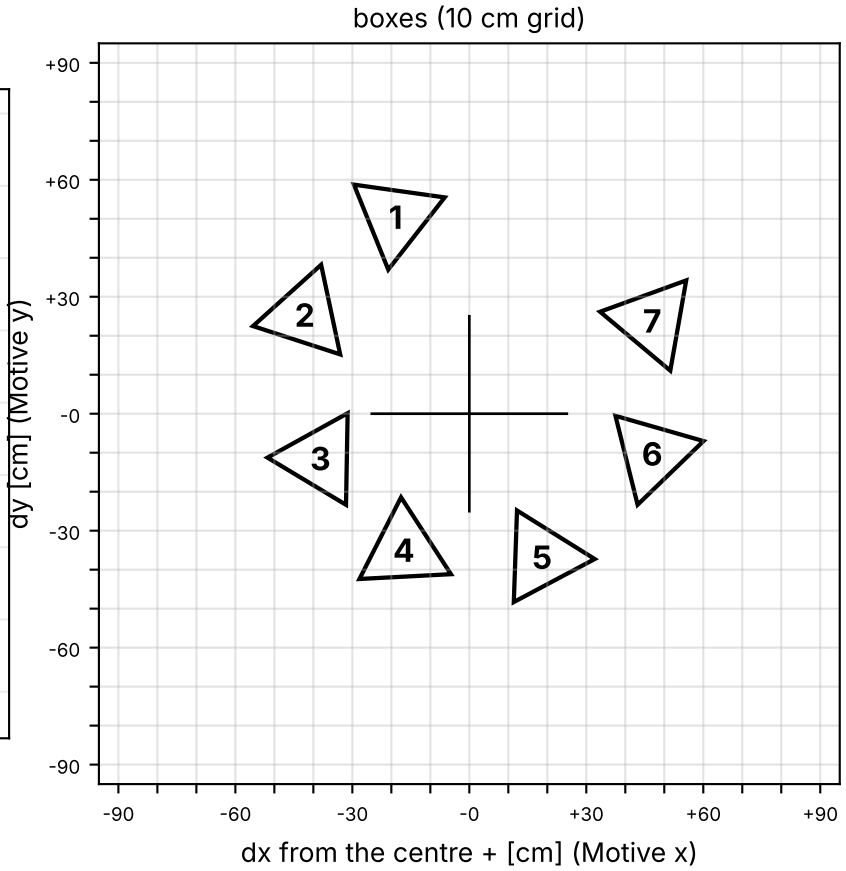
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R49 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R50 day 3 A base (test)

7 boxes, fila/arco, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-19	+50
2	-42	+25
3	-38	-12
4	-17	-35
5	+19	-37
6	+47	-10
7	+47	+24

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.91	+0.38
R2	-1.40	+0.51
R3	-1.20	-1.67
R4	+0.82	-1.00

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

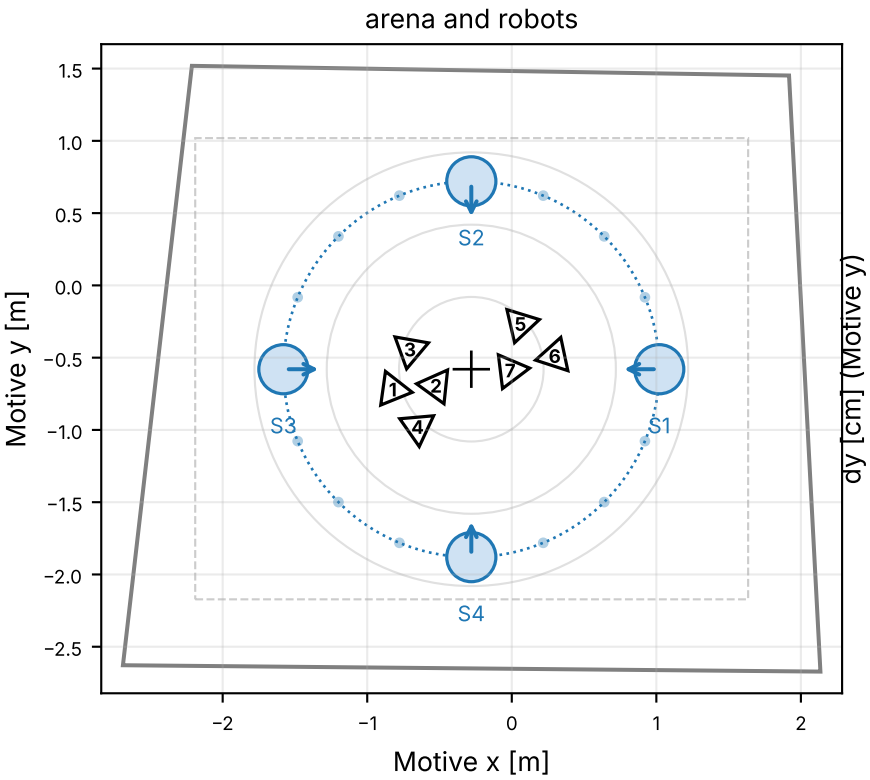
Place and capture: python -m tools.place_check --layout R50

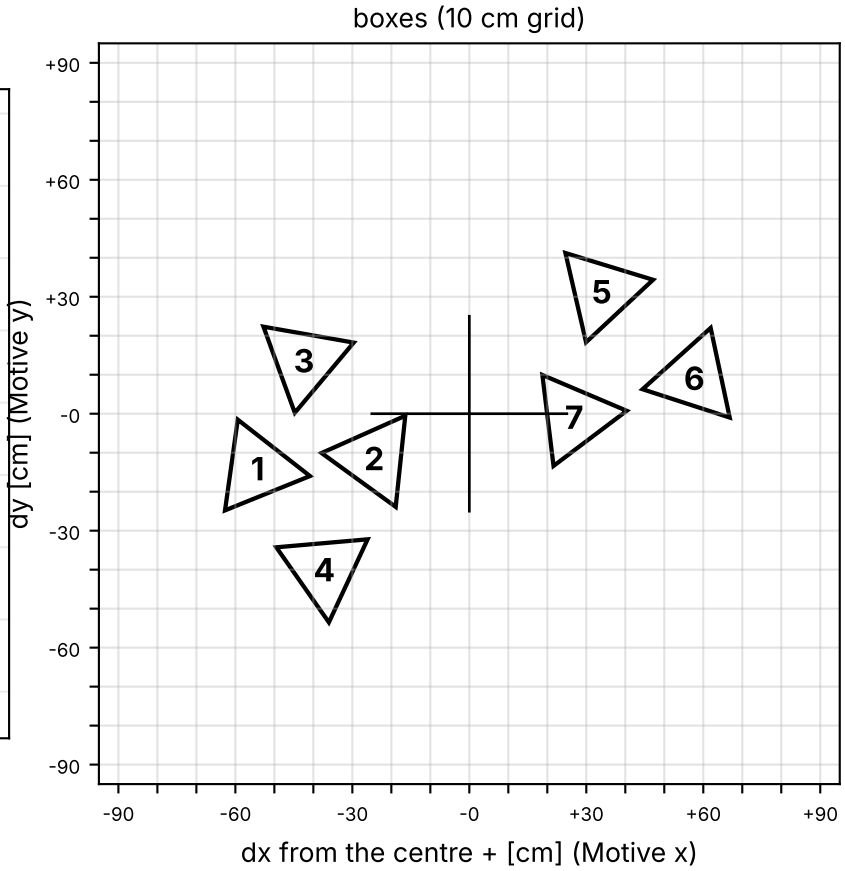
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R50 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R51 day 3 A base (test)

7 boxes, due gruppi, robots: cerchio 1.3 | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-54	-14
2	-24	-12
3	-42	+14
4	-37	-40
5	+34	+31
6	+58	+9
7	+27	-1

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

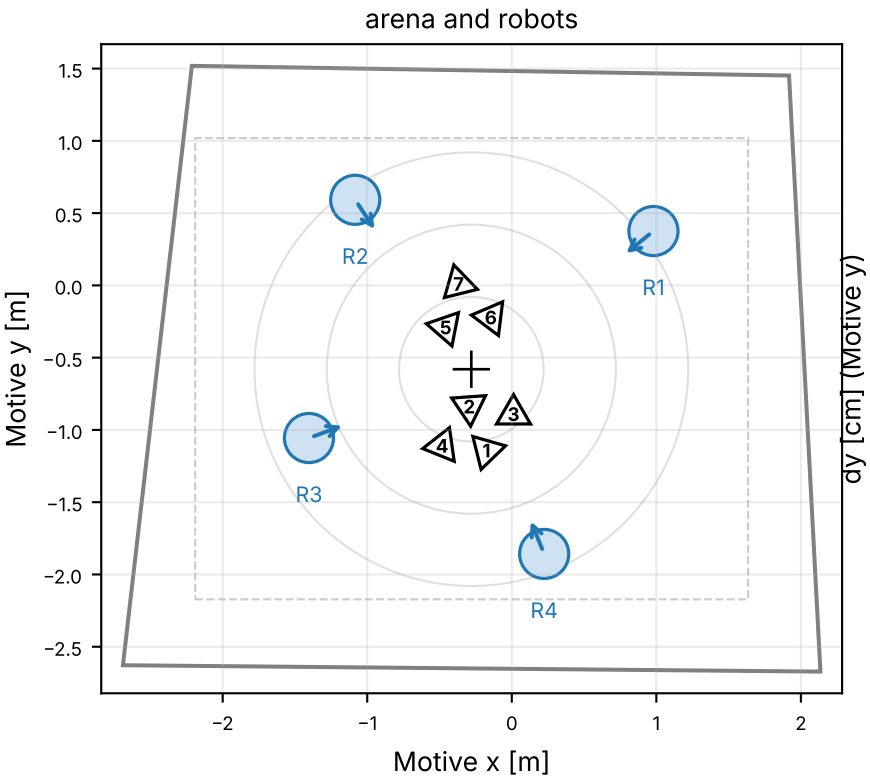
Place and capture: python -m tools.place_check --layout R51

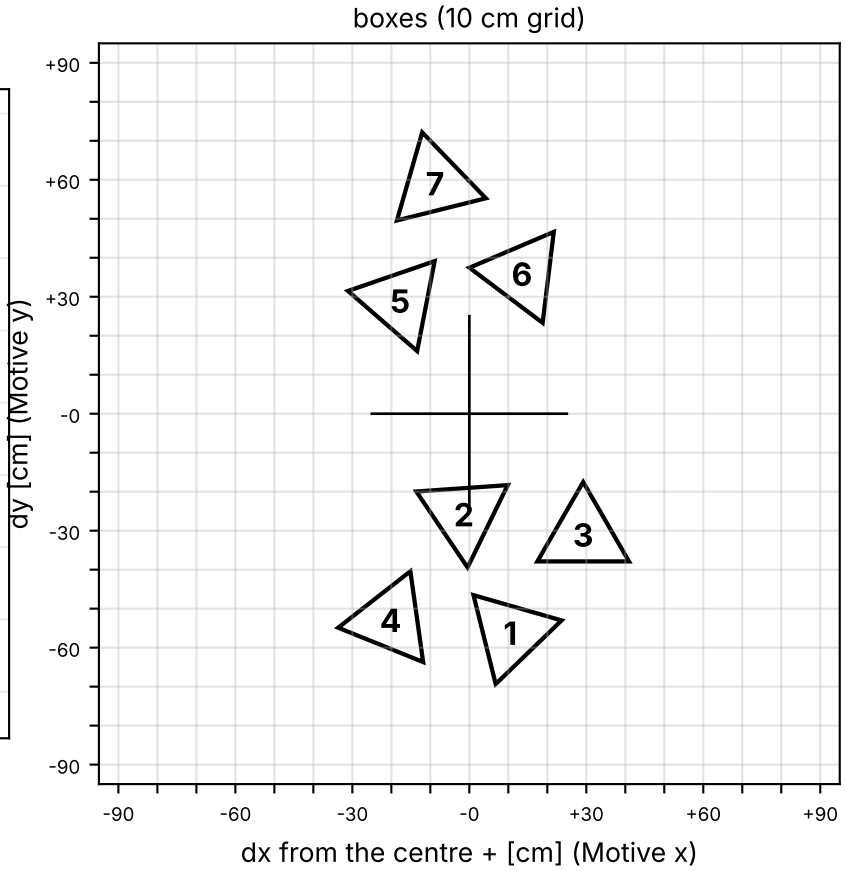
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R51 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R52 day 3 A base (test)

7 boxes, due gruppi, robots: casuale | replica b





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+11	-56
2	-1	-26
3	+29	-31
4	-20	-53
5	-18	+29
6	+14	+36
7	-9	+59

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.98	+0.38
R2	-1.08	+0.59
R3	-1.40	-1.06
R4	+0.22	-1.86

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

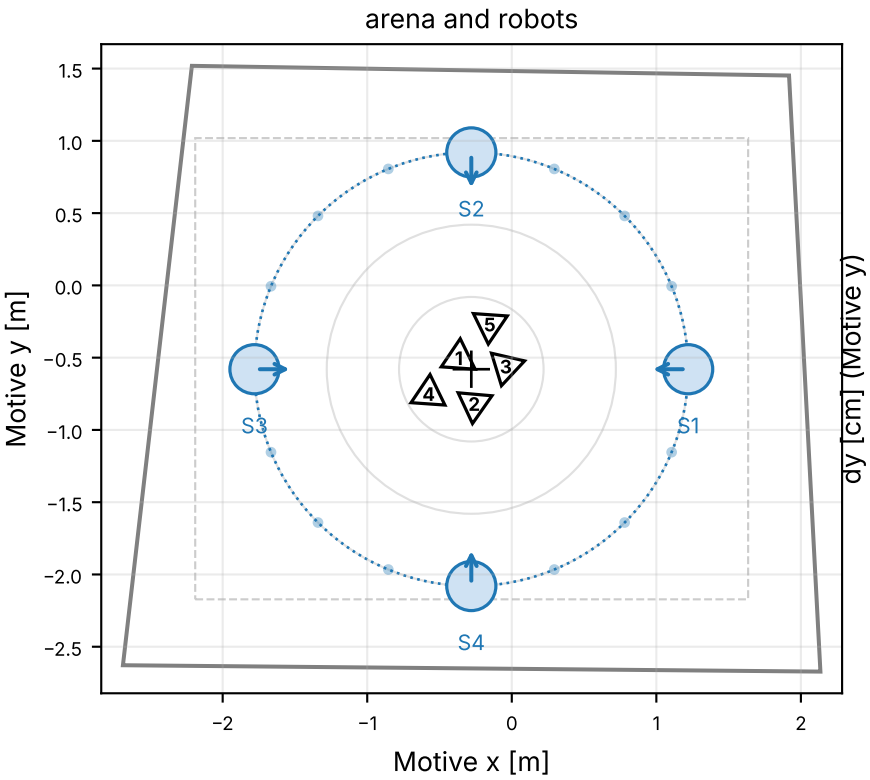
Place and capture: python -m tools.place_check --layout R52

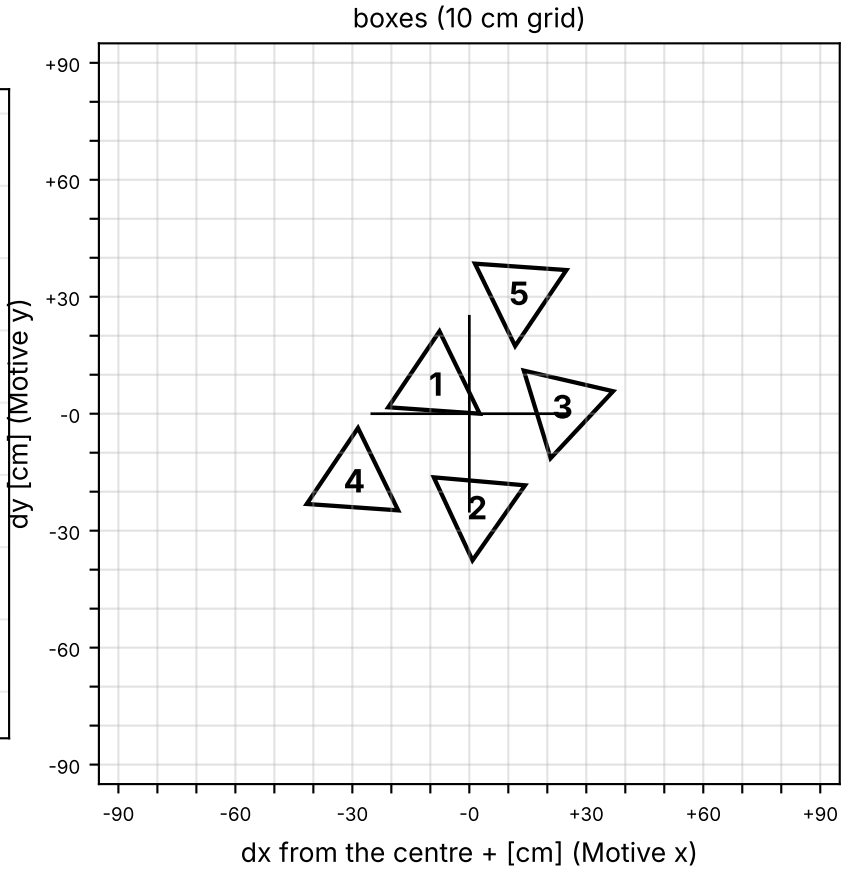
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R52 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R53 day 4 B distanza (test)

5 boxes, fitta, robots: cerchio 1.5





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-9	+8
2	+2	-24
3	+24	+2
4	-29	-17
5	+13	+31

Robots

anywhere near the 1.5 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

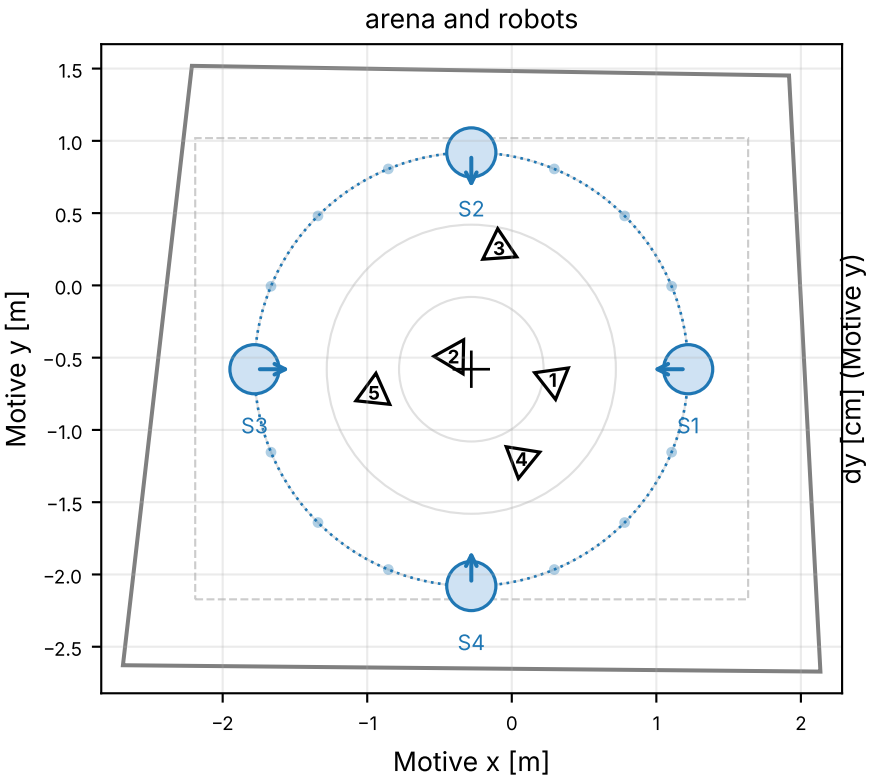
Place and capture: python -m tools.place_check --layout R53

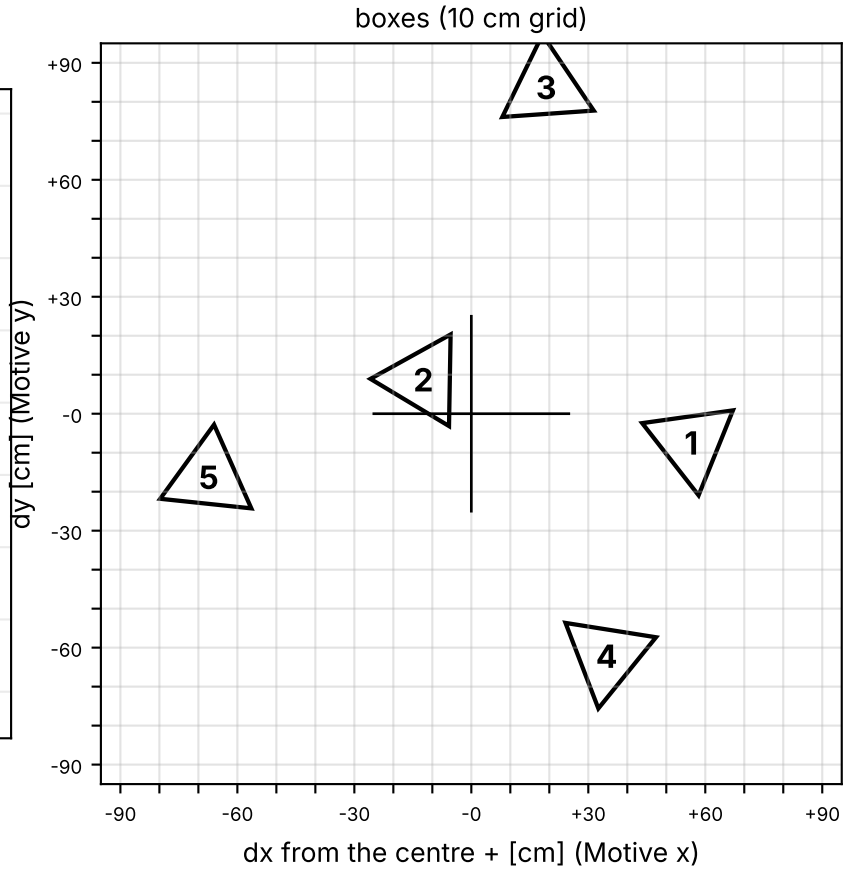
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R53 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.5)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R54 day 4 B distanza (test)

5 boxes, sparsa, robots: cerchio 1.5





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+56	-8
2	-12	+9
3	+19	+84
4	+35	-62
5	-67	-16

Robots

anywhere near the 1.5 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

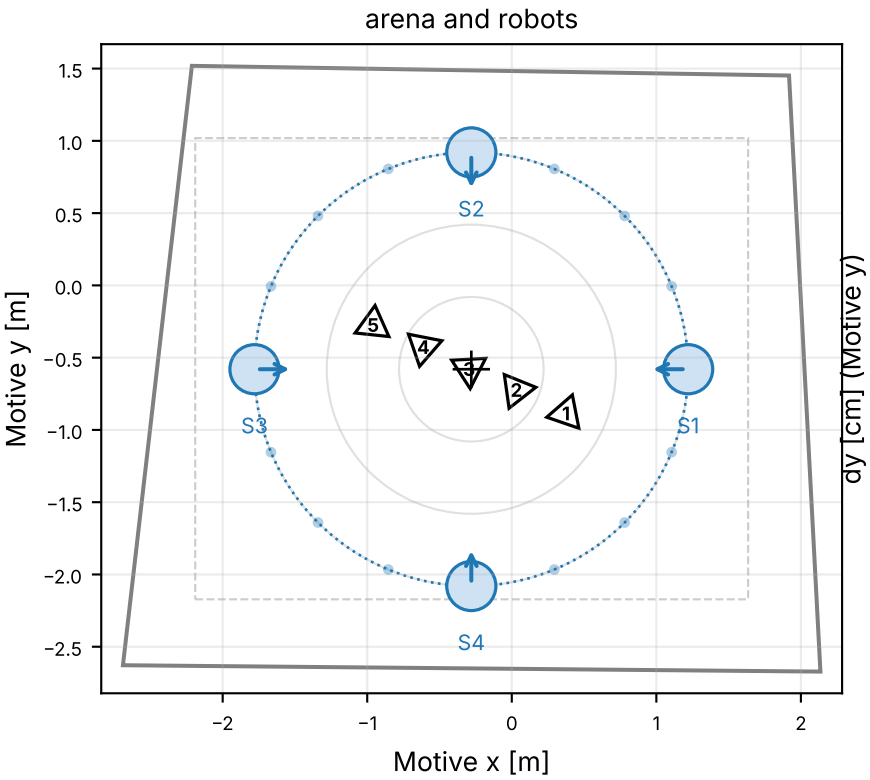
Place and capture: python -m tools.place_check --layout R54

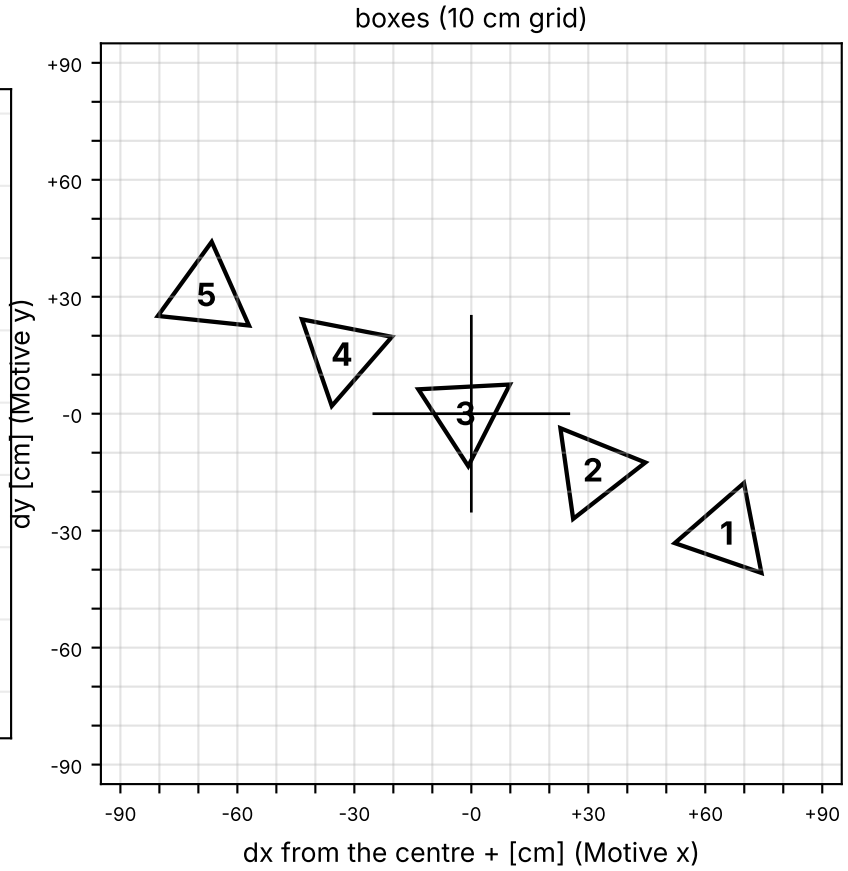
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R54 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.5)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R55 day 4 B distanza (test)

5 boxes, fila/arco, robots: cerchio 1.5





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+66	-31
2	+31	-14
3	-1	+0
4	-33	+15
5	-68	+31

Robots

anywhere near the 1.5 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

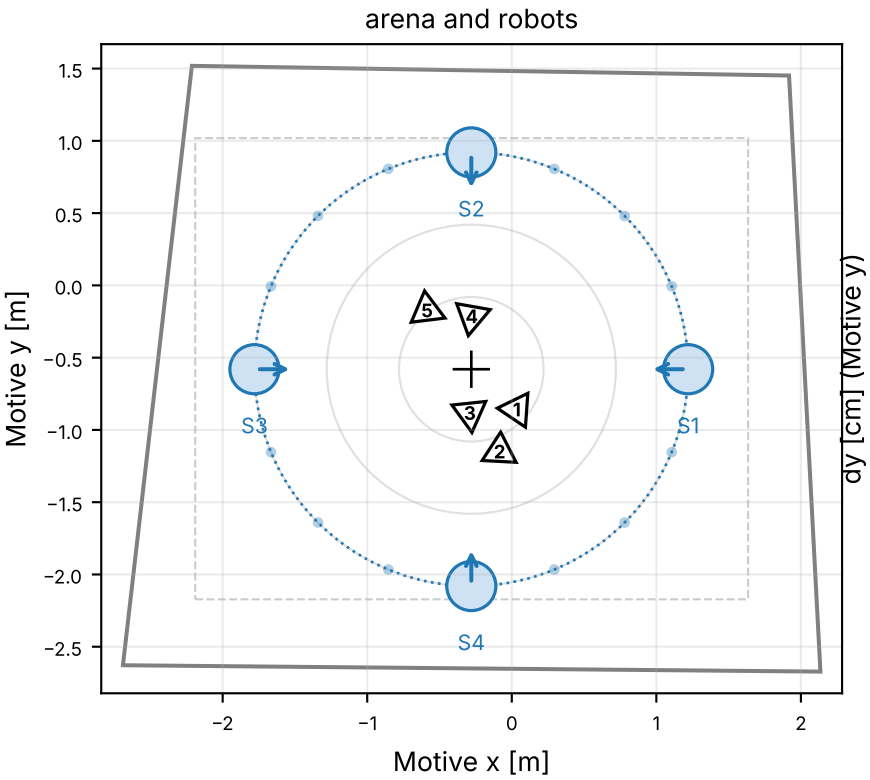
Place and capture: python -m tools.place_check --layout R55

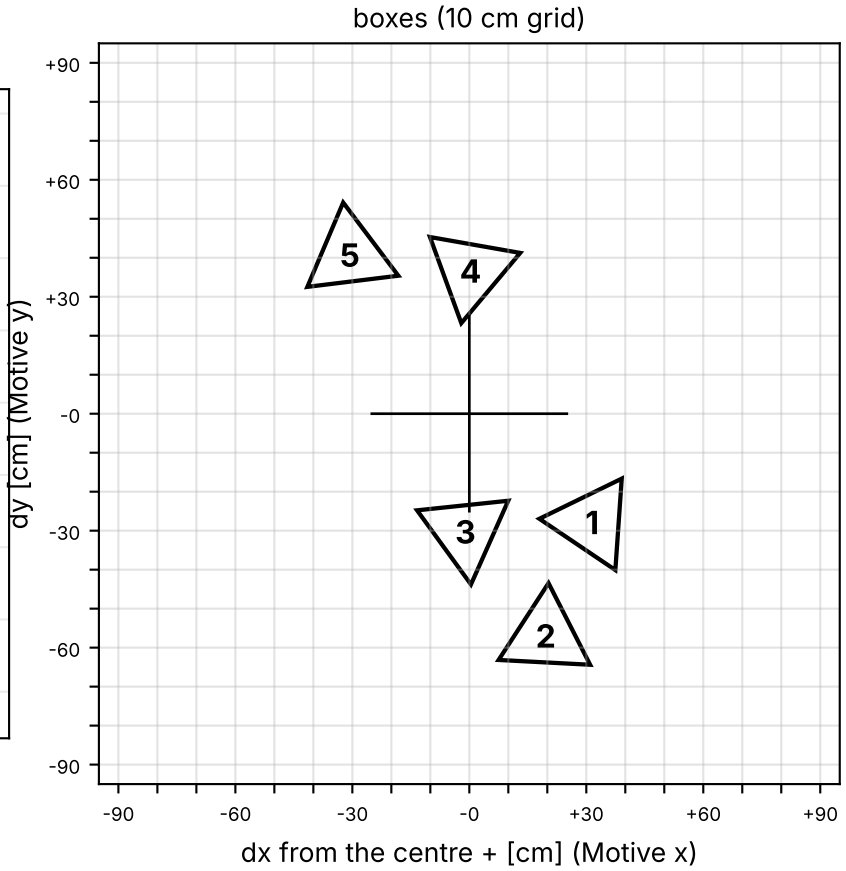
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R55 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.5)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R56 day 4 B distanza (test)

5 boxes, due gruppi, robots: cerchio 1.5





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+32	-28
2	+20	-57
3	-1	-30
4	+0	+37
5	-31	+41

Robots

anywhere near the 1.5 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

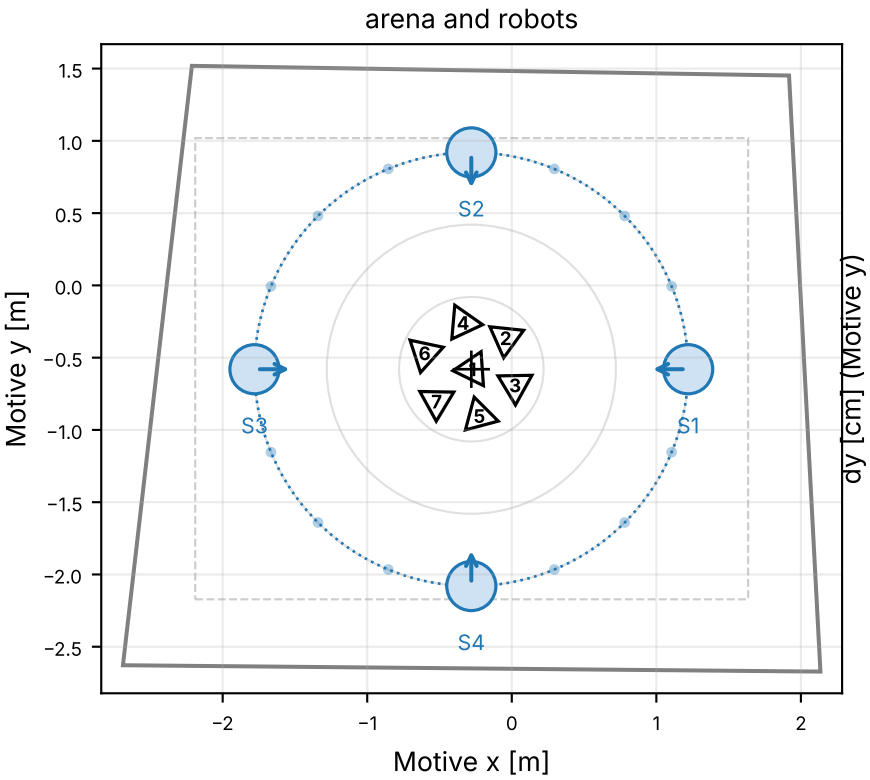
Place and capture: python -m tools.place_check --layout R56

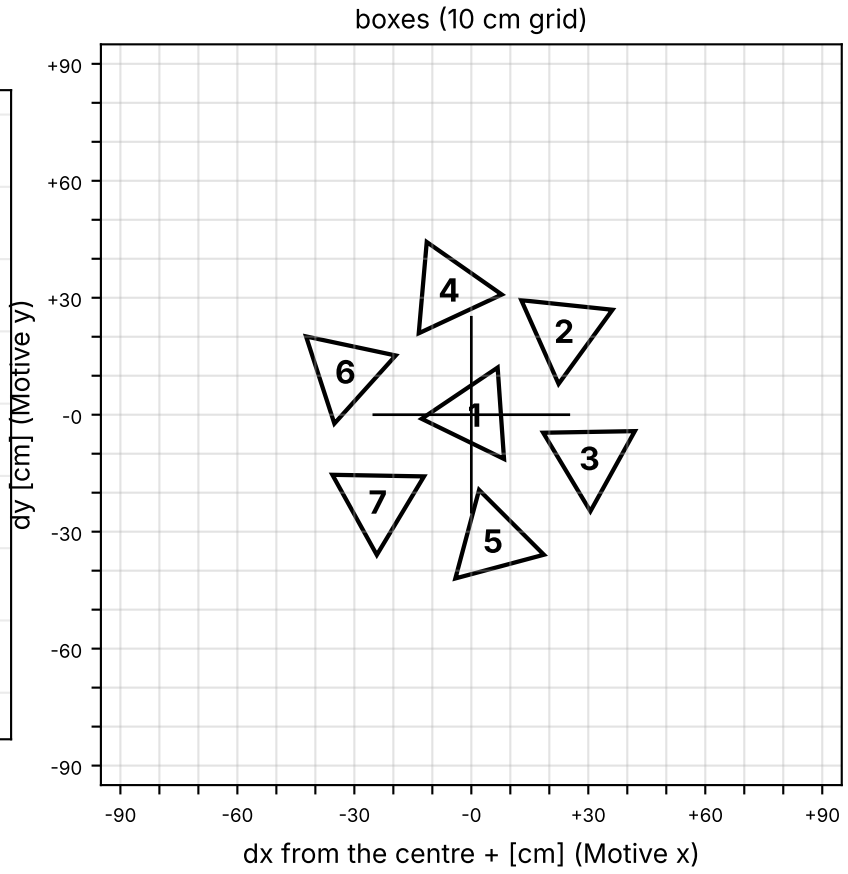
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R56 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.5)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R57 day 4 B distanza (test)

7 boxes, fitta, robots: cerchio 1.5





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+1	-0
2	+24	+21
3	+30	-11
4	-6	+32
5	+6	-32
6	-32	+11
7	-24	-22

Robots

anywhere near the 1.5 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

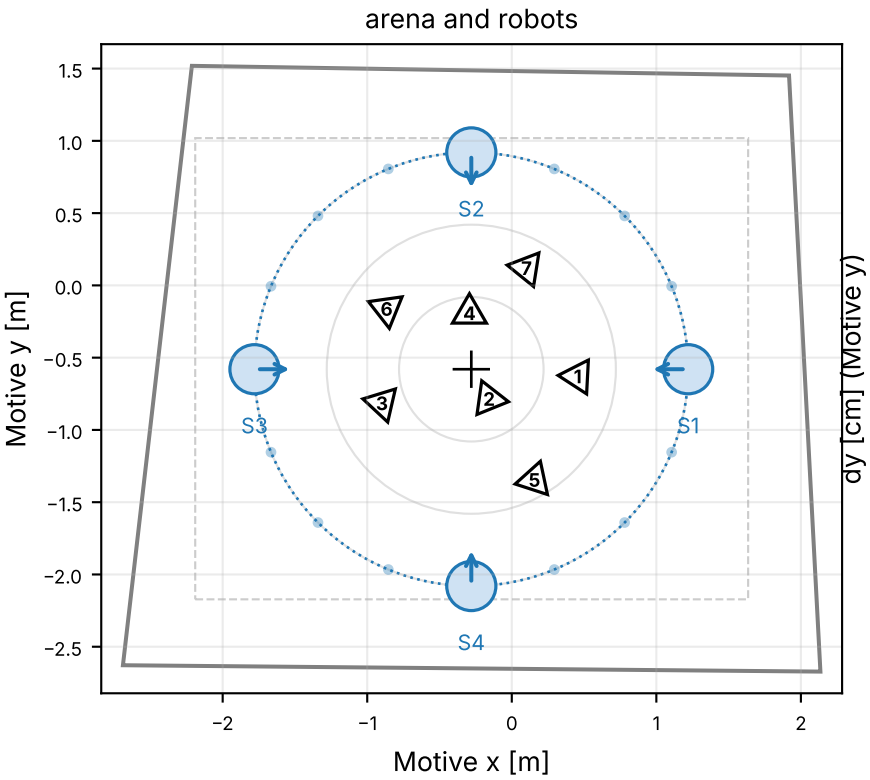
Place and capture: python -m tools.place_check --layout R57

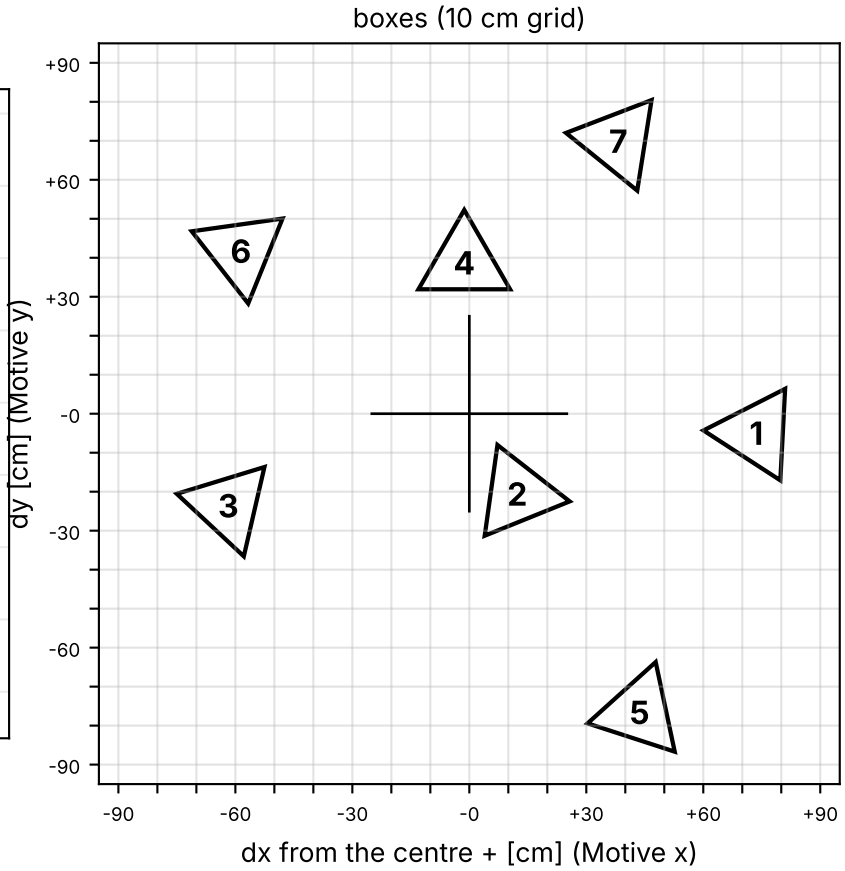
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R57 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.5)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R58 day 4 B distanza (test)

7 boxes, sparsa, robots: cerchio 1.5





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+74	-5
2	+12	-21
3	-62	-24
4	-1	+39
5	+44	-77
6	-59	+42
7	+38	+70

Robots

anywhere near the 1.5 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

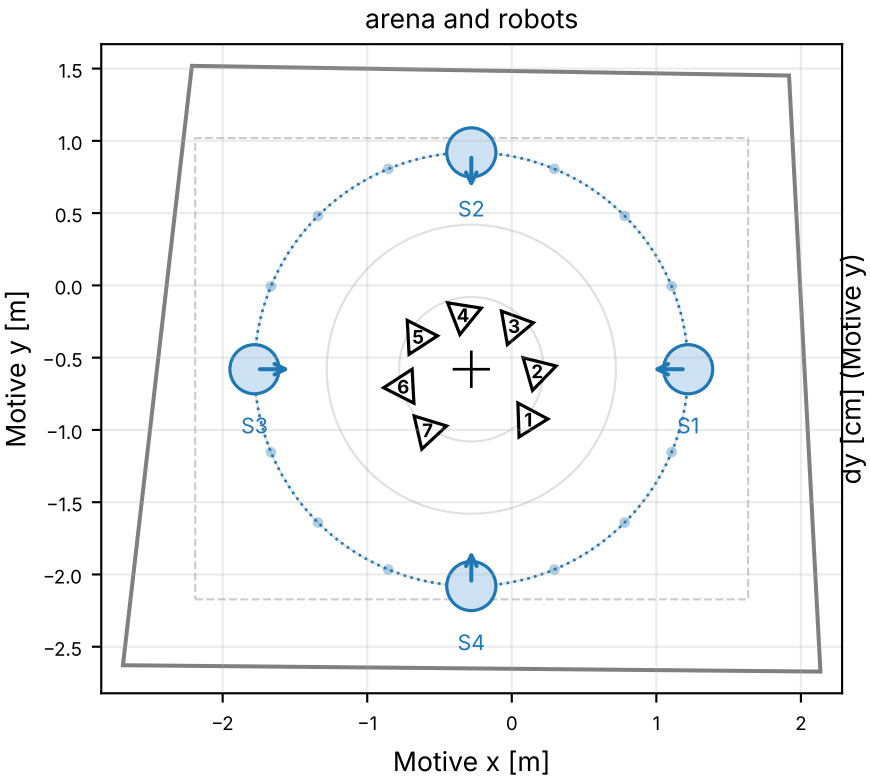
Place and capture: python -m tools.place_check --layout R58

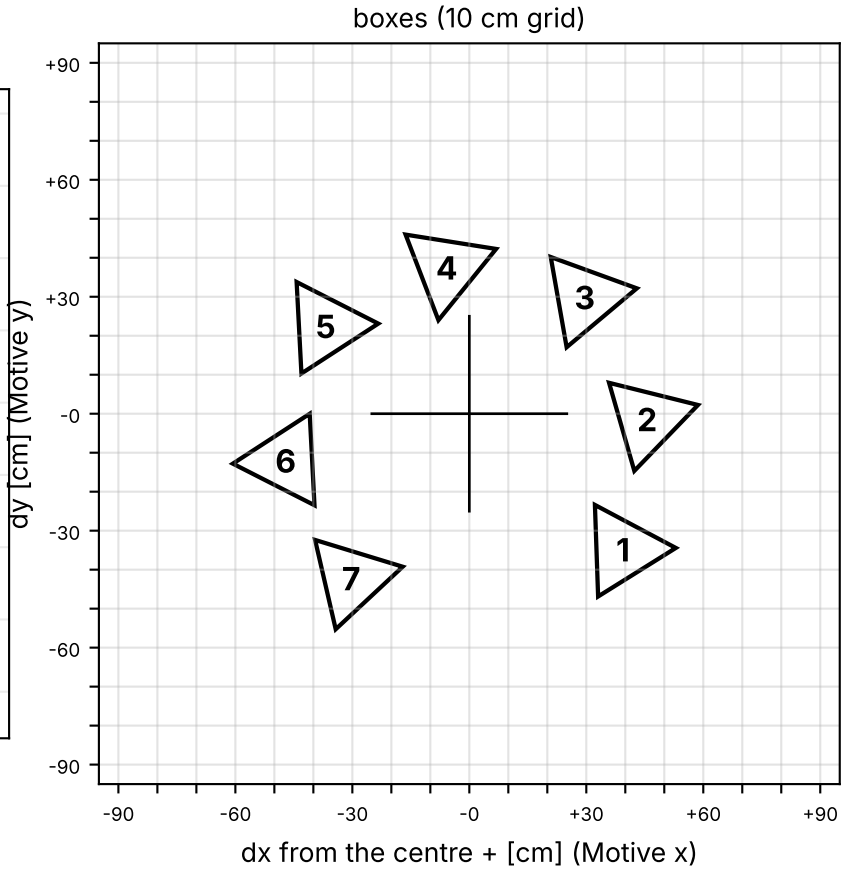
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R58 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.5)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R59 day 4 B distanza (test)

7 boxes, fila/arco, robots: cerchio 1.5





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+39	-35
2	+46	-2
3	+30	+30
4	-6	+37
5	-37	+22
6	-47	-12
7	-30	-42

Robots

anywhere near the 1.5 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

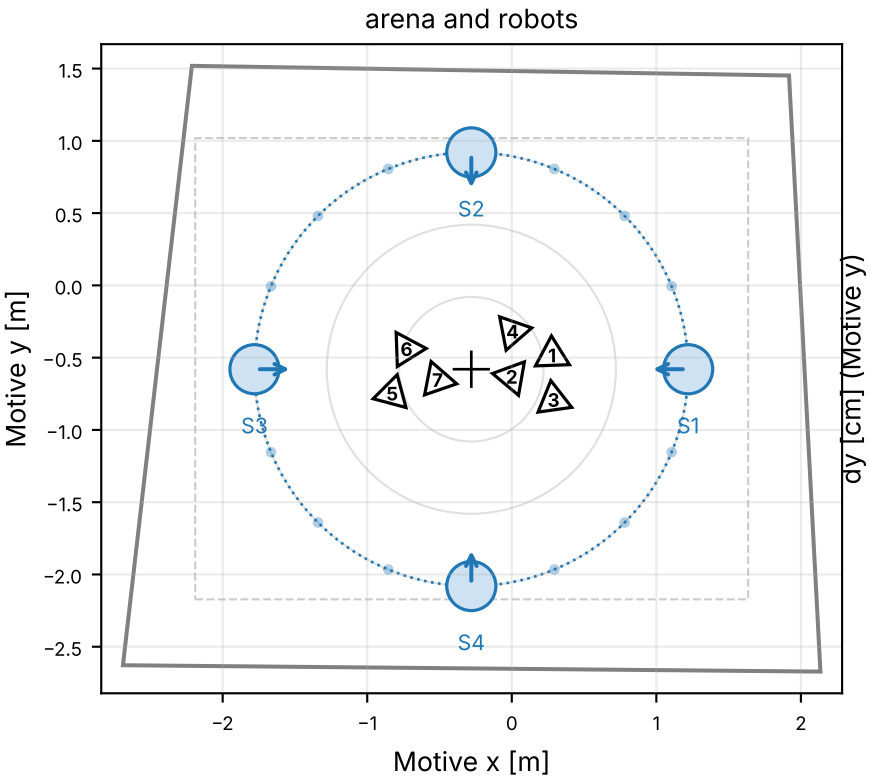
Place and capture: python -m tools.place_check --layout R59

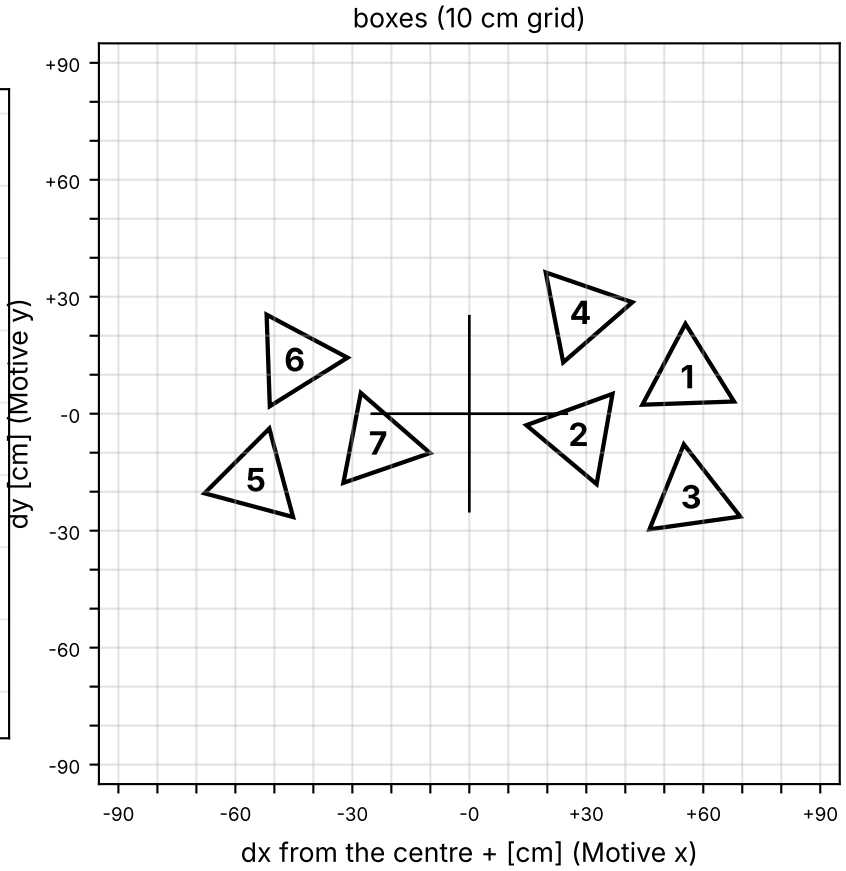
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R59 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.5)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R60 day 4 B distanza (test)

7 boxes, due gruppi, robots: cerchio 1.5





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+56	+9
2	+28	-5
3	+57	-21
4	+29	+26
5	-55	-17
6	-45	+14
7	-23	-8

Robots

anywhere near the 1.5 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

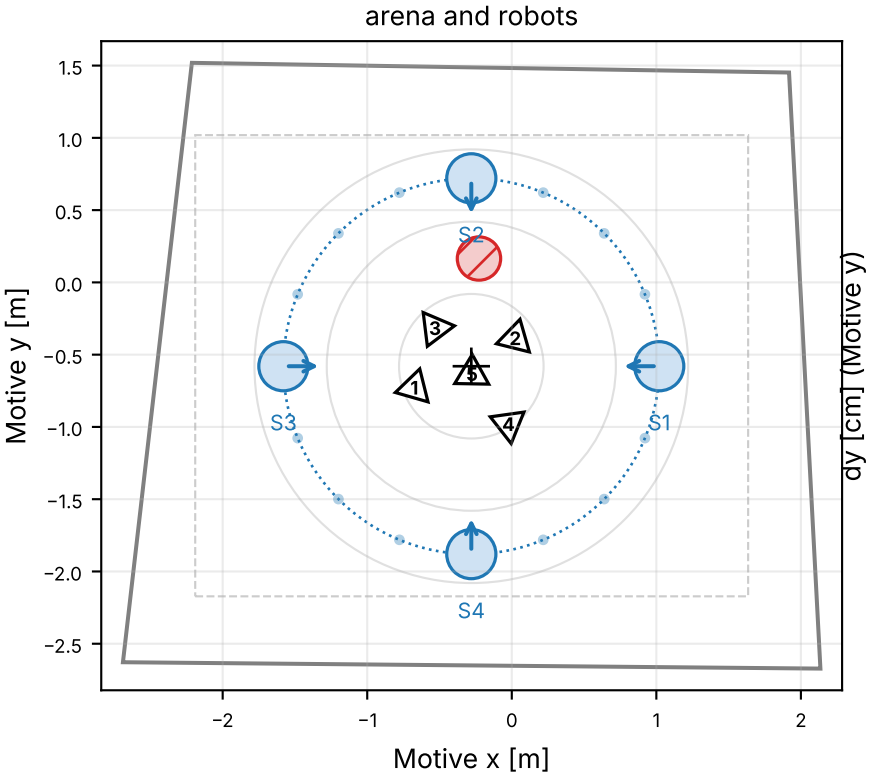
Place and capture: python -m tools.place_check --layout R60

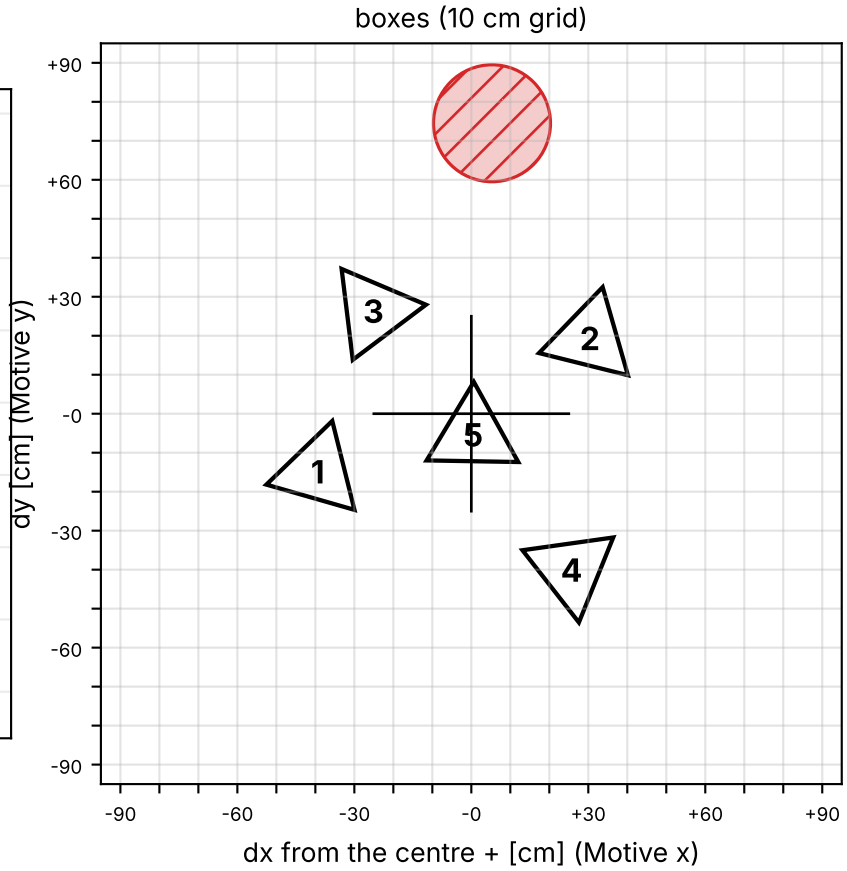
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R60 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.5)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R61 day 4 C occlusioni (test)

5 boxes, sparsa, robots: cerchio 1.3 | 1 ostacoli (cestino, sedia, zaino) tra scatole e robot





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-39	-15
2	+30	+19
3	-25	+26
4	+26	-40
5	+0	-5

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Other objects

obstacle	+5	+74

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

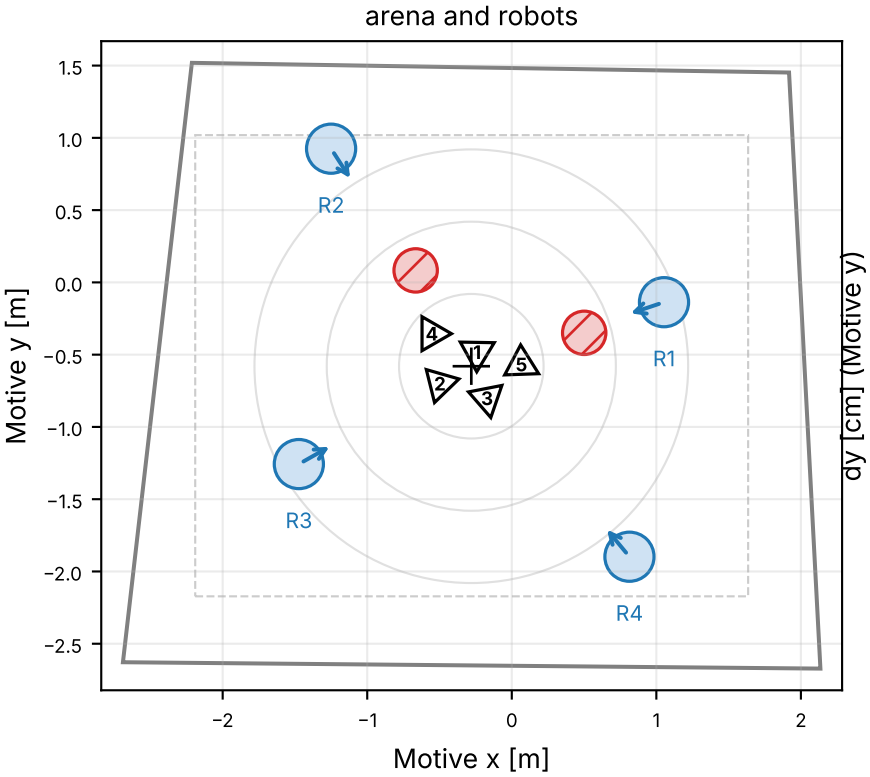
Place and capture: python -m tools.place_check --layout R61

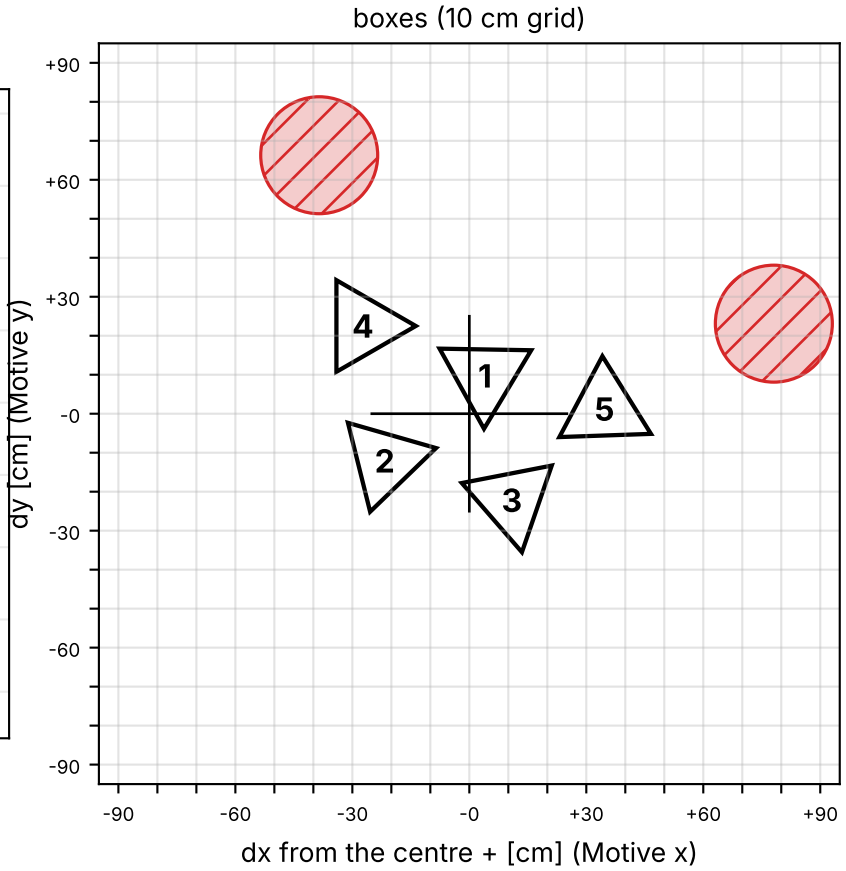
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R61 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R62 day 4 C occlusioni (dev)

5 boxes, fitta, robots: casuale | 2 ostacoli (cestino, sedia, zaino) tra scatole e robot





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+4	+10
2	-22	-12
3	+11	-22
4	-27	+22
5	+35	+1

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+1.05	-0.14
R2	-1.25	+0.93
R3	-1.47	-1.26
R4	+0.81	-1.90

Other objects

obstacle	+78	+23
obstacle	-38	+66

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

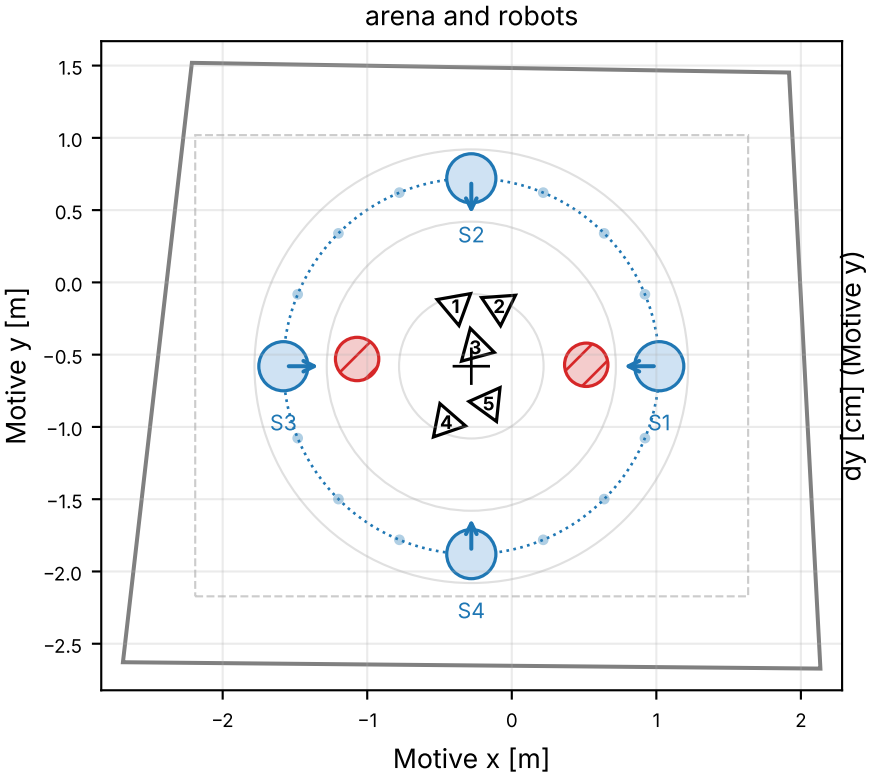
Place and capture: python -m tools.place_check --layout R62

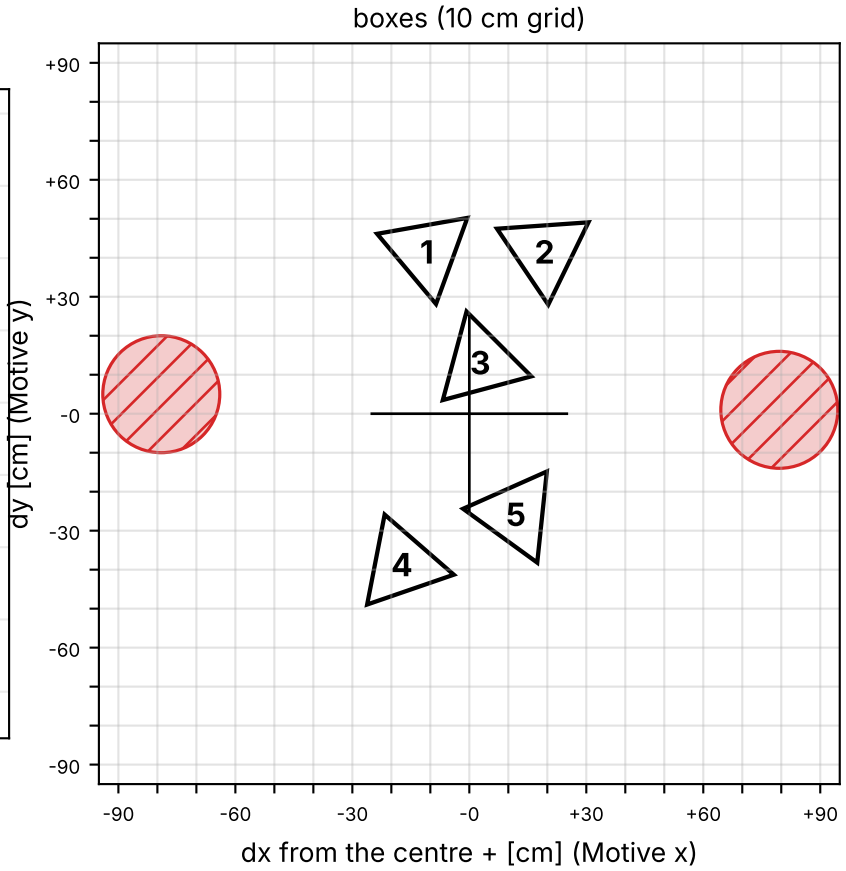
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R62 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R63 day 4 C occlusioni (test)

5 boxes, due gruppi, robots: cerchio 1.3 | 2 ostacoli (cestino, sedia, zaino) tra scatole e robot





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-11	+41
2	+19	+41
3	+3	+13
4	-17	-39
5	+12	-26

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Other objects

obstacle	-79	+5
obstacle	+80	+1

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

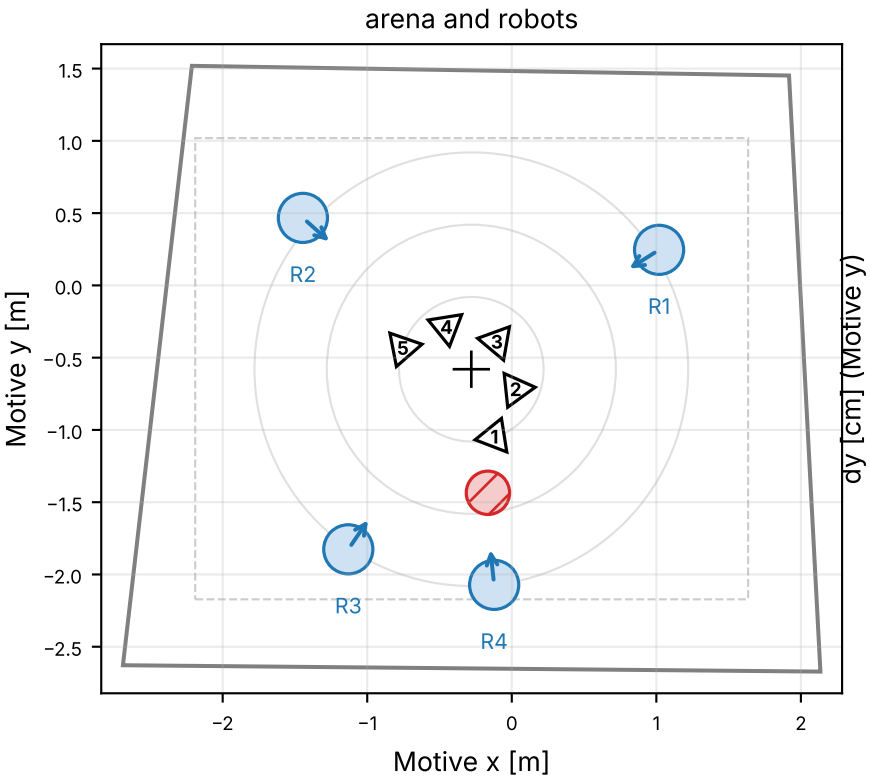
Place and capture: python -m tools.place_check --layout R63

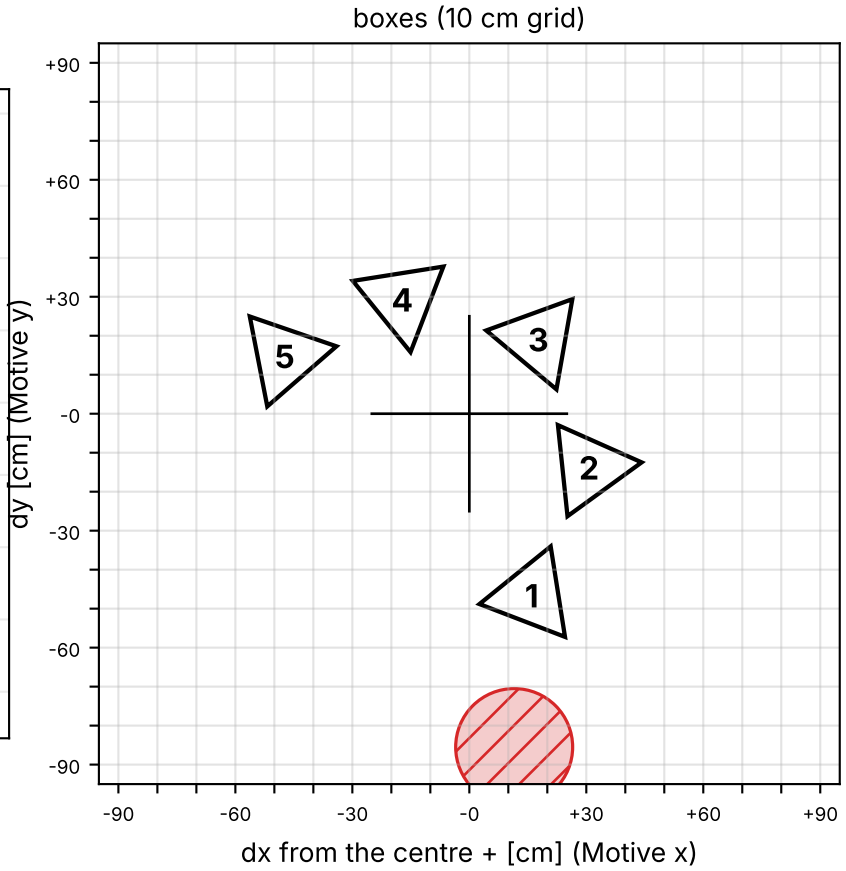
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R63 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R64 day 4 C occlusioni (test)

5 boxes, fila/arco, robots: casuale | 1 ostacoli (cestino, sedia, zaino) tra scatole e robot





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+16	-47
2	+31	-14
3	+18	+19
4	-17	+29
5	-47	+15

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+1.02	+0.25
R2	-1.45	+0.47
R3	-1.13	-1.83
R4	-0.12	-2.07

Other objects

obstacle	+12	-86

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

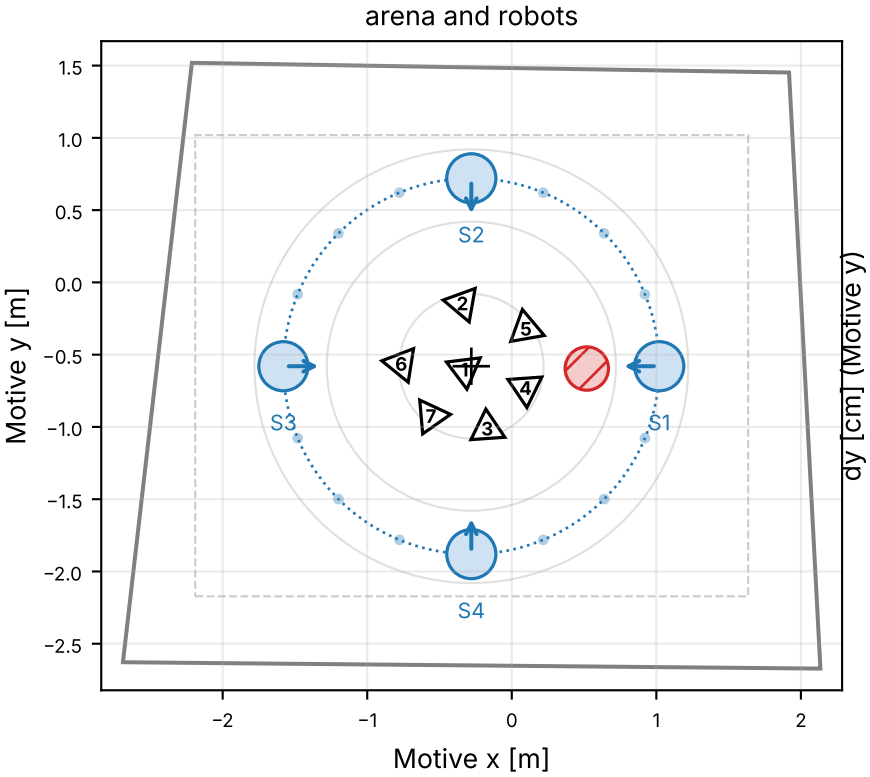
Place and capture: python -m tools.place_check --layout R64

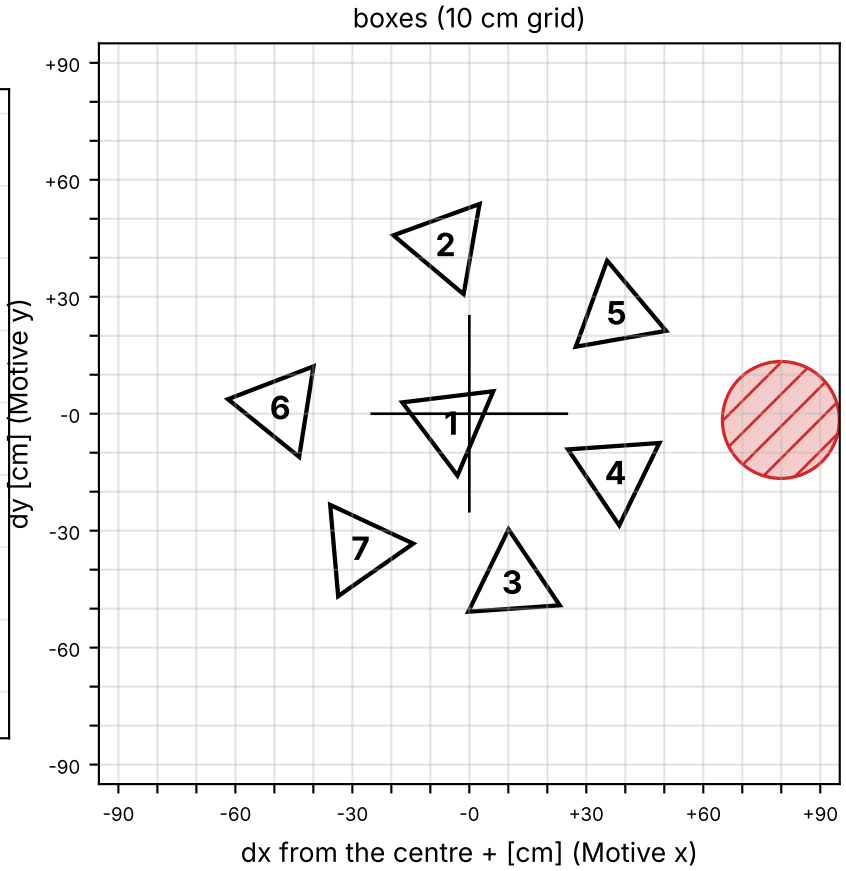
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R64 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth
config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080
--camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R65 day 4 C occlusioni (test)

7 boxes, sparsa, robots: cerchio 1.3 | 1 ostacoli (cestino, sedia, zaino) tra scatole e robot





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-5	-2
2	-6	+43
3	+11	-43
4	+38	-15
5	+38	+26
6	-48	+2
7	-28	-35

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Other objects

obstacle	+80	-2

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

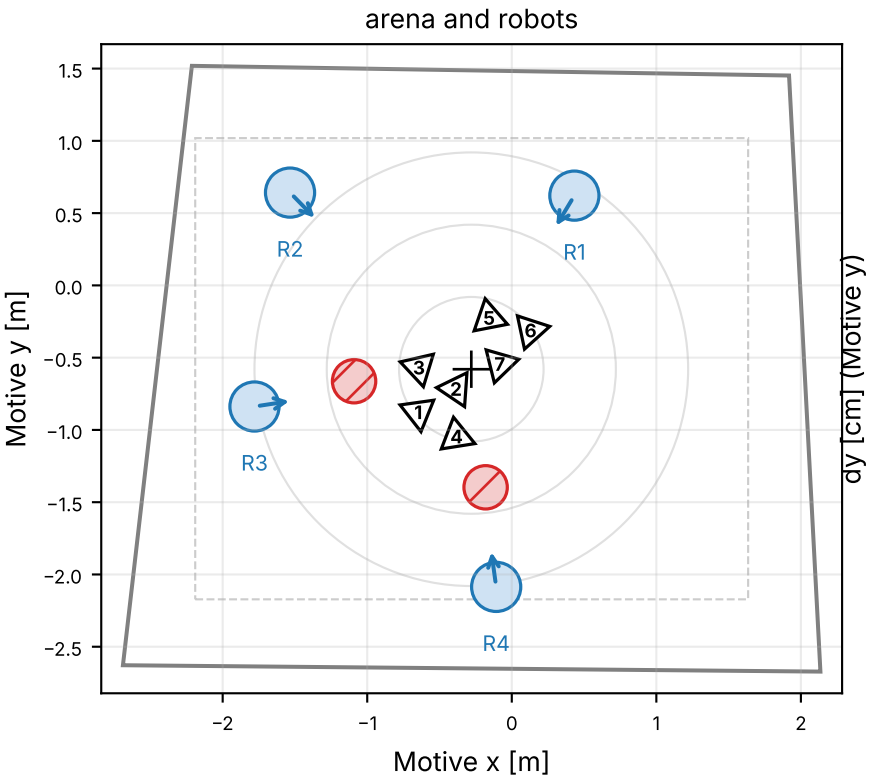
Place and capture: python -m tools.place_check --layout R65

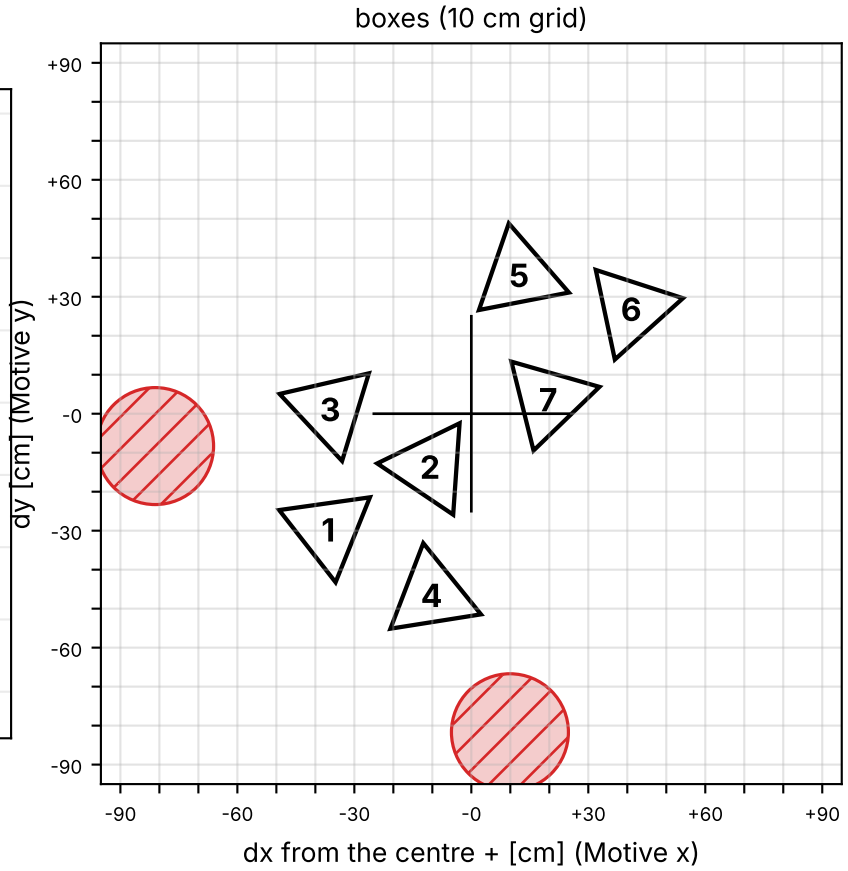
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R65 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R66 day 4 C occlusioni (test)

7 boxes, due gruppi, robots: casuale | 2 ostacoli (cestino, sedia, zaino) tra scatole e robot





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-37	-30
2	-11	-14
3	-36	+1
4	-10	-47
5	+12	+36
6	+41	+27
7	+20	+4

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.43	+0.62
R2	-1.53	+0.64
R3	-1.78	-0.84
R4	-0.11	-2.09

Other objects

obstacle	-81	-8
obstacle	+10	-82

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

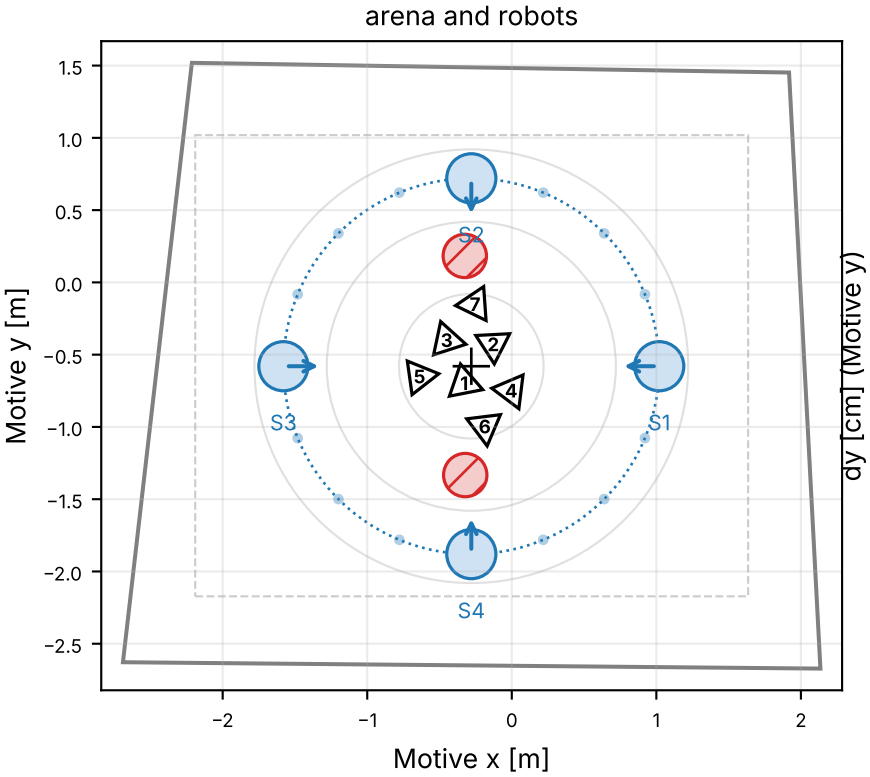
Place and capture: python -m tools.place_check --layout R66

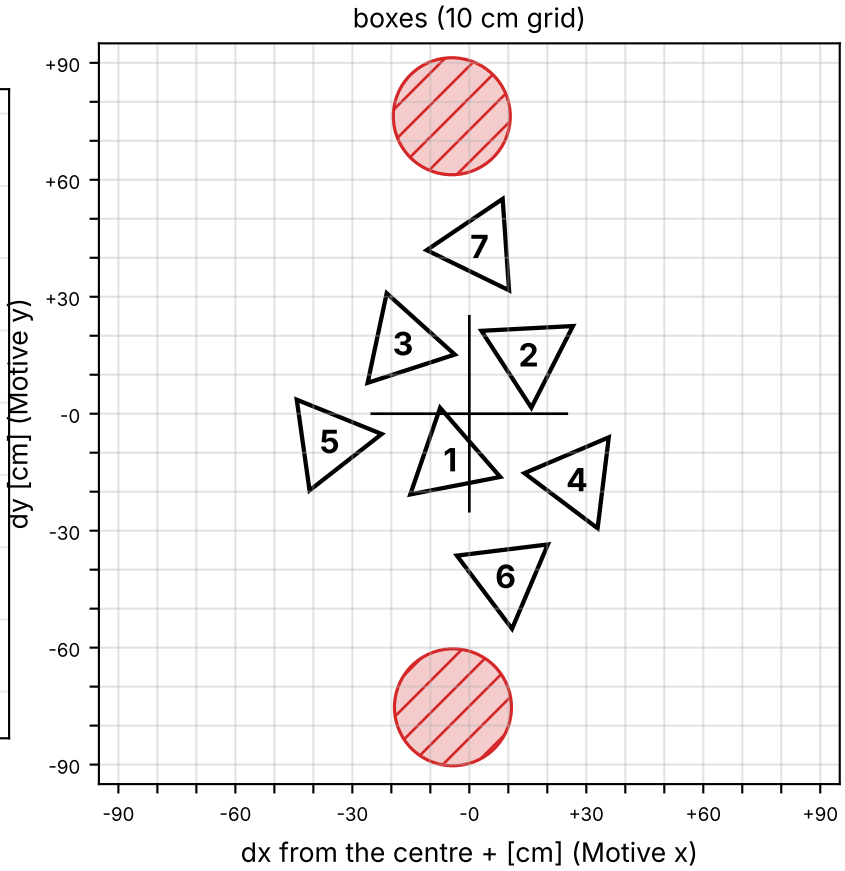
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R66 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R67 day 4 C occlusioni (test)

7 boxes, fitta, robots: cerchio 1.3 | 2 ostacoli (cestino, sedia, zaino) tra scatole e robot





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-5	-12
2	+15	+15
3	-17	+18
4	+28	-17
5	-36	-7
6	+9	-42
7	+3	+43

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Other objects

obstacle	-4	-75
obstacle	-4	+76

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

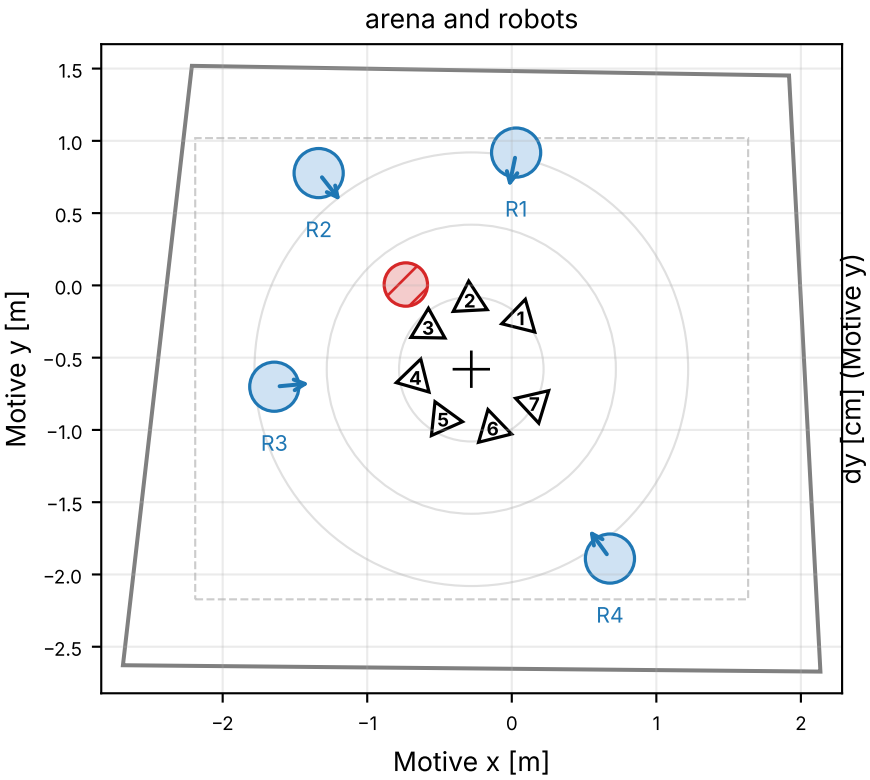
Place and capture: python -m tools.place_check --layout R67

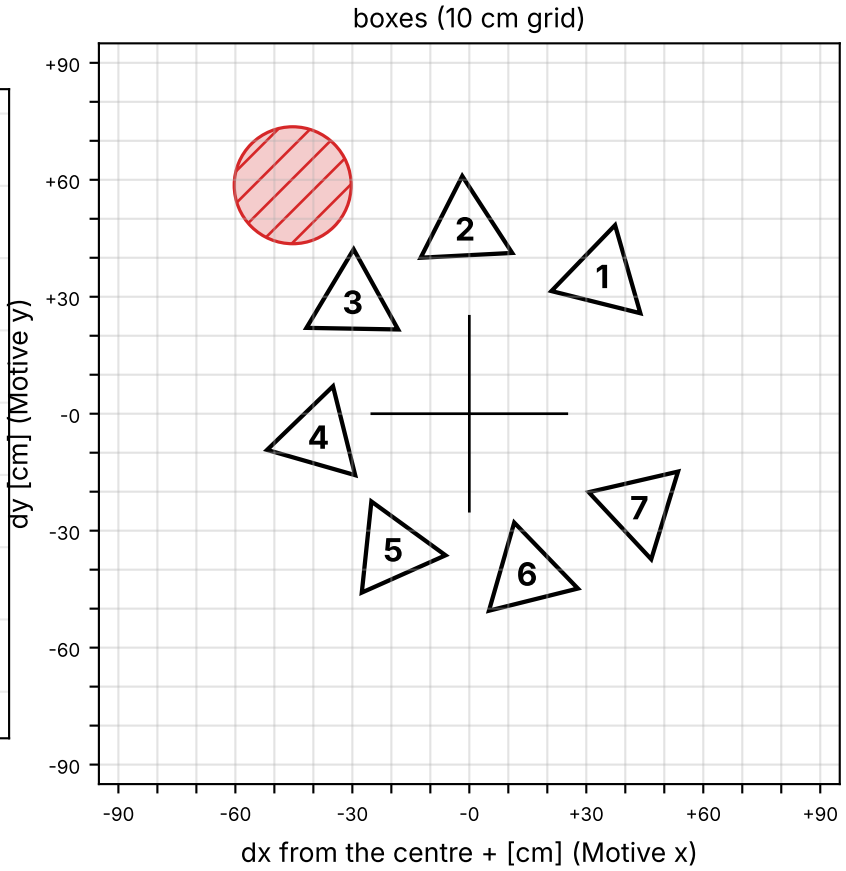
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R67 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R68 day 4 C occlusioni (test)

7 boxes, fila/arco, robots: casuale | 1 ostacoli (cestino, sedia, zaino) tra scatole e robot





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+34	+35
2	-1	+47
3	-30	+29
4	-39	-6
5	-20	-35
6	+15	-41
7	+44	-24

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.03	+0.92
R2	-1.34	+0.78
R3	-1.64	-0.70
R4	+0.68	-1.89

Other objects

obstacle	-45	+59

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

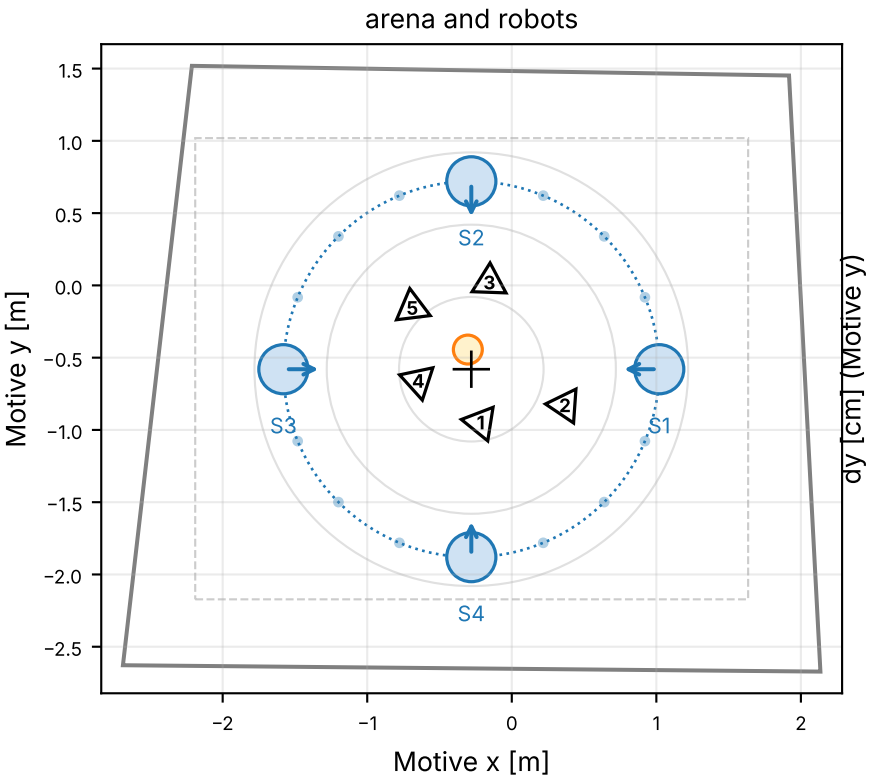
Place and capture: python -m tools.place_check --layout R68

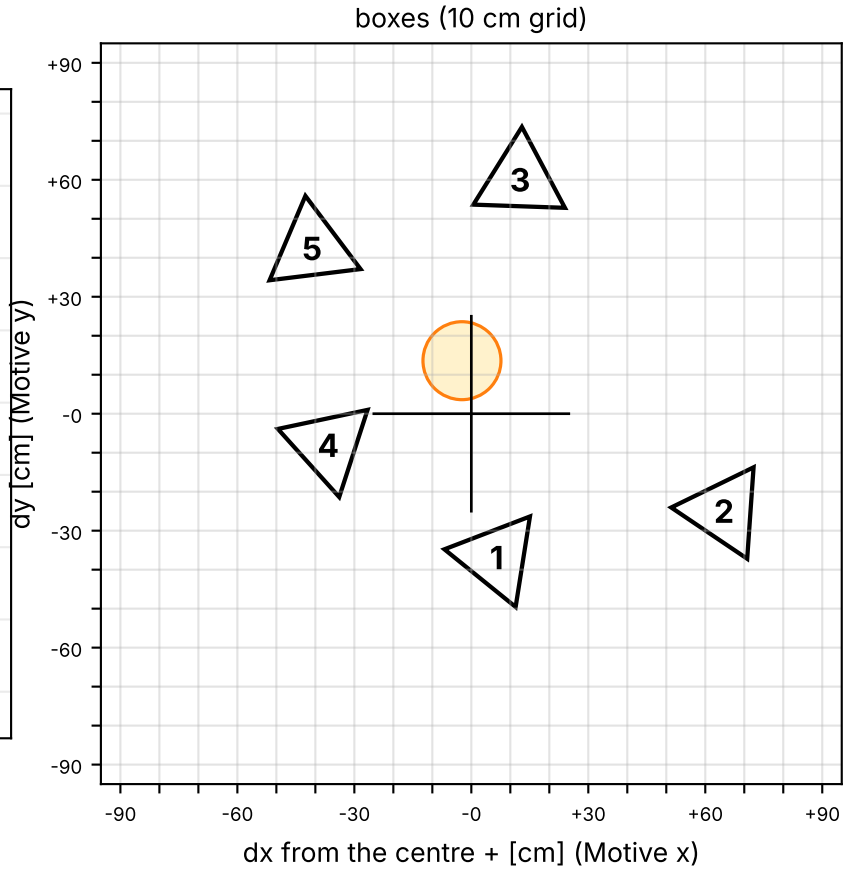
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R68 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box
--ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit
1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28
-0.58)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R69 day 5 D falsi positivi (test)

5 boxes, sparsa, robots: cerchio 1.3 | 1 oggetto bianco che non e' una scatola





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+7	-37
2	+65	-25
3	+13	+60
4	-37	-8
5	-41	+42

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Other objects

white object	-2	+14

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

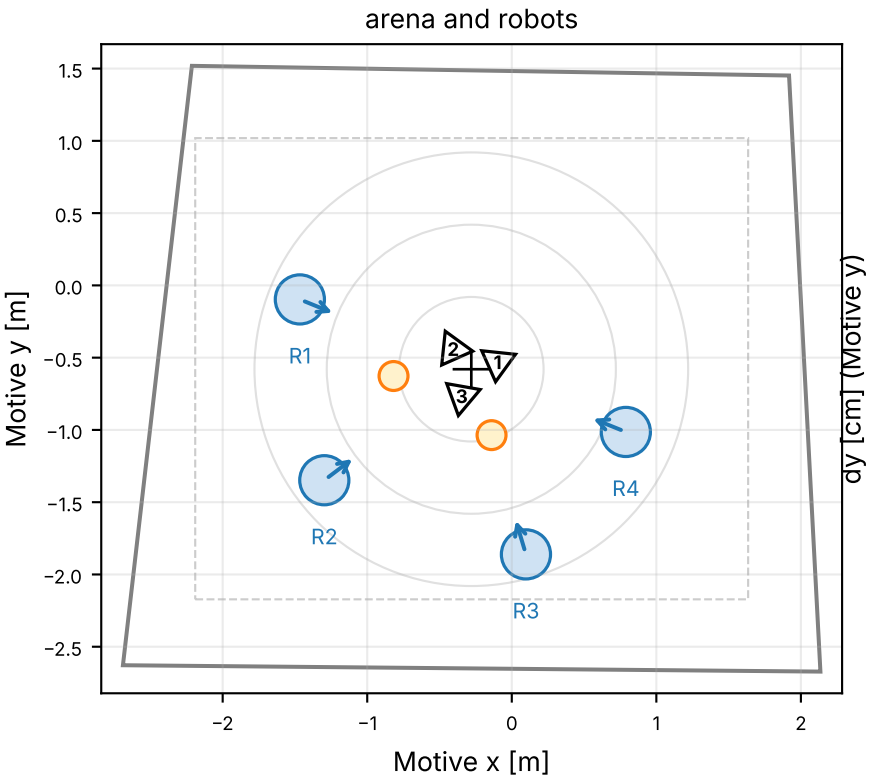
Place and capture: python -m tools.place_check --layout R69

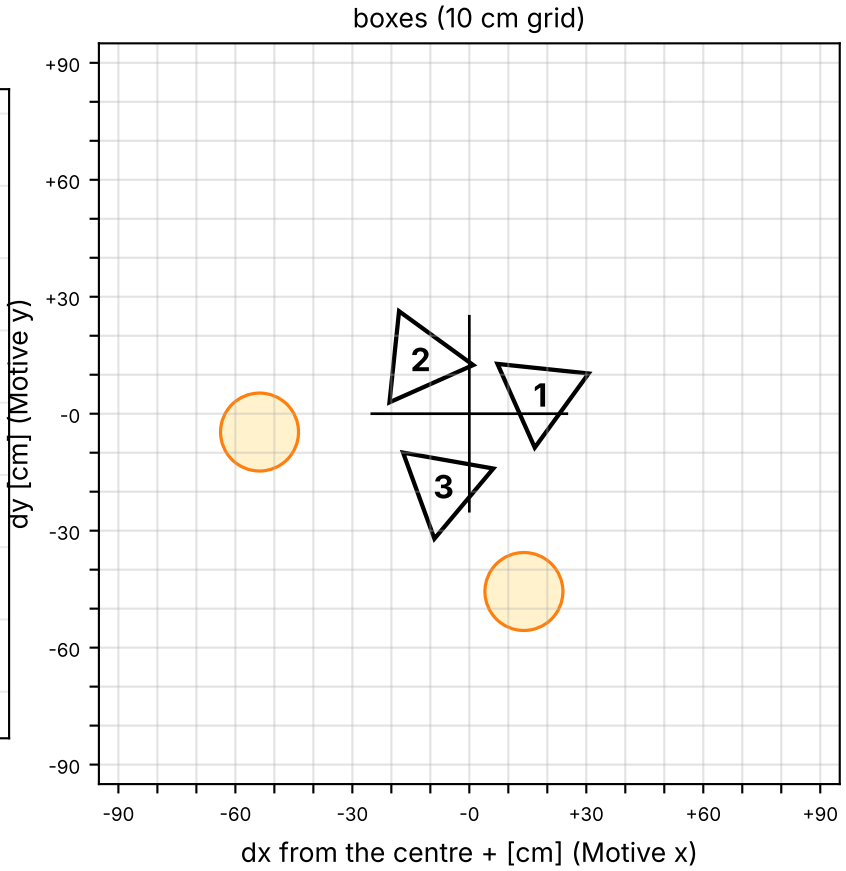
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R69 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R70 day 5 D falsi positivi (test)

3 boxes, fitta, robots: casuale | 2 oggetti bianchi non scatole





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+18	+5
2	-12	+14
3	-7	-19

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	-1.47	-0.10
R2	-1.30	-1.35
R3	+0.10	-1.86
R4	+0.79	-1.01

Other objects

white object	-54	-5
white object	+14	-46

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

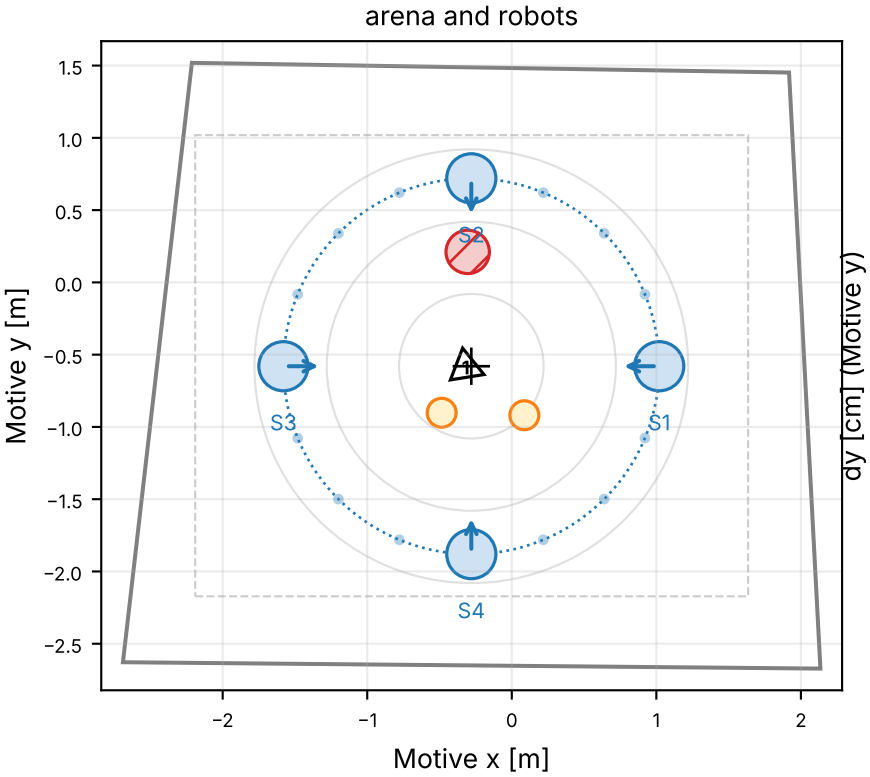
Place and capture: python -m tools.place_check --layout R70

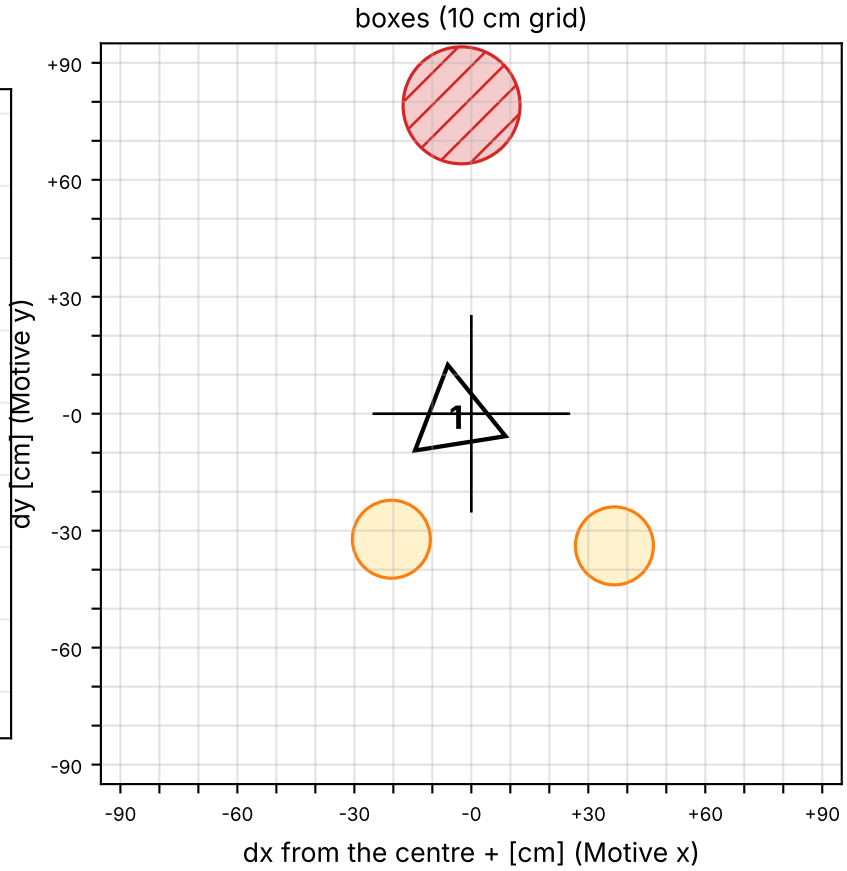
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R70 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R71 day 5 D falsi positivi (test)

1 boxes, al centro, robots: cerchio 1.3 | 1 scatola + ostacoli e oggetti bianchi sparsi





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-4	-1

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Other objects

obstacle	-2	+79
white object	-20	-32
white object	+37	-34

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

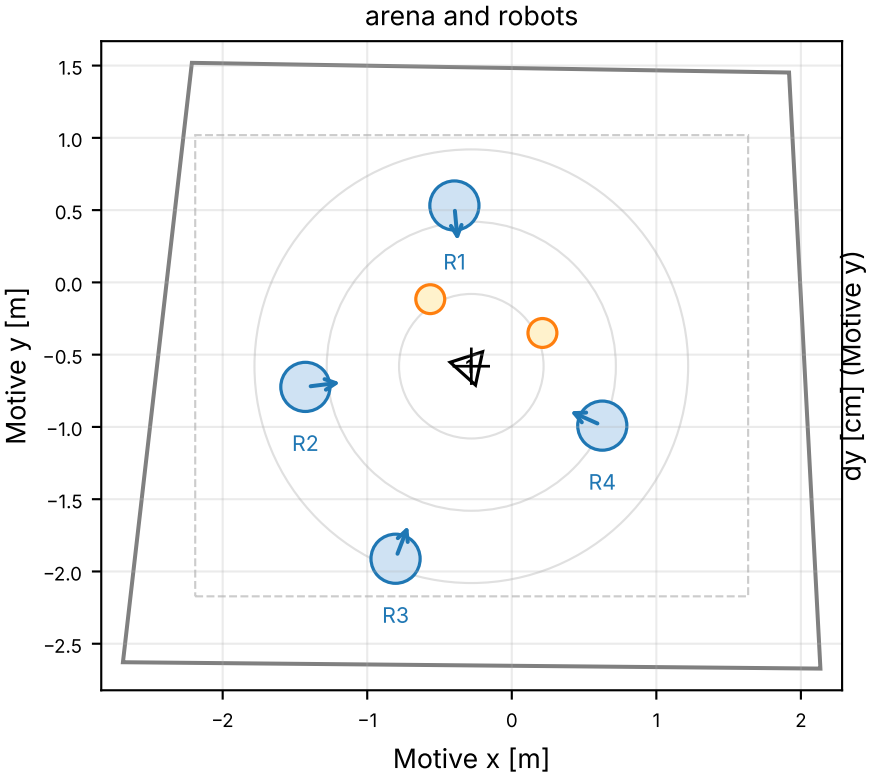
Place and capture: python -m tools.place_check --layout R71

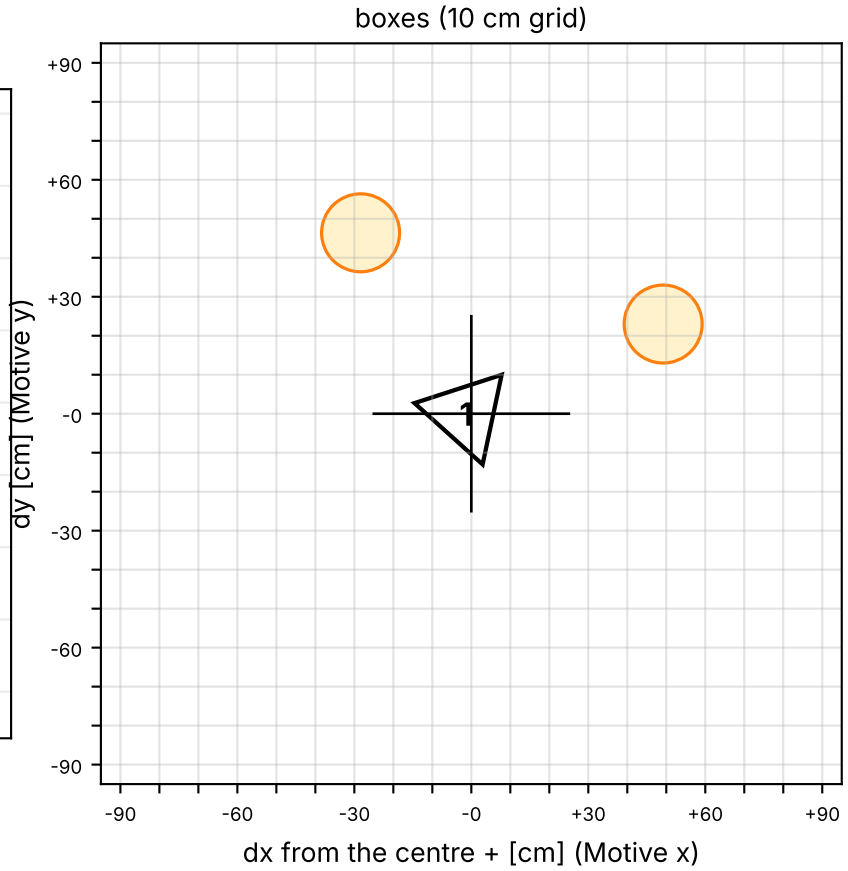
```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R71 --box 11:box --ground-truth config/ground_truth_positions.json --regions
config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-
threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R72 day 5 D falsi positivi (test)

1 boxes, al centro, robots: casuale | 1 scatola + oggetti bianchi sparsi





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-1	-0

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	-0.40	+0.53
R2	-1.43	-0.72
R3	-0.80	-1.91
R4	+0.63	-0.99

Other objects

white object	+49	+23
white object	-28	+46

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

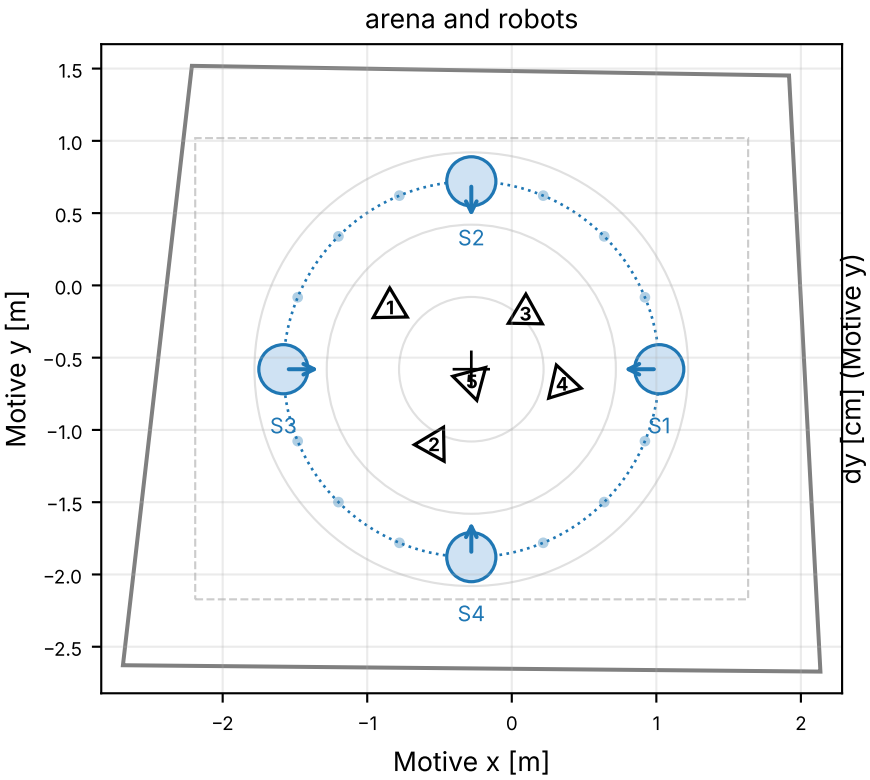
Place and capture: python -m tools.place_check --layout R72

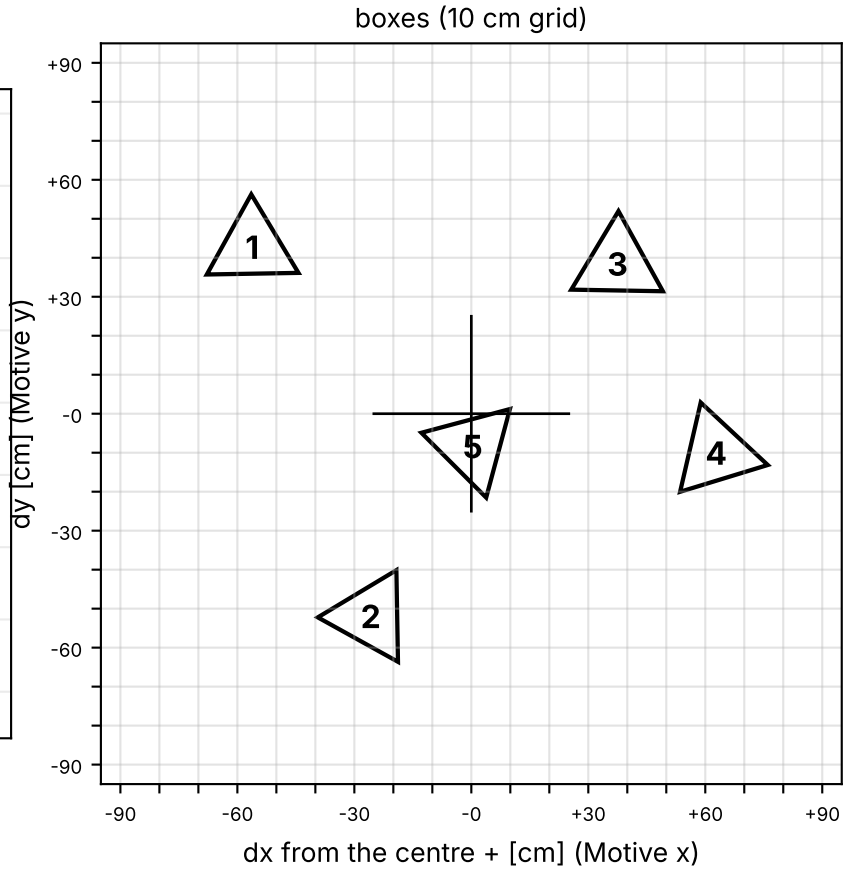
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R72 --box 11:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R73 day 5 E luce (test)

5 boxes, sparsa, robots: cerchio 1.3 | luce diversa (tende aperte o meta' luci spente)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-56	+43
2	-26	-52
3	+38	+38
4	+63	-10
5	+0	-8

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

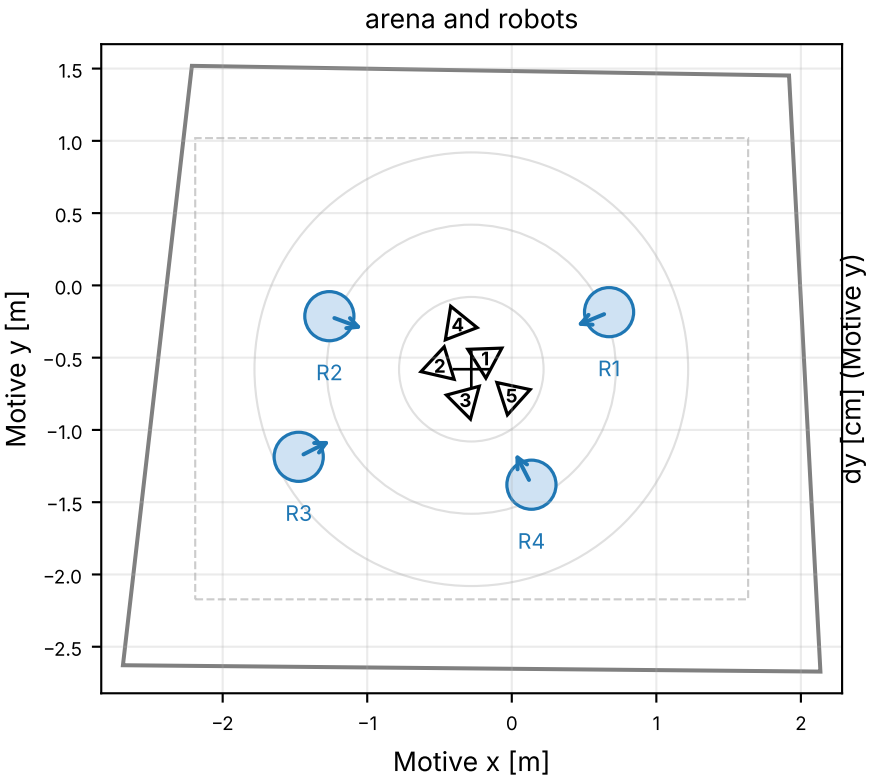
Place and capture: python -m tools.place_check --layout R73

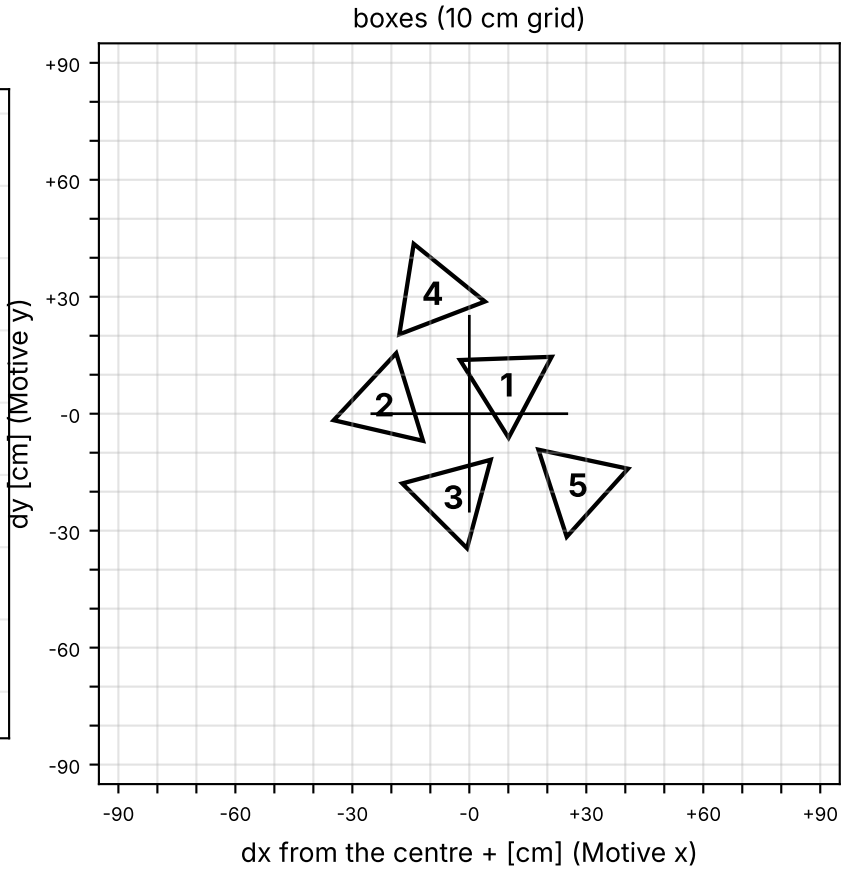
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R73 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R74 day 5 E luce (test)

5 boxes, fitta, robots: casuale | luce diversa (tende aperte o meta' luci spente)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+10	+7
2	-22	+2
3	-4	-21
4	-9	+31
5	+28	-18

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.67	-0.18
R2	-1.26	-0.21
R3	-1.47	-1.19
R4	+0.14	-1.38

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

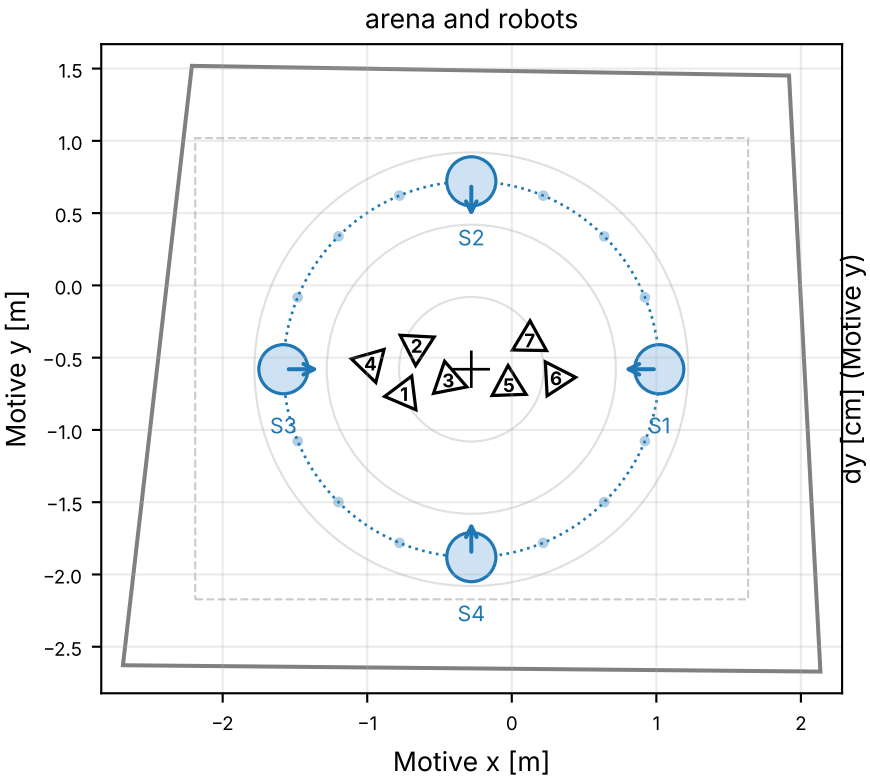
Place and capture: python -m tools.place_check --layout R74

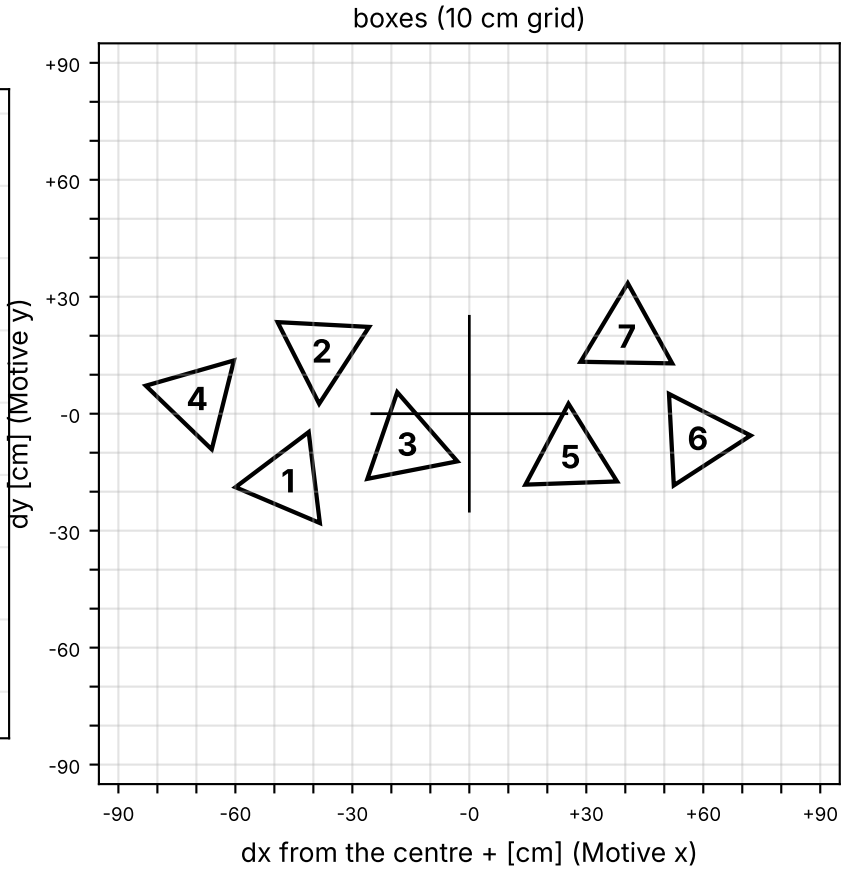
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R74 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R75 day 5 E luce (test)

7 boxes, due gruppi, robots: cerchio 1.3 | luce diversa (tende aperte o meta' luci spente)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-46	-17
2	-38	+16
3	-16	-8
4	-70	+4
5	+26	-11
6	+59	-6
7	+40	+20

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

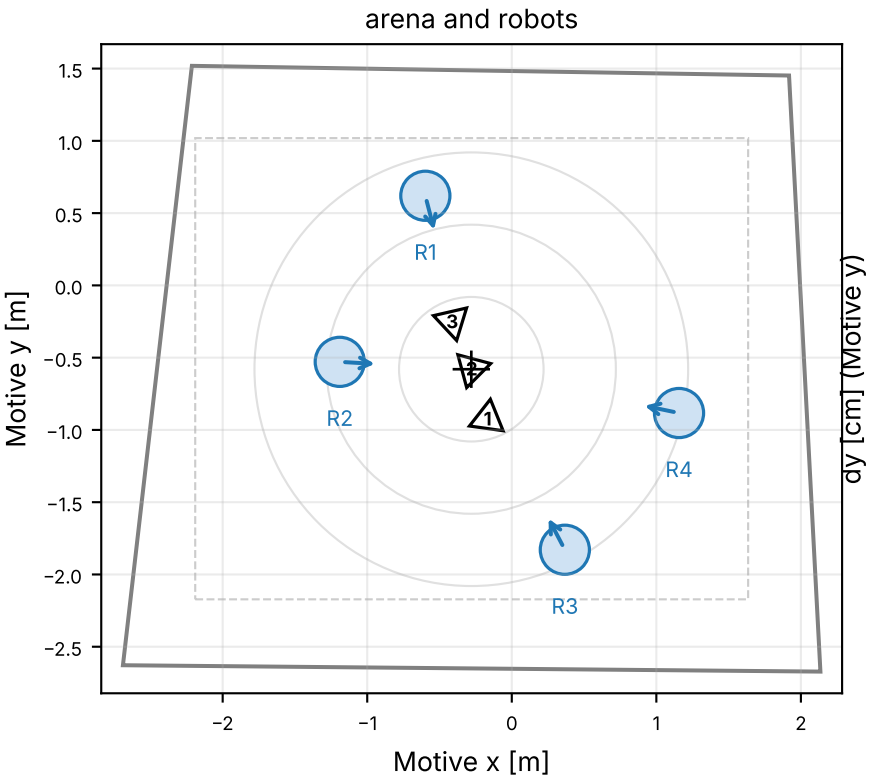
Place and capture: python -m tools.place_check --layout R75

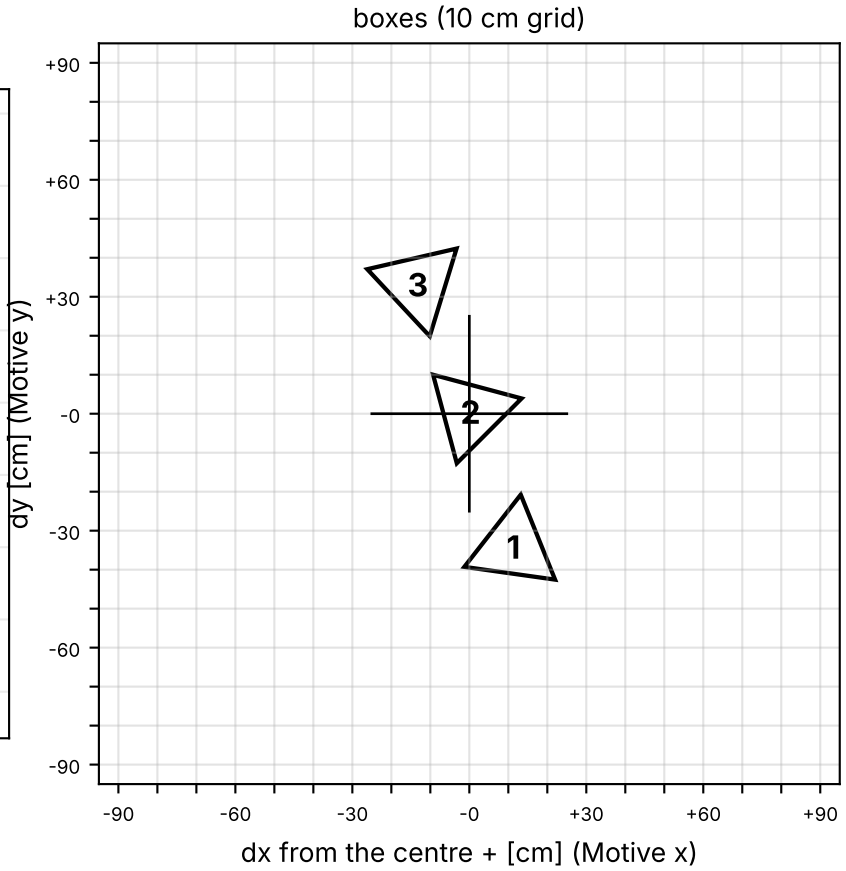
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R75 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R76 day 5 E luce (test)

3 boxes, fila/arco, robots: casuale | luce diversa (tende aperte o meta' luci spente)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+11	-34
2	+0	+0
3	-13	+33

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	-0.60	+0.62
R2	-1.19	-0.53
R3	+0.37	-1.83
R4	+1.16	-0.88

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

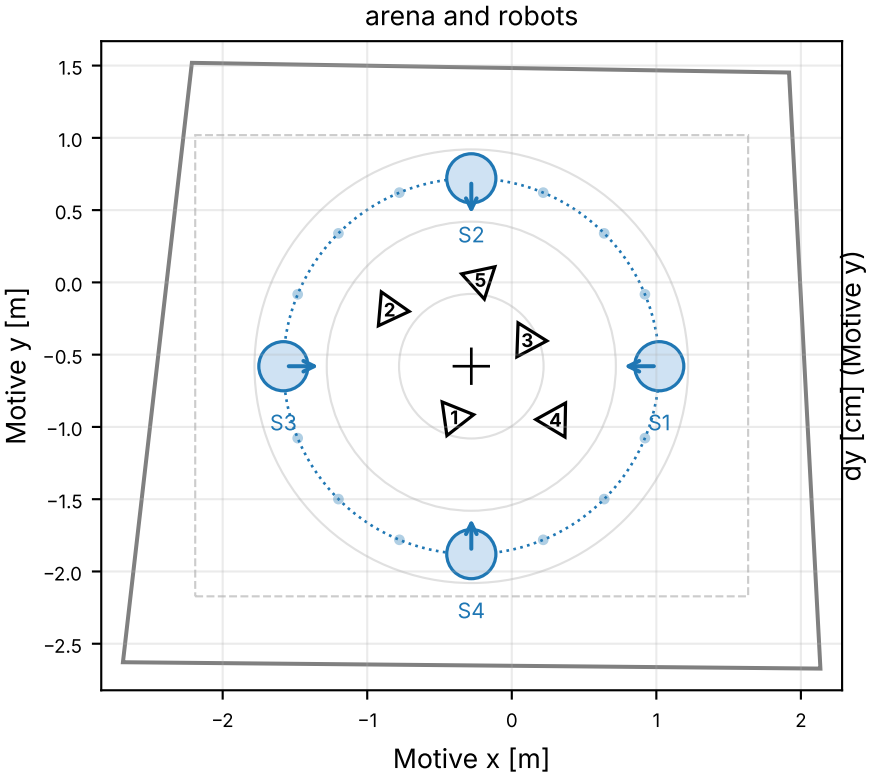
Place and capture: python -m tools.place_check --layout R76

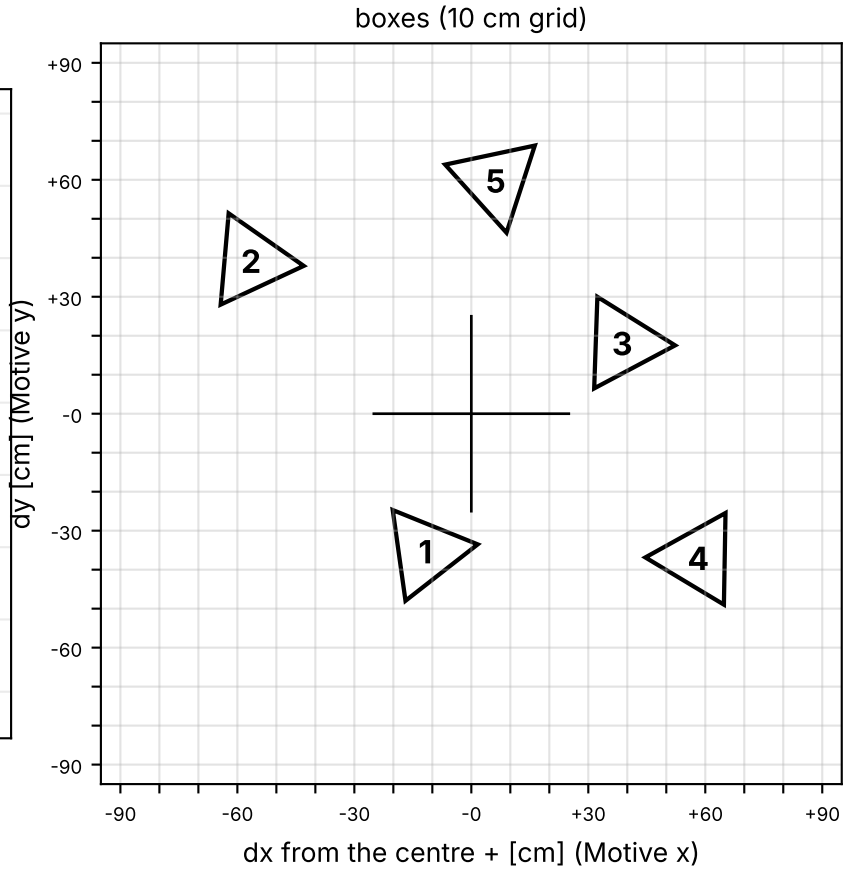
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R76 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R77 day 5 F camera girata (test)

5 boxes, sparsa, robots: cerchio 1.3 | --yaw-offsets-deg -20 0 20 (scatole anche ai bordi dell'immagine)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-12	-35
2	-56	+39
3	+39	+18
4	+58	-37
5	+6	+60

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

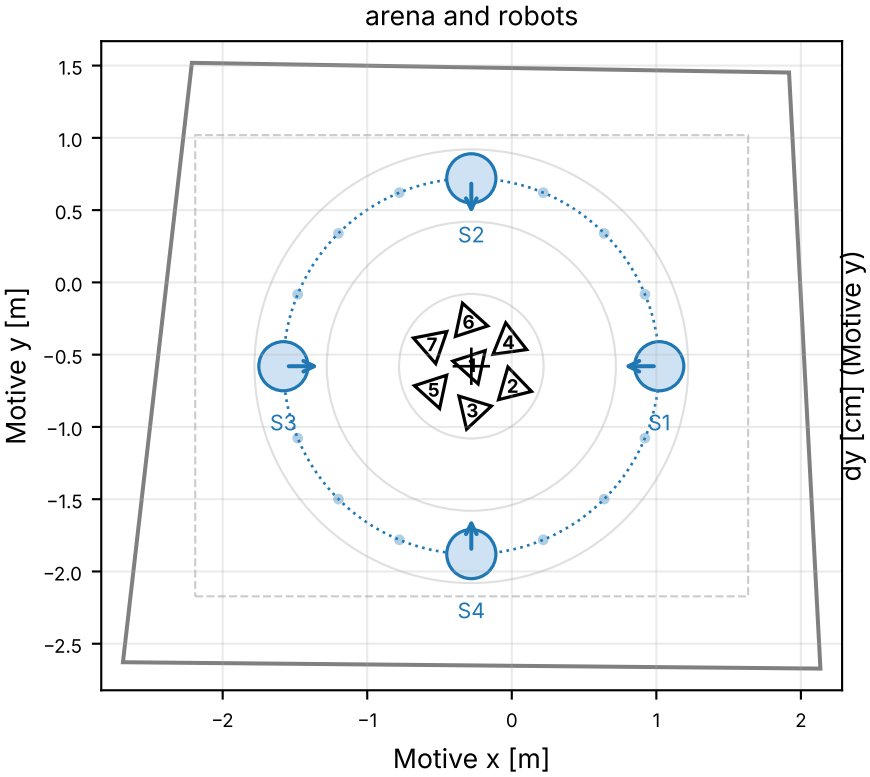
Place and capture: python -m tools.place_check --layout R77

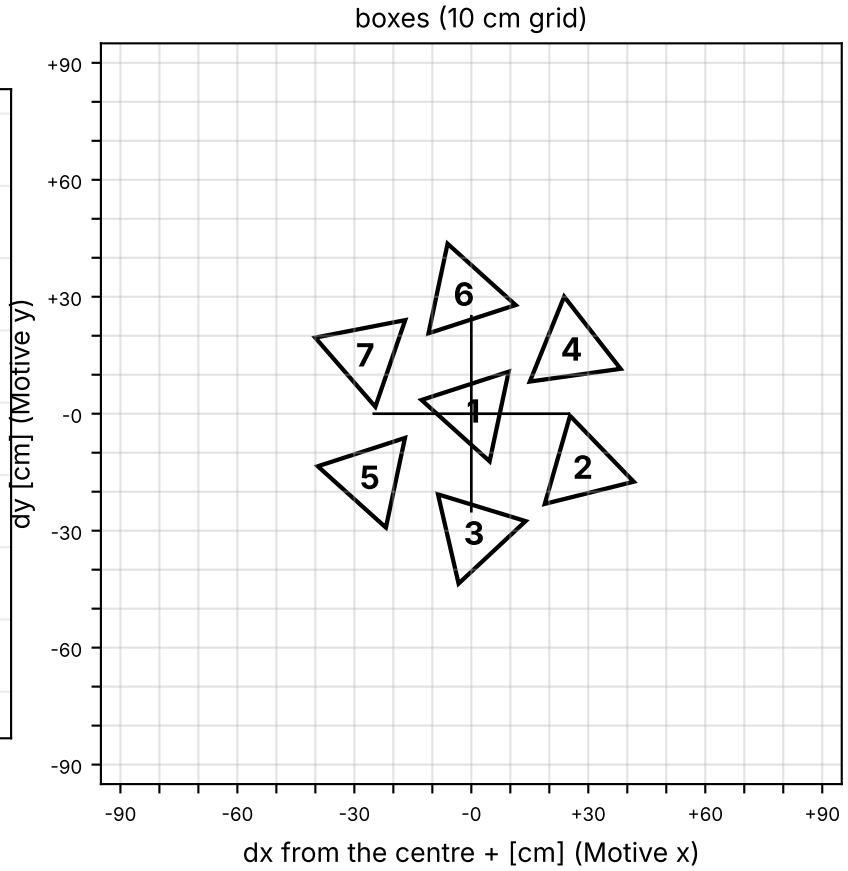
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R77 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3 --yaw-offsets-deg -20 0 20)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R78 day 5 F camera girata (test)

7 boxes, fitta, robots: cerchio 1.3 | --yaw-offsets-deg -20 0 20 (scatole anche ai bordi dell'immagine)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	+1	+1
2	+29	-14
3	+1	-31
4	+26	+17
5	-26	-16
6	-2	+31
7	-27	+15

Robots

anywhere near the 1.3 m circle: each drives to the nearest station S1-S4

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

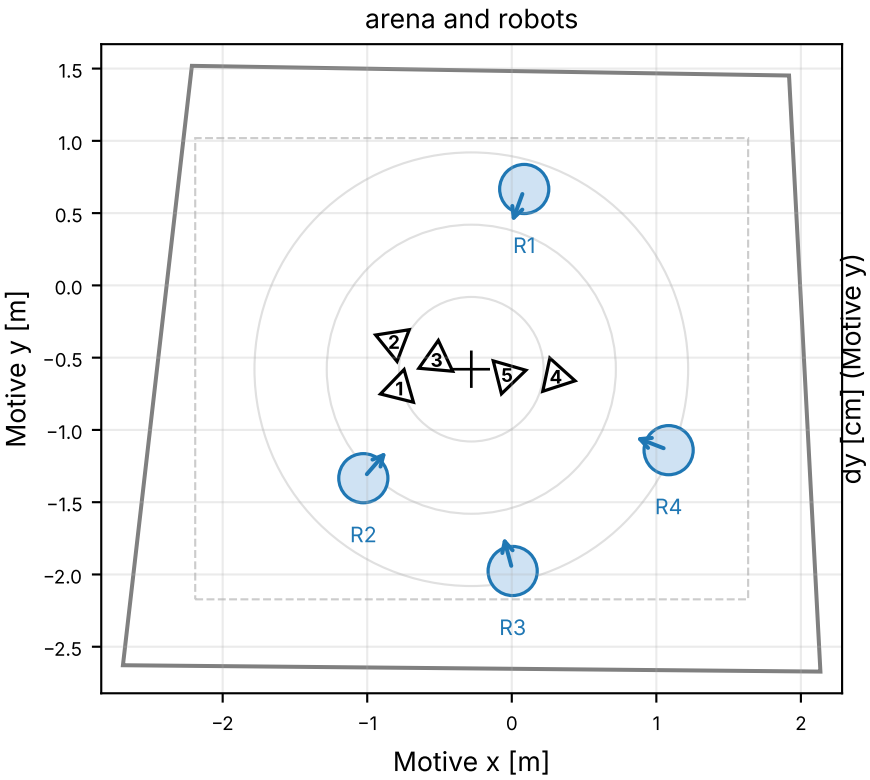
Place and capture: python -m tools.place_check --layout R78

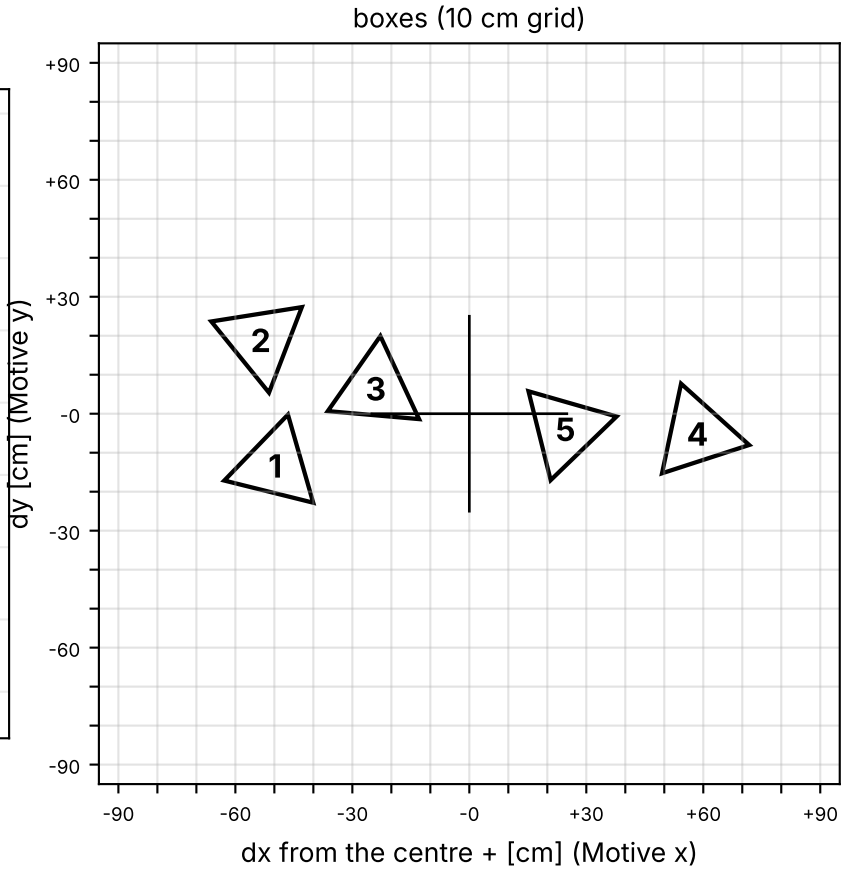
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R78 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --box 16:box --box 17:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --radius 1.3 --yaw-offsets-deg -20 0 20)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R79 day 5 F camera girata (test)

5 boxes, due gruppi, robots: casuale | --yaw-offsets-deg -20 0 20 (scatole anche ai bordi dell'immagine)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-50	-13
2	-53	+19
3	-24	+6
4	+58	-5
5	+25	-4

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.09	+0.67
R2	-1.03	-1.33
R3	+0.01	-1.98
R4	+1.08	-1.14

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

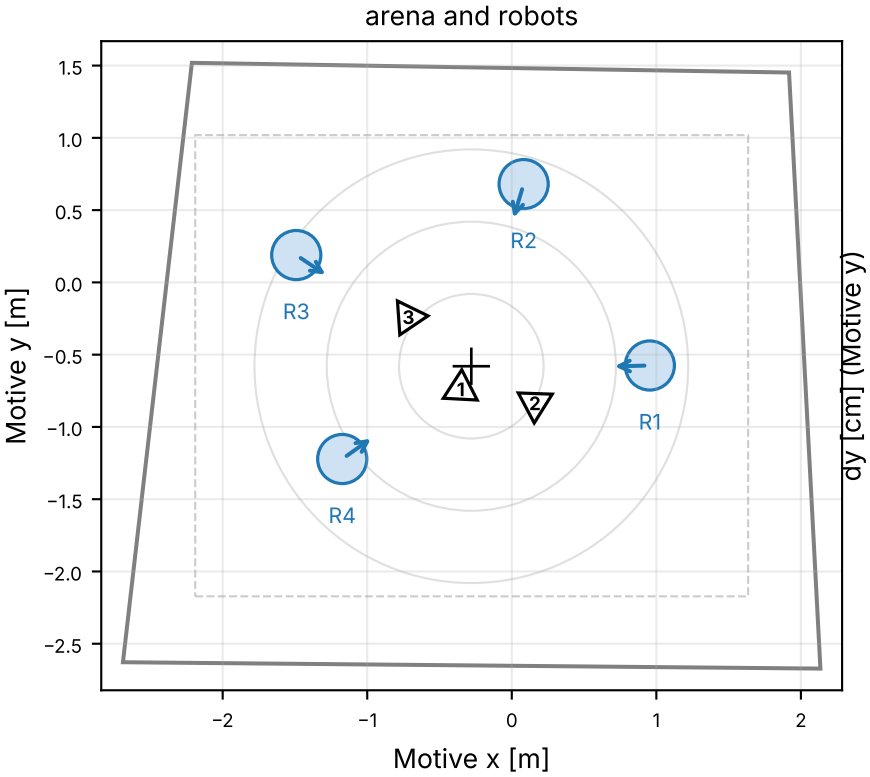
Place and capture: python -m tools.place_check --layout R79

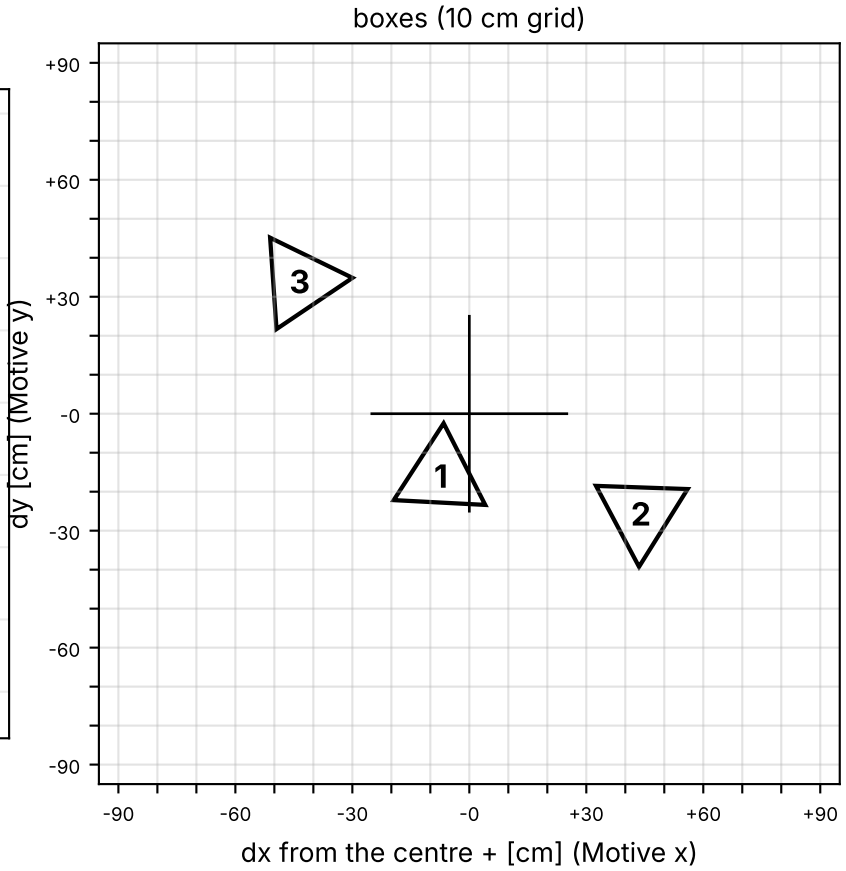
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot P5:6:config/checkerboard_calib_P1.json --layout-id R79 --box 11:box --box 12:box --box 13:box --box 14:box --box 15:box --ground-truth config/ground_truth_positions.json --regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12 --instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --yaw-offsets-deg -20 0 20)

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV

R80 day 5 F camera girata (test)

3 boxes, sparsa, robots: casuale | --yaw-offsets-deg -20 0 20 (scatole anche ai bordi dell'immagine)





Boxes from the centre +

#	dx [cm]	dy [cm]
1	-7	-16
2	+44	-26
3	-43	+34

Robots, round 0

by hand, facing the boxes

#	x [m]	y [m]
R1	+0.95	-0.57
R2	+0.08	+0.68
R3	-1.49	+0.19
R4	-1.17	-1.22

Rotate the boxes as drawn, by eye. Unused boxes out of the arena.

Place and capture: python -m tools.place_check --layout R80

```
(it runs: python -m acquisition.launch_all --robot P0:8 --robot P1:7 --robot P3:10:config/checkerboard_calib_P1.json --robot
P5:6:config/checkerboard_calib_P1.json --layout-id R80 --box 11:box --box 12:box --box 13:box --ground-truth config/ground_truth_positions.json
--regions config/optitrack_regions.json --out data/survey_v2/ --aruco-marker-size 0.077 --range-fit 1.064 -0.080 --camera-offset-forward-m 0.12
--instance-threshold-m 0.2 --step-deg 22.5 --sweep-deg 67.5 --no-robot-avoidance --circle-center -0.28 -0.58 --yaw-offsets-deg -20 0 20)
```

[ ] boxes placed [ ] robots placed [ ] BOXES rows show yaw [ ] 4 rounds done [ ] done in the CSV